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Z-residuals for Diagnosing Bayesian Models

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Response to the Associate Editor

July 6, 2026

We sincerely thank you for your time, valuable feedback, and general appreciation of the paper. Guided by the collective insights from you and other referees, we have substantially revised the manuscript. We have responded to your comments item by item, and your constructive suggestions have greatly improved both the methodology and the presentation. In this detailed response, we quote several updated passages for your reference (please note that some of these quoted excerpts may have been further refined in the final manuscript to ensure perfect flow and accuracy). In the revised manuscript, revised text is highlighted in orange; for substantially rewritten sections, only the section title is highlighted rather than the entire text.

In addition to those comments raised by the reviewers, please discuss when the Z-residuals would lead to a false detection.

Regarding your specific and important question about when Z-residuals might lead to false detections, we interpret this both specifically (practical scenarios leading to inflated Type I errors or misattributions) and broadly (inherent methodological limitations).

1. Multiple Testing Bias

Specifically, a primary source of false detection arises from multiple testing. If an analyst uses Z-residuals to hunt for correlated features among a large pool of candidate covariates without adjustment, the procedure is prone to false positives. We have emphasized this in Section 2.5, adding the following guidance:

> "if Z-residuals are used to hunt for correlated features among a large pool of unused covariates, or if a vast battery of pre-specified tests is deployed to criticize a model, the procedure remains subject to multiple testing bias. In such scenarios, the results must be interpreted with caution and accompanied by robust multiplicity corrections. Appropriate strategies include established false discovery rate controls (Benjamini & Hochberg 1995), E-value calibration techniques (Vovk & Wang 2021, Ramdas et al. 2023), or data-splitting methods that reserve a hold-out set for final model evaluation."

2. ISCV Variance Inflation and Dependence

A second cause of false detections (inflated Type I error rates) occurs under the importance-sampling cross-validatory (ISCV) framework in finite samples. Because ISCV Z-residuals suffer from structural variance inflation and do not completely eliminate dependence, they can falsely reject the null more often than posterior Z-residuals. We clarify this in the Sec. 2.4 of the revised manuscript:

> "In practice, we generally recommend posterior Z-residuals as the default tool for routine diagnostics of models without highly localized latent effects. While the cross-validatory construction is theoretically appealing because it alleviates the direct self-influence of y_i on its own predictive distribution, ISCV Z-residuals are not exactly standard normal in finite samples—their variance is greater than 1 for high-leverage observations, akin to why frequentist ordinary least squares relies on studentized residuals. Furthermore, this construction does not completely eliminate dependence in the resulting LOO Z-residuals, as the training sets $\boldsymbol{y}_{-i}$ and $\boldsymbol{y}_{-j}$ share $n-2$ observations. In our simulations, this variance inflation and residual dependence, combined with importance-sampling approximation errors, resulted in higher Type I error rates for ISCV Z-residuals. Conversely, posterior Z-residuals are surprisingly more stable at smaller sample sizes, and their double-use bias vanishes asymptotically as they converge to standard normal (as justified in the appendix). Nevertheless, because ISCV computation is fast, it is good practice to compute both, as a substantial discrepancy between the two is itself informative—typically flagging highly influential observations or unstable importance-sampling approximations (e.g., large Pareto k diagnostic values). Finally, ISCV Z-residuals remain distinctly superior for models with observation-specific latent variables (Li et al. 2016, Millar 2018), as including y_i during fitting exerts an overwhelming influence on its corresponding latent effect."

3. Cascading Misspecification (False Attribution)

An inadequate diagnostic procedure can lead to false conclusions about the *source* of the error. For example, an unmodeled nonlinear covariate effect can cascade into the marginal distribution, causing a rejection in the marginal Z-residual QQ plot and SW test. This could lead an analyst to falsely conclude that the distributional family needs changing. To prevent this, we added the following to Section 2.5:

> "It is worth noting that we do not prescribe a rigid order for model checking. Rather, effective model diagnostics should be regarded as a synthesis of diagnostic information aimed at identifying structural problems. *For example, severe mean-structure misspecifications can cascade and cause failures in the marginal QQ and SW checks.* By jointly evaluating these marginal metrics alongside covariate-specific plots and AOV tests, the analyst can better untangle these cascading effects and identify the true cause of the misfit."

4. Broad Methodological Limitations

Finally, interpreting your question more broadly, we recognize an inherent limitation of Z-residuals when compared to internal structural checks like PPC, HPC, or the new UPC. Because Z-residuals strictly check the observation-level predictive distribution against y_i, they can falsely "clear" a model (a false negative, or failing to detect structural flaws) if the misspecification lies deep within the model's internal hierarchy (e.g., poorly specified priors or latent structures) but does not strongly manifest at the observation level. We explicitly address this limitation in Section 5:

> "The core strength of the Z-residual framework lies in assessing the approximated predictive distribution (posterior or ISCV) against the actually observed y_i. Because this predictive distribution converges to the true data-generating distribution as the sample size increases, our method yields observation-level residuals that are asymptotically normal under the null. However, this strict observation-level focus entails an inherent **limitation**: the method is largely insensitive to non-observation-level components, such as latent variables, hierarchical effects, and priors. Consequently, the proposed Z-residuals are less effective at detecting misspecification within a model's internal structure.
>
> To address this limitation, a highly promising direction for future work is to extend the RSP framework to the evaluation of internal model components, drawing inspiration from Uniform Parametrization Checks (UPC) (Covington & Miller 2025). $\cdots$ Combining UPC's novel perspective on internal random components with the asymptotic stability of our "summarize-first" RSP framework offers an interesting avenue for diagnosing latent structures. For this combination to succeed, the primary methodological challenge will be to formally define what constitutes an "observed" value for these internal components, so that their corresponding RSPs can be validly constructed and assessed. "

References

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Response to Reviewer A

July 6, 2026

We sincerely thank you for your time, valuable feedback, and general appreciation of the paper. Guided by the collective insights from you and other referees, we have substantially revised the manuscript. We have responded to your comments item by item, and your constructive suggestions have greatly improved both the methodology and the presentation. In this detailed response, we quote several updated passages for your reference (please note that some of these quoted excerpts may have been further refined in the final manuscript to ensure perfect flow and accuracy). In the revised manuscript, revised text is highlighted in orange; for substantially rewritten sections, only the section title is highlighted rather than the entire text.

General Comments

General comment. This is an interesting piece of work. The authors propose a Z-residual concept that applies to general Bayesian models. The following issues need to be addressed. The major issues are #3, #4, #6, and #7.

1. Page 3, line 36. Can the authors explain why you use the term "Z-residual"?

Response: Thank you for this helpful suggestion. We use the term "Z-residual" because the proposed residual is represented on the standard-normal, or Z-score, scale. Specifically, each observation is first transformed to a predictive probability scale through the randomized survival probability, and this probability is then mapped to the standard-normal scale by the normal-score transformation, ($z_i = -\Phi^{-1} RSP_i(y_i)$). Under correct model specification, the parameter-wise randomized survival probability is uniform, so the corresponding residual has a standard normal reference distribution. Thus, the term "Z-residual" emphasizes the normal-score scale of the residual, rather than implying that it is a classical additive residual.

We have clarified this naming in Section 2.2, in the paragraph immediately following equation (2), by adding a sentence explaining that the term "Z-residual" refers to this standard-normal reference scale.

2. Page 3, line 40. Can the authors clarify if U_i is needed in the randomized survival probability when the outcome variable Y is continuous?

Response: Thank you for this helpful comment. When (Y) is continuous, the auxiliary random variable (U_i) is not needed. The randomization term is introduced to handle probability mass at observed values in discrete outcomes. For a continuous outcome, the probability mass at any exact observed value is zero, so ($p_i(y_i \mid \theta) = 0$) and the randomized survival probability reduces to the usual survival-probability transform. Equivalently, the construction becomes a probability-integral-transform type residual followed by the normal-score transformation.

We have clarified this point in Section 2.2, in the paragraph immediately following equation (1), by stating that (U_i) is only needed when the Response:has probability mass at the observed value and that, for continuous Response:, the construction reduces to ($S_i(y_i \mid \theta)$). We also reiterate this point in the first paragraph of Section 3.2 when introducing the continuous inverse Gaussian simulation.

3. **(How to position this paper?)** If U_i is not needed when Y is continuous, the authors need to clarify the main contribution of the paper. When Y is continuous, is your residual still novel in the Bayesian literature? Does the significance still remain? If not, the authors need to reposition this paper by saying that the novelty resides in the discrete case.

Response: Thank you for raising this important point. We agree that for continuous responses, the auxiliary randomization U_i is unnecessary, and the pointwise probability transform is simply the probability integral transform.

We entirely agree that the PIT and LOO-PIT are not novel concepts, and while Z-residuals (randomized quantile residuals) have been recently discussed much in frequentist statistics in the past decade, they remain largely absent from Bayesian model assessment. Therefore, our primary methodological contribution lies not in inventing these foundational concepts, but in synthesizing them into a framework for constructing **observation-level residual diagnostics** in applied Bayesian modeling. By combining randomized residuals with observation-level summaries over posterior or LOO predictive distributions, we address a critical, overlooked gap: the shift from overall model *checking* to granular residual *diagnostics*. Traditional goodness-of-fit (GOF) tests typically evaluate the overall distribution or rely on a single aggregate p-value to merely confirm or reject a model. In contrast, our expanded framework—supported by theoretical justification for its asymptotic calibration—facilitates rich exploratory analyses using standard graphical tools (e.g., scatterplots, boxplots, and quantile-quantile plots) alongside formal statistical tests. This enables practitioners to systematically pinpoint and understand specific structural flaws within the model. These tests include, but are not limited to: the ANOVA test for misspecified covariate functional forms (introduced in this paper); the Box-Ljung test for temporal autocorrelation; Moran's I statistic for global spatial dependencies; and the Breusch-Pagan and Bartlett tests for heteroskedasticity. Ultimately, the proposed Z-residuals enable the seamless incorporation of the full suite of frequentist diagnostic tools into Bayesian analysis.

To bring the rich suite of residual diagnostic tools into Bayesian statistics, we introduce the following **specific methodological innovations** in this paper:

- **A "summarize-first, then-test" paradigm:** We construct Z-residuals by integrating randomized survival probabilities (RSPs) over the posterior distribution prior to applying conventional diagnostic tools. This fundamentally replaces the traditional posterior predictive check (PPC) "test-first, then-summarize" approach, which evaluates discrepancy statistics for each MCMC draw and then summarizes the resulting p-values. Despite its conceptual simplicity, this reversal yields a diagnostic tool with solid theoretical foundations that unifies frequentist and Bayesian workflows. Notably, the construction of Z-residuals is independent of any specific test; therefore, they can be subsequently analyzed using any graphical or numerical diagnostic tool.

- **Frequentist asymptotic justification:** We provide a rigorous theoretical proof demonstrating that the posterior and LOOCV RSPs are asymptotically uniformly distributed, and the resulting Z-residuals are asymptotically normally distributed from a strictly frequentist perspective. This theoretical grounding validates the application of existing frequentist residual diagnostic tools to Bayesian models.

- **A distribution-free upper-bound p-value:** To overcome the auxiliary randomization introduced for discrete responses, we provide a novel, distribution-free upper bound to reliably summarize replicated p-values (or, more broadly, dependent p-values). This technique has broad applicability beyond the current context.

In terms of empirical findings, this paper establishes the following critical points:

- Relying solely on the marginal distribution of PIT or Z-residuals for overall goodness-of-fit (GOF) checking is fundamentally inadequate. As we demonstrate, covariate-specific tests—such as the Z-AOV—successfully detect functional form misspecifications, highlighting how Z-residuals can be leveraged to diagnose a wide array of structural flaws. These subtle deficiencies are often completely masked when inspecting only the marginal distribution of PITs or relying on simple summary statistics that ignore relationships with covariates.

- Graphical diagnostics utilizing Z-residuals serve as powerful exploratory tools for *discovering* unknown model deficiencies. Crucially, these visual insights guide the subsequent application of formal statistical tests, whether employing posterior/ISCV Z-residuals or conducting Z-residuals-PPC tests.

- Tests based on Z-residuals are well-calibrated and yield significantly higher statistical power than both traditional PPCs utilizing the χ^2 statistic and Z-residual-based PPCs. This dual advantage in calibration and power is attributed to two key factors: (1) unlike arbitrary discrepancy measures, the exact distribution of Z-residuals is known under the true predictive distribution; and (2) our "summarize-first, then-test" paradigm asymptotically approximates this true predictive distribution, as formally justified by the theory presented in this paper.

- Our simulations demonstrate that Z-residual-based PPCs mitigate the notorious conservatism of traditional χ^2 PPCs. Because they average over the MCMC draws, they remain highly stable despite the auxiliary $\boldsymbol{U}$ drawn at each iteration, serving as a reliable secondary confirmation of model misfit.

We have substantially revised Section 1 and Section 5, incorporating the points above to more clearly articulate our contributions and findings.

4. **(Lack of sensitivity analysis)** Does the choice of the prior influence diagnosis? In particular, when the sample size is small, an ill-chosen prior may lead to a wrong rejection of the model. Am I right? The authors should perform sensitivity analysis.

Response: Thank you for raising this important point. You are absolutely correct that if a very poor or ill-chosen prior is used, particularly with a small sample size, it can lead to the rejection of the model. However, from a Bayesian perspective, the prior is an integral component of the model specification. If an inappropriate prior severely degrades the model's predictive fit, rejecting the model is not a "wrong" rejection, but exactly a type of misspecification our diagnostic tools are devised to identify. Furthermore, because Z-residuals focus specifically on the predictive distribution of the observed data y_i, they are generally robust to routine prior choices. The diagnostic is not overly sensitive to prior misspecification unless the prior is highly influential and strictly conflicts with the likelihood. In most practical scenarios, the data will adequately inform the posterior predictive distribution, keeping the Z-residuals stable.

To illustrate the robustness of our diagnostic conclusions to the choice of prior, we conducted a sensitivity analysis on the dolphin-sighting data under baseline, tighter, and more diffuse prior specifications (Supplementary Table S5). After refitting the models and recomputing all diagnostic p-values (Z-residuals, PPC, and HPC), we found the qualitative patterns to be highly stable (Supplementary Figure S4). Most notably, the diagnostics consistently reject the linear HNB and NB models while accepting their nonlinear counterparts across all three prior regimes. We have added this full analysis to the Supplementary Material and noted the robustness of these findings in Section 4.

5. Page 17, Figure 1. I am not sure about the value of plotting Z-residuals against the index. What insights can the plots offer in addition to those offered by the QQ plots?

Response. Thank you for this helpful comment. We agree with your assessment and have removed the residual-versus-index plots, replacing them with residual-versus-covariate plots in the revised manuscript. While index plots can sometimes be useful in practice—since data indices are often non-random and may reveal order-dependent patterns or temporal clustering—we completely agree that for our specific simulation studies, the index contains no meaningful information.

6. **(Lack of numerical studies with respect to other Bayesian models)** The authors claim that they proposed "general-purpose residuals". But the numerical studies only focus on (hurdle) negative binomial models. This is relevant to my comment #3. If the authors position this paper as a general methodology, then more numerical studies using different models (for continuous and discrete data) should be presented.

Response. Thank you for this important suggestion. We agree that focusing solely on discrete count models limited our broader methodological claims. To address this and provide more diverse experimental scenarios, we have substantially broadened our numerical studies:

- **Added a Continuous Inverse Gaussian (IG) Simulation (Section 3.2):** We replaced the original count-data simulation in the main text with a continuous IG regression study. This example explicitly demonstrates the fundamental performance differences between PPCs and Z-residuals without the potential interference of the auxiliary randomization required for discrete outcomes.
- **Added a Mixed Misspecification Simulation (Section S1.2):** Addressing your advice for broader experimental scenarios, and motivated by another reviewer's feedback, we added a new simulation to the Supplementary Material that explores simultaneous misspecifications in both distributional families and covariate functional forms.
- **Relocated the Original Simulation (Section S1.1):** The original HNB-versus-NB wrong-family simulation has been moved to the Supplementary Material to ensure the main text focuses on the broader applicability of the methods.

7. **(Double use of the data)** I like the point the authors make about the double use of data in fitting and diagnosing the model. My question is: although you do not use y_i, you do use the rest of the data. Why does this not count as "double use of the data"? Furthermore, y_i is generated from the same model that generates other y's. So I believe your approach still has the same problem. The importance-sampling approach presented in Section 2.4 may alleviate the severity of the problem.

Response. Thank you for this insightful comment. You are absolutely correct: both the proposed posterior and cross-validatory (ISCV) Z-residuals technically reuse the data. The fundamental distinction between our approach and standard Posterior Predictive Checks (PPCs) lies not in the complete elimination of data reuse, but in the *severity* of the bias it causes, particularly in large samples.

To clarify this distinction, we have revised Section 2.7 to summarize the asymptotic differences:

- **Tests on Z-residuals:** Z-residuals are constructed observation-by-observation. As the sample size $n \to \infty$, the influence of any single observation y_i on the full posterior predictive distribution vanishes. Consequently, the full-data posterior predictive law and the LOO predictive law become asymptotically indistinguishable, both converging to the true data-generating law. Thus, the posterior and cross-validatory Z-residuals are asymptotically independent and follow a $N(0,1)$ distribution.

- **PPCs:** In contrast, a PPC evaluates a dataset-level discrepancy statistic $T(\mathbf{y}, \boldsymbol{\theta})$. Because the *entire* dataset $\mathbf{y}$ is used simultaneously to simulate the reference distribution, the simulated data is systematically pulled toward the observed data. This dataset-level double use causes PPC p-values to be inherently conservative even asymptotically (Robins et al. 2000). Specifically, as the sample size $n \to \infty$, the downward bias in the expected discrepancy does not wash out; rather, it converges to an $O(1)$ constant determined by the parameter count, p. Conceptually, this phenomenon parallels the optimism bias formalized by Wilks' theorem in frequentist statistics: just as maximum likelihood estimation absorbs data noise to produce a non-vanishing p-dimensional upward bias in the log-likelihood, the double use of data in PPCs systematically shrinks the expected discrepancy by p degrees of freedom. Because this penalty is permanent, it severely limits the statistical power of standard PPCs to detect subtle misfits.

Regarding your specific point about the cross-validatory approach: we agree that because y_i and y_j are generated from the same model, and their respective training sets (y_{-i} and y_{-j}) share $n-2$ observations, this overlapping data inherently introduces finite-sample correlation between the LOO Z-residuals. We have revised the text to explicitly state that ISCV alleviates the direct self-influence of y_i, but does not eliminate this correlation.

However, despite these finite-sample dependencies, our simulation studies demonstrate that the practical impact of data double-use in posterior Z-residuals is remarkably minor when aggregate diagnostic tests (like Shapiro–Wilk and ANOVA) are employed. The p-values from these tests remain nearly uniform under the null, showing vastly superior calibration and power compared to standard PPCs (especially those relying on the common χ^2 statistic).

Finally, we have added a note clarifying that when data is highly abundant, strict data-splitting to reserve a hold-out test set remains the most definitive way to completely eradicate data double-use.

8. **(Insufficient simulation replications)** Page 20, Figure 3. The authors should increase the simulation replications to 1000 if type I errors are part of the results.

Response. Thank you for this suggestion. While 1000 replications is ideal, the massive computational burden of Bayesian modeling—where a single MCMC fit can take 20 mins—makes this prohibitive across our expanded simulation suite. To practically estimate the rejection rates, we increased all scenarios to 500 replications. At $R = 500$, the Monte Carlo standard error for a nominal

0.05 Type I error rate is approximately $\sqrt{0.05(1-0.05)/500} \approx 0.010$, yielding a 95% margin of error of roughly ± 0.019. For a rejection rate near 0.50 (where variance is maximized), the standard error is $\sqrt{0.50(1-0.50)/500} \approx 0.022$, corresponding to a margin of error of about ± 0.044. This precision is sufficient for discerning diagnostic behaviors across different methods, and we have detailed these uncertainty calculations in the Supplementary Material.

9. **(Inflated type I error)** Page 20, Figure 3(a). Please report type I errors in numerical form. By reading from the graphical presentation, I feel that some of the type I errors are inflated, close to 0.1. The same can be said for Figure 6.

Response. Thank you for this valuable suggestion. We agree that graphical presentations alone are insufficient for accurately assessing Type I error inflation. We have fully taken your advice and compiled the exact numerical rejection rates for all simulation settings. Due to page limitations in the main manuscript, these detailed numerical summaries have been placed in the Supplementary Material. Please refer to Tables S1, S3 and S4, which explicitly report the empirical Type I error rates and statistical power, complete with Monte Carlo standard error calculations to clarify the Monte Carlo estimating errors.

10. **(Bin size in Figure 4)** How do the authors determine the bin size? The analysis of variance (and its p-value) is heavily dependent on how you bin the residual points.

Response. Thank you for pointing this out. We agree that the grouped ANOVA diagnostic depends on the binning rule, which must be made explicit. In the revised Section 2.5, we specify that for continuous covariates, we divide the observed range into k equally spaced intervals, recommending $k = 10$ as a moderate default to balance resolution against group-size stability.

To address your concern, we added a binning sensitivity analysis to the Supplementary Material (Table S7, Figure S6). Using the dolphin-sighting data, we evaluated $k \in \{5, 10, 15, 20\}$, summarizing 500 randomizations per k via the median upper-bound p-value. The qualitative conclusions remain remarkably stable: the linear HNB model consistently shows residual patterns against depth, while the nonlinear HNB model does not.

This stability likely stems from the automatic degrees-of-freedom (df) adjustment inherent in the ANOVA test. When k is small, the captured difference in group means is smaller, but the df for the Mean Squared Error (MSE) is larger. Conversely, when k is large, the mean differences may be more pronounced, but the MSE df decreases. In practice, we recommend fixing $k = 10$ as a default, though checking a few different values of k remains a sensible strategy to verify whether a functional form misspecification is present.

11. **(Tone down the self-promoting statements)** Page 15, line 40: "very promising". Page 30, line 26: "excellent performance". Please use objective statements.

Response. Thank you for this valuable suggestion. We agree that certain expressions in the original manuscript were overly promotional. Throughout the revised manuscript—particularly in the Abstract and Sections 1, 3, 4, and 5—we have replaced subjective phrases (e.g., "very promising," "excellent performance," "superior") with neutral, objective alternatives (e.g., "useful," "favorable," "better in the settings considered"). Furthermore, we carefully revised all comparative statements to ensure they are strictly grounded in our specific simulation or data-analysis settings, rather than reading as broad, generalized claims.

12. **(Reduce the excessive use of uncommon abbreviations)** The authors used GOF, PPC, PPP, RSP, LOO, ISCV, NB, HNB, MSP, SW, PIT, AOV, and other uncommon abbreviations. They do make the reading difficult. I have to flip the pages and spend a lot of time to find what they really mean. The journal *Biometrika* explicitly forbids the use of uncommon abbreviations. Please remove as many as possible.

Response. Thank you for this helpful comment. We agree that excessive abbreviations hindered readability, and we have significantly reduced their use throughout the revised manuscript. We now write out low-frequency abbreviations (e.g., CDF, PMF, LOOCV) and general terms (e.g., goodness-of-fit) in full. However, to prevent excessive length and maintain clean layouts, we retained a small set of core, high-frequency abbreviations (e.g., RSP, PPC, ISCV, HNB, SW). These are clearly defined at first use and, for the reader's convenience, *we have now added a list of these core abbreviations to the first page of the manuscript.* Finally, we updated figure and table captions to explicitly spell out key tests (such as the Shapiro–Wilk and ANOVA tests) so they function as self-contained references without requiring readers to search for definitions.

References

Robins, J. M., Van der Vaart, A. & Ventura, V. (2000), 'Asymptotic distribution of p values in composite null models', *Journal of the American Statistical Association* **95**(452), 1143–1156.

Response to Reviewer B

July 6, 2026

We sincerely thank you for your time, valuable feedback, and general appreciation of the paper. Guided by the collective insights from you and other referees, we have substantially revised the manuscript. We have responded to your comments item by item, and your constructive suggestions have greatly improved both the methodology and the presentation. In this detailed response, we quote several updated passages for your reference (please note that some of these quoted excerpts may have been further refined in the final manuscript to ensure perfect flow and accuracy). In the revised manuscript, revised text is highlighted in orange; for substantially rewritten sections, only the section title is highlighted rather than the entire text.

Overall Comments

1 Summary. The manuscript deals with the problem of diagnosing Bayesian models using randomised survival probabilities (RSPs) based on full posterior draws, as well as leave-one-out (LOO) posterior draws. The authors develop theoretical guarantees relating to the asymptotic uniformity of these RSPs under both full posterior and LOO posterior draws, and present some numerical demonstrations on simulated and real data demonstrating their promise.

I will quickly give some context on my own expertise, and thus my interests and limitations as a reviewer. My work has primarily focussed on understanding the theoretical characteristics of Bayesian prediction, prior elicitation, and model selection. While I have come across some of the methods discussed in this manuscript, and some other literature which I believe to be related but unreferenced by the authors, my own related expertise lies in using cross-validation for the same ambitions. As such, I apologise in advance should some of my comments be misguided.

1.1 Overall evaluation. Bayesian model diagnostics, and in particular predictive diagnostics, are becoming increasingly important in our field. And many of the approaches I have seen used in practice exist without the theoretical guarantees the authors have developed here. As such, I believe the ambition of this manuscript to be very well placed. The writing of this manuscript, however, leads me to be unclear as to where the authors' contributions lie. In particular, I believe much of the manuscript could be trimmed and cut down to reveal more clearly the fundamental notions and intuition which are missing for a reader such as myself. And given the apparent empirical benefits of this direction, I would like to see a greater variety in numerical examples. This, in combination with my principal concern that much of what the authors claim as novelty appears to me to be pre-existing, at least in spirit, in unreferenced literature and software, leads me to be unconvinced as to the manuscript's readiness for publication in its current form. I believe that many of my points below could be handled by the authors, and with some clarification and editing, this manuscript could potentially fill a gap which I feel is indeed missing in the literature and would be valuable to the readership of the Journal of the American Statistical Association and beyond.

Response: We sincerely thank the reviewer for their time, careful reading, and constructive feedback. We are highly encouraged by your recognition that "Bayesian model diagnostics, and in particular predictive diagnostics, are becoming increasingly important in our field." We also deeply appreciate your encouraging assessment that this work "could potentially fill a gap which I feel is indeed missing in the literature and would be valuable to the readership of the Journal of the American Statistical Association."

Regarding your concern about novelty, we agree that the specific techniques used to construct Z-residuals may appear plain, particularly to those familiar with statistical learning. However, the core contribution of this paper lies in demonstrating that the *combination* of these techniques yields exceptionally well-calibrated and powerful diagnostic tools. While the individual mathematical components may be established, the development of a cohesive, theoretically justified residual diagnostic framework as a whole is highly novel to applied Bayesian statistics.

We have thoroughly revised the manuscript to streamline the exposition, clarify our fundamental contributions, and address your specific points regarding missing literature and numerical examples. Please find our detailed responses below.

Major comments

2.1a Clarity of novelty and historical context. My principal concern with the manuscript in its current state is the clarity of its novelty. In particular, the authors point out that the RSP transform is distributionally equivalent to the probability integral transform (PIT). Well, the use of PIT and the leave-one-out version of the PIT (LOO-PIT) for model diagnostics is not a novel idea: it was first discussed, to the best of my knowledge, in a reference which I would consider a somewhat critical omission from the manuscript (Gelfand et al., 1992, and in particular the discussion provided by Prof. Francoise Seillier-Moiseiwitsch). And the theoretical results the authors present seem to me to be closely related to pre-existing results mentioned therein: see, e.g., Theorem 2.1 of O'Reilly and Quesenberry (1973), Theorem 1 of Seillier-Moiseiwitsch and Dawid (1993), and the review paper of Seillier-Moiseiwitsch and Seillier-Moiseiwitsch (1993). While these results are slightly different from what the authors present—namely some of them assume a natural ordering of data, though not all—it is not clear to me how different they are, and the importance of these differences in terms of the fundamental message and novelty of the manuscript at hand. Do I understand correctly that the primary contribution of the manuscript is a synthesis and extension of these previous contributions to new theoretical results which are missing in the literature? Or is the p-value-style testing itself the authors feel is novel? [...] At the very least, a thorough literature review making clear what results are already available, what tools are already implemented in software, synthesising these into a unified framework, and thus clarifying the gaps in the literature which the authors would endeavour to fill would greatly aid the manuscript.

Response: We thank the reviewer for this insightful critique and for pointing out the historical references. We agree that the manuscript needs to be clearer about its exact contributions, especially regarding the rich history of the probability integral transform (PIT) and its leave-one-out (LOO) variants. **Your understanding is entirely correct.** Our primary contribution is synthesizing and extending these foundational concepts into a versatile residual framework for modern Bayesian regression diagnostics.

We entirely agree that the PIT and LOO-PIT are not novel concepts, and while Z-residuals (randomized quantile residuals) have been recently discussed much in frequentist statistics in the past decade, they remain largely absent from Bayesian model assessment. Therefore, our primary methodological contribution lies not in inventing these foundational concepts, but in synthesizing them into a framework for constructing **observation-level residual diagnostics** in applied Bayesian modeling. By combining randomized residuals with observation-level summaries over posterior or LOO predictive distributions, we address a critical, overlooked gap: the shift from overall model *checking* to granular residual *diagnostics*. Traditional goodness-of-fit (GOF) tests typically evaluate the overall distribution or rely on a single aggregate p-value to merely confirm or reject a model. In contrast, our expanded framework—supported by theoretical justification for its asymptotic calibration—facilitates rich exploratory analyses using standard graphical tools (e.g., scatterplots, boxplots, and quantile-quantile plots) alongside formal statistical tests. This enables practitioners to systematically pinpoint and understand specific structural flaws within the model. These tests include, but are not limited to: the ANOVA test for misspecified covariate functional forms (introduced in this paper); the Box-Ljung test for temporal autocorrelation; Moran's I statistic for global spatial dependencies; and the Breusch-Pagan and Bartlett tests for heteroskedasticity. Ultimately, the proposed Z-residuals enable the seamless incorporation of the rich suite of frequentist diagnostic tools into Bayesian analysis.

To bring the rich suite of residual diagnostic tools into Bayesian statistics, we introduce the following **specific methodological innovations** in this paper:

- **A "summarize-first, then-test" paradigm:** We construct Z-residuals by integrating randomized survival probabilities (RSPs) over the posterior distribution prior to applying conventional diagnostic tools. This fundamentally diffs from the "test-first, then-summarize" approach used in traditional posterior predictive check (PPC), the newly proposed HPC and UPC, which evaluate discrepancy statistics for each MCMC draw and then summarizes the resulting p-values. Despite its conceptual simplicity, this reversal yields a diagnostic tool with solid theoretical foundations that unifies frequentist and Bayesian workflows. Notably, the construction of Z-residuals is independent of any specific test; therefore, they can be subsequently analyzed using numerous graphical or numerical diagnostic tools.

- **Frequentist asymptotic justification:** We provide a rigorous theoretical proof demonstrating that the posterior and LOOCV RSPs are asymptotically uniformly distributed, and the resulting Z-residuals are asymptotically normally distributed from a strictly frequentist perspective. This theoretical grounding validates the application of existing frequentist residual diagnostic tools to Bayesian models.

- **A distribution-free upper-bound p-value:** To overcome the auxiliary randomization introduced for discrete responses, we provide a novel, distribution-free upper bound to reliably summarize replicated p-values (or, more broadly, dependent p-values). This technique has broad applicability beyond the current context.

In terms of empirical findings, this paper establishes the following critical points:

- Relying solely on the marginal distribution of PIT or Z-residuals for overall goodness-of-fit (GOF) checking is fundamentally inadequate. As we demonstrate, covariate-specific tests—such as the Z-AOV—successfully detect functional form misspecifications, highlighting how Z-residuals can be leveraged to diagnose a wide array of structural flaws. These subtle deficiencies are often completely masked when inspecting only the marginal distribution of PITs or relying on simple summary statistics that ignore relationships with covariates.

- Graphical diagnostics utilizing Z-residuals serve as powerful exploratory tools for *discovering* unknown model deficiencies. Crucially, these visual insights guide the subsequent application of formal statistical tests, whether employing posterior/ISCV Z-residuals or conducting Z-residuals-PPC tests.

- Tests based on Z-residuals are well-calibrated and yield significantly higher statistical power than both traditional PPCs utilizing the χ^2 statistic and Z-residual-based PPCs. This dual advantage in calibration and power is attributed to two key factors: (1) unlike arbitrary discrepancy measures, the exact distribution of Z-residuals is known under the true predictive distribution; and (2) our "summarize-first, then-test" paradigm asymptotically approximates this true predictive distribution, as formally justified by the theory presented in this paper.

- Our simulations demonstrate that Z-residual-based PPCs mitigate the notorious conservatism of traditional χ^2 PPCs. Because they average over the MCMC draws, they remain highly stable despite the auxiliary $\boldsymbol{U}$ drawn at each iteration, serving as a reliable secondary confirmation of model misfit.

We have substantially revised Section 1 and Section 5, incorporating the above points to more clearly articulate our contributions and findings. We acknowledge the omission of the foundational references mentioned (Gelfand et al., 1992; O'Reilly & Quesenberry, 1973; Seillier-Moiseiwitsch & Dawid, 1993). We have extensively updated the literature review to explicitly discuss these earlier works.

2.1b The "order of operations" difference. The authors mention the difference in "the order of operations" (p. 12) between PPP and Z-residuals; it may be that I am confused about this difference and am referring above only to the former, in which case I believe a more concrete example (theoretical and numerical) would clarify this confusion.

Response: The computational algorithms that explicitly outline this "order of operations" difference are detailed side-by-side in Table 1. To prevent further confusion, we have revised the title of Section 2.6 and the caption of Table 1 to reflect the true focus of this section: its primary purpose is to explain how to conduct PPCs using Z-residuals. Contrasting this approach with traditional PPCs and posterior/ISCV Z-residuals is now framed as the secondary focus, serving specifically to highlight the mechanical differences between the methods.

This difference is the mathematical crux of why Z-residuals avoid the conservative bias of PPPs.

- **Test-first, then summarize (PPP):** For a traditional PPP, one computes a test statistic for each posterior draw $\boldsymbol{\theta}_k$, compares it to the observed data, and averages the results:

$$\text{PPP} = \frac{1}{K}\sum_{k=1}^{K} I\big(T(\boldsymbol{y}_k^{\text{rep}}, \boldsymbol{\theta}_k) > T(\boldsymbol{y}, \boldsymbol{\theta}_k)\big)$$

  Because $\boldsymbol{\theta}_k$ is drawn from a posterior already conditioned on $\boldsymbol{y}$, the replicated data $\boldsymbol{y}_k^{\text{rep}}$ is artificially close to $\boldsymbol{y}$, dragging the resulting p-value distribution toward 0.5.

- **Summarize-first, then test (Z-residuals):** In our framework, we first average the observation-level probabilities over the posterior to get a single, stable estimate of the marginal survival probability, transform this to a normal scale, and *then* compute a test statistic:

$$z_i = -\Phi^{-1}\left(\frac{1}{K}\sum_{k=1}^{K} \text{RSP}_i(y_i \mid \boldsymbol{\theta}_k)\right)$$

  The diagnostic test statistic $T(\boldsymbol{z})$ is then calculated exactly once on this finalized set of residuals. As we prove in the manuscript, this approach integrates out the parameter-level bias asymptotically, leading to well-calibrated uniform p-values.

2.1c Distinction from existing software. A subsequent concern is that the use of PIT and LOO-PIT predictive diagnostics, at least in terms of testing their uniformity, is already common in applications and has already been implemented in open-source software including ArviZ in Python and bayesplot in R. As such, while the empirical promise of these tools is naturally very exciting, it does not seem to be specific to anything the authors have presented in their manuscript.

Response: While tools like ArviZ and `bayesplot` offer LOO-PIT diagnostics, their focus on marginal uniformity often masks structural regression errors, and they lack the essential randomization step (U_i). To address this, we introduce the **Zresidual** package (CRAN/GitHub). It computes, graphs, and formally tests Z-residuals to detect subtle misspecifications—such as nonlinearity and heteroskedasticity—going well beyond simple goodness-of-fit checks. The package natively supports standard models (`glm`, `coxph`, `survreg`), Bayesian models (`brmsfit`), and custom models. We have expanded the "Software" section to detail these capabilities.

2.2a Breadth of examples. The authors present numerical demonstrations only for the negative binomial and hurdle negative binomial family. A more complete selection of model families would render these sections more convincing of the methods' empirical prowess. For instance, the authors show an example where one can diagnose the use of the wrong model family between two discrete families: could they extend this to choosing between a continuous and discrete family? For instance, in the following case study: https://users.aalto.fi/~ave/casestudies/Nabiximols/nabiximols.html

Response: We thank the reviewer for suggesting the Nabiximols case study. However, our primary focus remains on diagnosing structural flaws (e.g., functional forms) rather than traditional goodness-of-fit checking, as we believe this offers greater practical impact. To address other reviewers' requests for diverse model families, we added a continuous inverse Gaussian simulation instead, and moved the original NB versus HNB family-misspecification example to the appendix.

To strengthen our numerical evaluations, all simulations now use 500 replications to reduce Monte Carlo error. Furthermore, we incorporated holdout predictive checks (HPCs) across all settings—including the new inverse Gaussian, nonlinear HNB, and appended wrong-family simulations. This ensures a comprehensive, side-by-side comparison of posterior Z-residuals, ISCV Z-residuals, PPCs, and HPCs across diverse misspecification scenarios.

2.2b LOO-CV in the real-world experiment. The authors write in the theory section that the LOO-PIT (or LOO-RSP) approach can resolve many of the problems with the standard PIT (RSP) diagnostics. Yet the real-world experiment uses only RSP diagnostics and no LOO-CV equivalent. Can the authors please clarify why this is?

Response: We have expanded the dolphin-sighting analysis so that Table 2 now reports both posterior and LOO-based diagnostics side by side. The reason we initially focused solely on the posterior Z-residuals is grounded in finite-sample stability. Although LOO is theoretically less biased, LOO residuals **are not exactly standard normal**—their variance is greater than 1 for high-leverage observations, akin to why frequentist OLS relies on studentized residuals. In our simulations, this property, combined with the approximation error of importance sampling (IS), resulted in higher Type-1 errors for the LOO Z-residuals. We found that posterior Z-residuals are actually more stable at smaller sample sizes (surprising to us). Nevertheless, adding the LOO results clarifies their relationship in the applied example. For this dataset, both methods yield similar conclusions: the linear models show residual patterns against depth, while the nonlinear models do not. Note, however, our other studies have found that LOO Z-residuals are definitely superior than posterior Z-residuals in the models with observation-specific latent variables.

2.2c Summary statistic for predictive performance. Further, in their Table 2, the Δelpd and se columns related to the elpd relate to the mean and standard deviation of the normal approximation of the difference in predictive performance. One could produce a summary statistic from these two columns by computing $P(\Delta\text{elpd} \geq 0)$ in R using `pnorm(0, delta_elpd, se_elpd, lower.tail = FALSE)`. Showing this as an additional column would help articulate their conclusions.

Response: We have added a new column to Table 3 reporting the normal-approximation probability $\Pr(\Delta_{\text{elpd}} \geq 0)$, computed from the leave-one-out Δ_{elpd} and its standard error using `pnorm(0, delta_elpd, se_elpd, lower.tail = FALSE)`. Here Δ_{elpd} is measured relative to the nonlinear HNB model, which is the reference model. The resulting probability is 0.002 for the linear HNB model and less than 0.001 for the other non-reference models. This additional summary further supports the nonlinear HNB model as the best predictive candidate among the fitted models.

2.2d Clarification of power numbers. In Section 3.2 (p. 20) the authors write that the PPC test has "significantly lower power (e.g., ~ 0.05 vs. ~ 0.4 for $n = 50$)"; could the authors highlight where these numbers come from in the plot above, since it shows a great many different methods?

Response: Regarding the numerical comparison mentioned by the reviewer from the earlier count-model simulation, we have clarified these values by adding legend symbols directly into the text: "In contrast, the corresponding PPC is overly conservative, with Type I error rates near zero and significantly lower power (e.g., ~ 0.036 ($-\times-$) vs. ~ 0.310 ($-\triangle-$) for $n = 50$)." We have also supplemented the figures with tables in Supplementary S1.3 to facilitate the reading of these numbers. Please note that this entire simulation of this example has now been moved to Supplementary Section S1.1.

2.2e Interpretation of replicated p-values. A small numerical example at the beginning of Section 2.8 with visualisations would be very helpful in understanding what it would mean for "a large proportion of small p-values" to represent (p. 14).

Response: We acknowledge that summarizing dependent p-values may be the most significant technical challenge when auxiliary randomization is used. Due to space limitations, we did not include an additional illustrative plot for this; instead, we refer readers to the histograms of replicated p-values provided in the real-data example (Figure 8). We agree that the term "large number" is somewhat vague, as we lack a known distribution for dependent p-values generated from the same dataset to determine a formal cutoff based on a probability metric.

To provide rigorous alternatives, Section 2.8 details an upper-bound p-value that is valid for any distribution, which is exceptionally useful in applied Bayesian modeling for decisively flagging severe model deficiencies. For situations requiring more precision, Section 5 outlines a bootstrap-style methodology to calculate an exact p-value directly from the replicated p-values. Furthermore, Z-residual-based PPCs also provide a highly satisfactory, single-number summary of these checks.

2.3a Sensitivity and subjectivity of the test statistic. On this point, I think that the authors should make clear that while it is true that p-values based on PPCs 'depend heavily on the choice of test statistic" (p. 1), the same seems also to be true for their Z-residuals. Ensuring that the language they employ does not hide this sensitivity, i.e. does not suggest that their approach entirely does away with the 'subjectivity" (p. 3) of choice of test statistic, seems important to me.

Response: We thank the reviewer for highlighting this nuance. We have revised the manuscript to replace "subjective" with "arbitrary." The word "subjective" is often misconstrued; great scientific discoveries are frequently subjective but rooted in empirical evidence and appropriate reasoning. The true weakness of standard PPCs is not subjectivity, but rather that the lack of guidance from the data modeling results makes the choice of test statistic fundamentally arbitrary. Our Z-residual framework solves this by making the choice of test statistic an evidence-based decision, driven directly by what is observed during the graphical assessment of the residuals.

2.3b 'Well-chosen" test statistics and reasonable defaults. In their numerical examples, the authors show how in some instances some tests are 'more powerful" than others, and subsequently that there could exist some 'well-chosen test statistic" (p. 19) for a given problem. On this point I want to offer two comments: the first is whether we can tell ahead of time which test statistics are indeed 'well-chosen"; [...] And if we are to pre-assign a set of tests, then which tests in particular do the authors propose as reasonable defaults?

Response: We agree with the reviewer that selecting a "well-chosen" test statistic is non-trivial. We have revised the manuscript (Section 2.5) to clarify our underlying diagnostic philosophy: ideally, the choice of a test statistic should be informed by the observed residual patterns. A robust workflow begins by graphically inspecting the Z-residuals to uncover the nature of the structural flaws, and then applying the specific test that formally quantifies the observed discrepancy.

However, when a specific type of model departure is suspected—or when practitioners require a pre-assigned set of baseline diagnostic checks—we do recommend specific tests as reasonable defaults based on our theoretical and empirical findings.

For the task of **marginal checking** to assess overall distributional fit (e.g., suspecting an incorrect response family), we explicitly recommend the Shapiro-Wilk (SW) test as the default. To address the reviewer's point, we have added the following clarification to Section 2.5 regarding why the SW test is superior to a standard χ^2 goodness-of-fit approach:

> "This iterative workflow typically begins with marginal checking to assess the overall distributional fit. Practitioners can check whether the Z-residuals approximately follow a

normal reference distribution visually using a QQ plot, and numerically using a normality test such as the SW test. For marginal checking, we also explored a χ^2 test, adjusting the degrees of freedom to account for posterior variance shrinkage. However, we do not recommend this test. As shown in our simulation studies, it might perform poorly due to severe sensitivity to residual variance deflation and inflation. The SW test is a better-calibrated choice."

For the task of detecting **functional form misspecification** (e.g., covariate non-linearity), we recommend the grouped ANOVA test (Z-AOV) as the default. Because this grouped approach does not assume a specific non-linear functional form, it provides a broadly powerful diagnostic test for arbitrary non-linearity.

Finally, we agree with the reviewer that formalizing diagnostic power is a critical pursuit. We have added a note to Section 5 clarifying that devising maximally powerful tests for specific data tasks, as well as designing structured combinations of tests to effectively differentiate between multiple overlapping sources of model deficiency, remain compelling topics for future work.

2.3c Multiple testing, hold-out folds, and E-values. [...] the second is, in the event that we first choose a test statistic which yields no actionable information, whether we are at risk of some multiple testing concerns by performing a new test on the same residuals. In particular, do the authors suggest that we should enumerate all possible tests we would wish to perform ahead of time? Or that we should use different hold-out folds for different tests? If not, is there some intuition as to why this does not represent a concern for type I error guarantees? Could this be mitigated by some correction? Or the use of E-values (see Section 4.1 below)?

Response: Thank you for raising this insightful question regarding multiple test bias. We have added the following paragraph in Sec 2.5 to address this issue:

"Finally, we wish to clarify that multiple comparison bias is generally not a primary concern in this context. The use of statistical tests in model diagnostics differs fundamentally from traditional scientific hypothesis testing. In scientific discovery, a significant result is the goal, and multiple testing inflates the risk of false discoveries. In model diagnostics, the goal is a well-fitting model, indicated by *non-significant* p-values. An inflated Type-I error rate in this exploratory phase merely makes the diagnostic scrutiny more rigorous. If a test flags a "significant" model deficiency, it does not stand as a final conclusion, but rather triggers an iterative refinement process aimed at resolving the issue. Once a refined model yields non-significant p-values, it should then be further compared against previous iterations using independent predictive criteria, such as WAIC or LOOCV-IC (Vehtari et al. 2017), in addition to diagnostic p-values, to ensure genuine improvement. However, we note one important exception: if Z-residuals are used to hunt for correlated features among a large pool of unused covariates, or if a vast, pre-specified battery of tests is deployed to criticize a model, the procedure remains fully subject to multiple testing bias. In such scenarios, the results must be interpreted with caution and accompanied by robust multiplicity corrections. Appropriate strategies include established false discovery rate controls (Benjamini & Hochberg 1995), E-value calibration techniques (Vovk & Wang 2021, Ramdas et al. 2023), and data-splitting methods that reserve a hold-out set for final model evaluation."

2.3d Impact of noise variables. In Section 3.3 the authors write that "the inclusion of noise variables in both models aims to provide a more realistic and robust testing environment" (p. 21). Could they please comment on how the form of this noise affects their conclusions? That is, if the form of the model is correct but the noise is heavy-tailed, asymmetric, or comprised of point outliers, how do the conclusions of this experiment change?

Response: We have changed the terminology from "noise variables" to 'additional covariates" to clarify that these are additional predictors used to increase dimensionality, rather than additive errors on the response (such as heavy-tailed noise, asymmetry, or point outliers). Furthermore, because we are working with conditional regression models, the specific probability distributions of these predictors have no effect on our overall conclusions, which is why we opted not to detail this minor point in the manuscript. As expected, having fewer non-relevant predictors generally leads to better statistical power and Type I error rates. We included these covariates purely to simulate a more realistic, higher-dimensional testing environment.

Minor comments

3.1 Assumptions for theoretical results. In Theorem S2.2, the authors make the assumption that the model is well specified, yet their proof technique seems only to make use of the usual conditions required for posterior concentration, which occurs independently of model well specification. As such, I am not sure where they require this assumption for their result. In particular, in Step 1 of their proof of Theorem S2.3 (which overall does of course require the assumption of well specification), it may be that in the case of misspecification we would instead have

$$\|P_i^{(n),CV} - P_i^{(*)}\|_{TV} \to \varepsilon > 0.$$

If I have understood this correctly, could we extend their results to make this irreducible error appear in the final bound? That is, do the authors believe that it would be possible to derive a bound where the degree of misspecification, ε, is made explicit, and subsequently interpreted from a p-value?

Response: We thank the reviewer for this insightful theoretical observation. The fundamental goal of model diagnostics is to seek a well-fitting model where, ideally, no misspecification exists. The reason we focus exclusively on deriving the theoretical properties under the null hypothesis (i.e., when the model is correctly specified) is that this null distribution serves as the essential reference distribution required to formally assess the observed Z-residuals. While we agree that extending the bounds to quantify the irreducible error ε under misspecification is an interesting mathematical direction, our primary objective is practical model criticism. For this purpose, establishing the exact asymptotic behavior under the null is what provides practitioners with the rigorous baseline needed to calculate diagnostic p-values and flag structural departures.

3.2a Intuition for randomization. Adding some intuition for why (p. 8) "the randomisation in the RSP is essential for ensuring a truly uniform distribution for discrete data" would have helped my understanding.

Response: We agree that providing this intuition is highly beneficial for readers. Accordingly, we have expanded Section 2.2 to include the following detailed explanation clarifying the necessity of the randomization:

> "The randomization is needed when the response distribution has non-zero probability mass at the observed y_i. Discrete y_i can be regarded as an incomplete observation of a continuous random variable, hence, an injected U_i is necessary to recover it. Specifically, the survival function has a jump at y_i, and without randomization all observations with the same discrete value would be mapped to the same survival probability. The auxiliary variable U_i randomly maps y_i to a random number in the probability interval at this jump and thereby restoring the uniform reference distribution under the true model. A key property of this transformation is that if the model and parameters $\boldsymbol{\theta}$ are correct and the observations are independent, the resulting RSPs are independent standard uniform distribution; see Theorem S3.1 for a formal proof and a graphical illustration using a toy example. Detailed illustrations of the uniformity can be found in Dunn & Smyth (1996), Feng et al. (2020)."

3.2b Partitioning covariates. Similarly (p. 10) it is not immediately clear to me, and perhaps some other less familiar readers, why "partitioning Z-residuals into groups with respect to a covariate" is required. Is this to mitigate multiple testing concerns?

Response: We clarified in Section 2.5 that grouping is not intended for multiplicity correction. Instead, it serves as a specific formal test for non-linearity against a continuous covariate. We have added the following text:

"For a discrete covariate, any systematic trend can be formally evaluated by directly applying an AOV test across its natural categories to check for grouped mean changes, or Bartlett's test for heteroskedasticity. For a continuous covariate, we achieve this by first dividing its range into k equally spaced intervals and then applying the same tests. Because this grouped AOV approach does not assume a specific non-linear function, it provides a broadly powerful, formal test for arbitrary non-linearity."

3.2c Approximating standard normality. And (p. 10) when the authors mention that "Z-residuals are approximately standard normal", is this approximation present only in the importance-sampling case? Or is this a pre-asymptotic statement? Again some clarification on where this bias emerges, and whether it can be quantified, would help greatly.

Response: We have added this sentence to clarify the "approximation" after presenting the two asymptotic theorems (added in the revised version):

"The same argument extends to the posterior construction: posterior RSPs are likewise asymptotically independent and uniformly distributed (Theorems S3.4 and S3.6). Consequently, in large samples both posterior and ISCV Z-residuals behave approximately as an *i.i.d. sample from the* $N(0,1)$ *distribution*, so that well-behaved diagnostic procedures applied to them are asymptotically well-calibrated. We say "approximately" because both constructions incur pre-asymptotic error: the RSPs are averaged over the finite-sample posterior or LOO distribution of $\boldsymbol{\theta}$ rather than evaluated at the true $\boldsymbol{\theta}$, and ISCV incurs additional error from importance sampling."

3.2d Degrees of freedom adjustment. When discussing the adjustment to the degrees of freedom in the χ^2 statistic (p. 11), the authors adjust n (the number of data points) by p_{eff} (the effective number of parameters). Some further discussion on why this forms a valid adjustment, and the relationship between the number of data and the number of parameters, would again greatly aid comprehension.

Response: We have added discussion in Section 2.5 framing this as a heuristic test rather than an exact distributional result.

We've added the following text:

For marginal checking, we also explored a χ^2 GOF test, adjusting the degrees of freedom to account for posterior variance shrinkage. However, we do not recommend this test. It performed poorly in our simulations due to severe sensitivity to residual variance deflation and inflation, demonstrating that the SW test is a far better-calibrated choice.

Indeed, our simulation studies confirms your conjecture of their poor performance due to sensitivity to variance inflation and deflation. We believe that such negative findings are valuable too.

3.2e Logical flow in Section 2.6. In Section 2.6, the sentence starting with "therefore" (p. 11) does not follow immediately to me from what came before. Just another sentence here might help me and other readers understand more clearly.

Response: I think the confusion is caused by the use of $\boldsymbol{\theta}$ rather than $\boldsymbol{\theta}^{(t)}$. We have modified this part as follows:

> A key simplification for computing PPP follows from a basic testing principle (Meng 1994). Conditional on a parameter value $\boldsymbol{\theta}^{(t)}$, a replicate $\boldsymbol{y}^{\text{rep}}$ is generated from the model specified by $\boldsymbol{\theta}^{(t)}$, which is the true parameter for $\boldsymbol{y}^{\text{rep}}$. Therefore, if $\mathcal{P}(\boldsymbol{y}, \boldsymbol{\theta}^{(t)})$ is a well-calibrated test p-value conditional on $\boldsymbol{\theta}^{(t)}$, then $\mathcal{P}(\boldsymbol{y}^{\text{rep}}, \boldsymbol{\theta}^{(t)})$ is *exactly* uniform on $(0, 1)$. For a fixed observed value $\mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)})$, this implies that the conditional probability of the replicated p-value being smaller than the observed p-value is simply equal to the observed p-value itself: $P\big(\mathcal{P}(\boldsymbol{Y}^{\text{rep}}, \boldsymbol{\theta}^{(t)}) < \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)}) \,|\, \boldsymbol{\theta}^{(t)}\big) = \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)})$. By the law of iterated expectations,
>
> $$P\big(\mathcal{P}(\boldsymbol{Y}^{\text{rep}}, \boldsymbol{\Theta}) < \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\Theta}) \,|\, \boldsymbol{y}^{\text{obs}}\big) = \mathbb{E}_{\boldsymbol{\theta} \,|\, \boldsymbol{y}^{\text{obs}}}\left[\mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta})\right]. \tag{1}$$
>
> Therefore, we need not replicate datasets for computing PPC p-value with the RSP Z–residuals. The p-value is simply the posterior average of the p-values computed on the observed data,
>
> $$\widehat{\text{PPP}}^{\text{RSP}} = \frac{1}{T}\sum_{t=1}^{T} \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)}). \tag{2}$$

3.3 Plotting. As mentioned above, many of the tools the authors propose seem to me to have already been implemented in software (e.g. by Gabry et al., 2019; Sailynoja et al., 2021, which could be referenced alongside Gelman et al., 2013). As such, presenting plots which are inline with pre-existing styles, or motivating the difference, would help some readers like me to understand the connection to this other, currently missing side of the literature.
In Figure 1, plotting the marginal distribution of the Z-residuals would help articulate the point the authors make.

Response: Thank you for this suggestion. To address your point regarding Figure 1, we have updated it (now Supplementary Figure S1) by adding a third row that displays the marginal distributions of the Z-residuals against a standard normal reference density. In the introduction section, we have also added references to Gabry et al. (2019) and Säilynoja et al. (2021) to better connect our work with existing software like bayesplot and ArviZ.

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However, we wish to clarify an important distinction. While those existing tools provide valuable visualizations, they primarily focus on checking the marginal distributions of PITs. By evaluating Z-residuals on a standard normal scale, our framework expands upon this by embracing the full suite of classical frequentist residual diagnostics—combining these new marginal plots with observation-level scatterplots and QQ plots for a much more thorough evaluation of Bayesian models.

3.4 Acronyms. While the list of abbreviations is helpful, it is also very long and indicative perhaps of an opportunity to trim them down. At some points I found myself returning to it to ensure I understood what I was reading, which distracts from the flow of the manuscript. Further, some acronyms including "QQ plots" are never introduced nor are they present in this list.

Response: Thank you for this feedback. We have revised the manuscript to improve readability by minimizing abbreviations:

- **Expanded terms:** Low-frequency terms (e.g., cross-validation, goodness-of-fit) are now written out in full.
- **Retained core acronyms:** High-frequency terms (e.g., RSP, PPC, ISCV) are kept to prevent formula and table clutter, but are clearly defined at first use.
- **QQ Plots:** We explicitly define quantile-quantile (QQ) plots at their first appearance, as requested.
- **Self-contained visuals:** Figure and table captions now spell out key tests (e.g., Shapiro-Wilk, AOV, Bartlett) so readers do not have to search for definitions.

3.5 Flowery wording. The second sentence of the manuscript (p. 1), "Statistical modelling without rigorous checking is akin to building a high-rise on sand—elegant in appearance but prone to collapse under scrutiny," is perhaps overly romantic in style, disjointed from the rest of the paper, and distracts from the rest of the introduction in my personal opinion.

Response: Thank you for the feedback. We have deleted this sentence from the revised manuscript to maintain a more direct academic tone. We note, however, that the original intent of this analogy was to strongly emphasize a critical and pervasive challenge within the broader statistical community: the frequent prioritization of complex model building over rigorous model checking.

3.6 Typos and notation.

1. (p. 2) the data are denoted in bold; specifying the space in which they live would help justify this.

2. (p. 3) the number 0.5 comes out of thin air when discussing the bias of PPPs.

3. (p. 3) defining the survival probability function here would be very helpful in introducing the concepts of the paper, as opposed to only defining it in Appendix S2.2.

4. (p. 6) "((in a measure-theoretic sense)" has extra parentheses.

5. (p. 8) adding a reference to where the authors formally show that "[u]nder the true model [...] Z-residuals are independent and identically distributed standard normal variables" would help.

6. (p. 9) the notation around Equation 5 could be refined; it is currently quite heavy.

7. (p. 15) "(supplementary))" could be deleted, and has extra parentheses.

Response: Thank you for your careful reading and for identifying these typos and notational issues; we have corrected them throughout the revised manuscript. To briefly summarize the key changes:

- We explicitly defined the sample space $\mathbf{y} \in \prod_{i=1}^{n} \mathcal{Y}_i$ and the survival function $S_i(y \mid x_i, \theta) = P(Y_i > y \mid x_i, \theta)$ in Section 2.1.

- We clarified the concentration of posterior predictive p-values around 0.5.

- We explicitly linked RSP Z-residual standard normality to Theorem S3.1.

- We removed redundant parentheses and wording errors.

While we acknowledge the notation around Equation (5) is dense, we retained the explicit superscripts and subscripts to prevent ambiguity. Because our framework formally compares four distinct residual constructions (posterior RSP/MSP and ISCV RSP/MSP), this explicit bookkeeping is essential for clarity later in the manuscript. To assist readers, we have added an introductory sentence immediately before Equation (5) to clearly unpack the components of this notation.

Curiosity

4.1 E-values and multiple predictive testing. Is there space for E-values (Grunwald et al., 2024) to replace the p-value-style tests the authors propose here? Do the authors believe this could resolve my question of multiple testing in Section 2.3?

Response: Thank you for this insightful suggestion. We agree that the safe testing framework using E-values (Grünwald et al. 2024) offers a highly promising alternative. As discussed previously, multiple comparison bias is generally not our primary concern for routine model diagnostics, because a "significant" p-value is not a terminal scientific discovery, but rather an internal signal to iteratively refine the model. However, if Z-residuals are used to systematically hunt for correlated features among a large pool of unused covariates, multiple testing bias becomes a critical issue. We believe E-values could elegantly resolve the multiplicity concerns in that specific exploratory scenario, avoiding the rigidities of traditional p-value corrections. We have added a discussion of this potential integration to Section 5.

4.2 Rates of convergence. The authors make the point at the beginning of Section 2.7 that "for both the cross-validated and posterior constructions, the RSPs are asymptotically independent and uniformly distributed." Do we have any understanding of whether this occurs at the same rate for both constructions? And subsequently, at what rate they are indistinguishable from one another?

Response: Thank you for this insightful theoretical question. While establishing the exact finite-sample rates of convergence is an interesting problem, we did not pursue this mathematical direction in depth, as our primary focus remains on the practical diagnostic utility of these tools. However, our intuition is that both the leave-one-out and full posterior predictive distributions should converge to the true predictive distribution at the exact same asymptotic rate. Because the difference between the full posterior and the leave-one-out posterior inherently diminishes as n grows, the two resulting RSP constructions should couple and become indistinguishable from one another at that same rate.

4.3 Beyond perfect model specification. Related to Section 3.1 above, the authors have shown their theoretical results under correct model specification. While this is clearly the standard in the literature, I am sure the authors will agree that perfect model specification is a fantasy in reality. Do they believe that the sorts of results they have developed (e.g. Theorem S2.3 as well as Theorem S2.2), and those pre-existing in the literature, could be extended to notions of ε-misspecification as previously detailed? That is, where the model is misspecified, but perhaps not so gravely misspecified as to make a noticeable difference in reality?

Response: Thank you for this insightful question. While perfect specification is rare, this reality is precisely why robust diagnostics are needed: rather than passively accepting misspecification, we strive to iteratively find models closer to the real data. To build valid diagnostics for this search, we must first establish exact reference distributions under the null hypothesis (correct specification) for proper calibration. Characterizing ε-misspecification concerns a test's power rather than its baseline calibration, which is why our theoretical focus remains restricted to the exact null setting.

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Response to Reviewer C

July 6, 2026

We sincerely thank you for your time, valuable feedback, and general appreciation of the paper. Guided by the collective insights from you and other referees, we have substantially revised the manuscript. We have responded to your comments item by item, and your constructive suggestions have greatly improved both the methodology and the presentation. In this detailed response, we quote several updated passages for your reference (please note that some of these quoted excerpts may have been further refined in the final manuscript to ensure perfect flow and accuracy). In the revised manuscript, revised text is highlighted in orange; for substantially rewritten sections, only the section title is highlighted rather than the entire text.

Summary. The paper presents an approach to Bayesian model checking based on the notion of Z-residuals, which are defined for a given observation i as

$$z_i^{\mathrm{RSP}}(y_i \mid \theta) = -\Phi^{-1}\big(\mathrm{RSP}_i(y_i \mid \theta)\big),$$

where $\Phi(\cdot)$ is the cdf of a standard normal and $\mathrm{RSP}_i(y_i \mid \theta)$ is the randomized survival probability, defined as

$$\mathrm{RSP}_i(y_i \mid \theta) = S_i(y_i \mid \theta) + U_i p_i(y_i \mid \theta),$$

where $U_i \sim \mathrm{Unif}(0,1)$, $S_i(y_i \mid \theta)$ is the survival function of Y, and $p_i(y_i \mid \theta)$ is its probability mass function (equal to zero if the random variable is continuous). Specifically, the authors show that if the model is correctly specified, then $\mathrm{RSP}_i(y_i \mid \theta) \sim \mathrm{Unif}(0,1)$. In turn, this makes $z_i^{\mathrm{RSP}}(y_i \mid \theta)$ normally distributed. In particular, the authors use this property to construct posterior Z-residuals, obtained by averaging $\mathrm{RSP}_i(y_i \mid \theta^{(t)})$ for each posterior sample $\theta^{(t)}$. They also introduce a leave-one-out cross-validated version paired with importance sampling, to further correct the problematic double use of the data (which undermines calibration of the p-values). Their approach is useful for testing goodness-of-fit, e.g., via a χ^2 test, as well as a tool to capture potential misspecifications in the functional form of a covariate. The authors showcase their approach in a variety of simulated settings and in a real dataset of dolphin sighting.

General comment. This is an interesting and potentially helpful contribution. While the notion of RSP is equivalent to the probability integral transform, which has been used for model checking in the Bayesian literature, what the author presents is a general strategy for checking both the correct specification of the likelihood and also which covariates might be misspecified. Moreover, the theoretical statements and proof appear correct. However, I have the following comments and questions.

Response: We sincerely thank the reviewer for this excellent and highly accurate summary of our manuscript. The reviewer's understanding of the proposed Z-residual framework, the role of randomized survival probabilities (RSPs), and the posterior and ISCV implementations is perfectly correct.

We are particularly grateful for the reviewer's recognition of the core innovation of this work. As the reviewer acutely pointed out, while the mathematical foundation of RSPs is equivalent to the probability integral transform in the context of continuous response, our primary contribution is elevating this concept into a comprehensive, general strategy for devising versatile normally-distributed residuals that enable the diagnosis of overall likelihood specification, localized covariate misspecifications, and many other model deficiencies of the likelihood. We also greatly appreciate the positive assessment of our theoretical statements and proofs.

Inspired by this summary, we have substantially revised the Section of Introduction to articulate these contributions in a clearer way: it now more clearly distinguishes the PIT foundation shared with the existing literature from the new elements of our framework—the summarize-first construction of a single residual set, its asymptotic calibration, and the resulting suite of targeted graphical and numerical diagnostics.

Question 1. The author presents differences between the RSP, MSP, and PPC. However, they should also compare their approach against the more recent holdout predictive checks (HPC) of Moran et al. (2023), which also avoid using the data twice and provide calibrated p-values, at the cost, however, of lower power. Since the proposed approach does not suffer from this loss of power, it would be great to see examples where your method works better (or comparatively).

Response: Thank you for this excellent advice. We have included holdout predictive checks (HPC) in all of our empirical studies, and we believe that adding this benchmark has greatly strengthened the paper.

Interestingly, our empirical studies additionally revealed that HPC is not only less powerful but can also exhibit slightly worse-calibrated Type I errors in small-sample settings. While initially surprising, this behavior is reasonable upon reflection: the finite-sample posterior of θ (trained on the observed data) is not identical to the true parameter governing the isolated test data. Because the posterior predictive distribution integrates over this finite-sample posterior uncertainty, it is structurally different from the true data-generating distribution. Consequently, evaluating holdout data against this approximate predictive distribution can distort the exact uniformity of the resulting p-values in small samples.

We have revised Sections 3 and 4, as well as Supplementary Section S1, to include the HPC comparisons and to discuss these finite-sample power and calibration trade-offs explicitly.

Question 2. It would be interesting to discuss your Z-residuals with the recent U-values posed by Covington and Miller (2025). Their approach introduces uniform distributions for each of the model parameters, while yours adds a z variable for each observation. Under which settings should your approach be preferred?

Response: Thank you for this helpful suggestion. Discrete response is essentially an incomplete observation of an underlying continuous random variable. The U-value approach of Covington and Miller (2025) is closely related in spirit to ours: both transform random elements onto a uniform reference scale for recovering the underlying continuous random variables. Indeed, for a single posterior draw $\boldsymbol{\theta}^{(t)}$, their observation-level uniform variables are distributionally identical to $1 - \mathrm{RSP}_i(y_i \mid \boldsymbol{\theta}^{(t)})$. The two frameworks differ in two key respects. First, in posterior aggregation: UPC follows a "test-first, then-summarize" strategy, combining MCMC-wise p-values via the Cauchy combination rule (Liu & Xie 2020), whose approximation error may not fully vanish asymptotically, whereas we summarize RSPs over the posterior or leave-one-out predictive distribution first, yielding a single residual set that is asymptotically i.i.d. $N(0, 1)$. Second, in diagnostic target: U-values can be attached to any random element of the generative specification—parameters, priors, or latent variables—making them well suited to locating which internal component of the model is problematic. Z-residuals, in contrast, are observation-level predictive residuals on a standard normal scale, supporting regression-style diagnostics such as QQ plots, residual-versus-covariate plots, grouped boxplots, and functional-form tests.

Accordingly, our approach is preferable when the goal is to assess the fitted predictive distribution at the observation level—for example, tail behavior, nonlinear or omitted covariate effects, or zero versus nonzero patterns in count models—while U-values are more appropriate for questions about parameter-, prior-, or latent-variable components. We view the two approaches as complementary rather than competing, and we have added a detailed discussion of Covington and Miller (2025) to Section 5 and a discussion of how they can be combined to achieve the diagnosis of internal constructs as are demanded in applied Bayesian modelling.

Question 3. The authors indicate that the Z-residuals may detect potential misspecification in the functional form of some of the covariates (e.g., the simulation in Section 3.3). By looking at Figure 4(c), it looks like the trend between the z-residuals and x_1 in the model that assumes x_1 as linear has approximately a log-shape, which is the true functional form.

Response: Thank you for this acute observation, which is consistent with our experience. When the response family is correctly specified, the scatterplot of Z-residuals against a covariate under a misspecified linear model often reflects the shape of the true functional form—in this case, the logarithmic shape visible in Figure 4(c). Intuitively, the Z-residuals capture the discrepancy between the true covariate effect and its best linear approximation, so the residual trend traces the unmodeled portion of the true curve. This property makes the residual-versus-covariate plot not only a detector of functional-form misspecification but also a practical guide for choosing a corrected specification, as illustrated in our dolphin-sighting application, where the residual pattern guided the revision to the nonlinear HNB model, resulting in a more predictive model.

3(a) The HNB family here is correctly specified. What would happen if one were to model the data as an NB model, under both the correct functional form for x_1 (logarithmic) and the linear one? It would be important to clarify if the potential functional form between a given x and the z-residuals is impacted by the misspecification of the likelihood. A hint on what happens is suggested in Table 2 in the application (NB vs HNB in the nonlinear part), but since it is real data, it is hard to grasp the true relationship.

Response: Thank you for this very thought-provoking question. Following the reviewer's suggestion, we have added simulation studies in the new Supplementary Section S1.2, using the same HNB-generated data as in Section 3.3 and fitting two additional NB models: NB-log (misspecified response family) and NB-linear (both misspecified). The Z-residual scatterplots against X_1 and the diagnostic test results for a representative dataset are presented in Section S1.2; the p-values are reproduced in the table below.

Table 1: Z-residual diagnostic p-values for the response-family sensitivity analysis using the same HNB-generated representative dataset as in Section 3.3 of the main text. The HNB-log and HNB-linear rows correspond to the models considered in the main text, while the NB-log and NB-linear rows are added here to assess the effect of response-family misspecification. Because the response is discrete and RSP Z-residuals depend on auxiliary randomization, we report median upper-bound p-values (Eq. (16)) for the posterior and ISCV RSP Z-residual tests. The AOV and Bartlett (BL) tests are applied against X_1. The logarithmic form is marked "Wrong" for the NB family because the X_1-dependent hurdle probability is improperly absorbed into the NB mean structure (see text).

			Posterior RSP Z-residuals				ISCV RSP Z-residuals			
Model	Family	Functional form	SW	AOV-X_1	BL-X_1	χ^2	SW	AOV-X_1	BL-X_1	χ^2
HNB-log	HNB (True)	log (True)	0.904	0.630	0.120	0.884	0.595	0.767	0.089	0.606
HNB-linear	HNB (True)	linear (Wrong)	0.282	< 0.001	0.002	0.165	0.700	< 0.001	0.017	0.140
NB-log	NB (Wrong)	log (Wrong)	0.003	0.137	0.001	0.363	0.044	0.281	0.001	0.178
NB-linear	NB (Wrong)	linear (Wrong)	< 0.001	0.007	< 0.001	0.415	< 0.001	0.004	< 0.001	0.368

The short answer to the reviewer's question is that a functional-form deficiency remains detectable under a misspecified response family. For the NB-linear model, both the SW test and the AOV test against X_1 reject decisively (p-values < 0.001 and 0.007 for posterior Z-residuals, respectively, with similar ISCV results), and the residual scatterplot against X_1 still displays a visible and pronounced trend.

The situation is more nuanced for the NB-log model, which clearly illustrates how a likelihood misspecification can impact functional form. While this model uses the logarithmic form that was correct for the HNB process, fitting a single NB family absorbs the X_1-dependent hurdle probability into the NB mean structure. Consequently, the log-linear form alone fails to correctly specify the conditional mean for the NB model. As expected, the SW test flags the overall distributional mismatch strongly (p-values 0.003 and 0.044). The AOV test against X_1 yields moderate median upper-bound p-values (0.137 and 0.281). While these do not strictly cross the 0.05 threshold here, they are conservative by design: the replicated p-values actually concentrate between 0.05 and 0.10 (and as an aside, frequently drop below 0.05 in supplementary simulations with other datasets). This moderate AOV signal correctly detects the induced mean distortion rather than acting as a false alarm, a finding corroborated by a clear near-linear upward trend in the corresponding residual scatterplot.

More broadly, the reviewer's question highlights the intricacies of disentangling different sources of model error, a topic we regard as a very important avenue to pursue. Our general finding is that the SW test is highly sensitive to response-family misspecifications that alter the global residual distribution, whereas residual-versus-covariate scatterplots and grouped AOV tests are tuned to localized functional-form errors. However, these sources are not entirely separable: as demonstrated by the NB-log example, a misspecified response family can distort the implied mean structure, causing family errors to legitimately propagate into covariate-specific tests. Evaluating these diagnostics jointly helps practitioners pinpoint the nature of a model's deficiency, but isolating the exact source remains challenging. A robust diagnostic workflow thus requires combining various graphical and numerical residual tools, supplemented by model comparison metrics such as information criteria. We are also working on extending the proposed Z-residuals to compute residuals for different submodels (e.g., the zero-count component of hurdle models). We have added a discussion of these points in Section S1.2. Considering page limits, the main text remains focused on comparing the proposed Z-residual methods with PPC, HPC, and SW-based tests.

Question 3(b) Would it make sense to compare the z-residuals against a covariate that has not been used by the model?

Response: Thank you for this helpful question. Yes, examining Z-residuals against a covariate not included in the fitted model is a valid and highly useful strategy for discovering omitted predictors, in direct analogy with classical regression diagnostics for detecting omitted predictors. A systematic trend, curvature, or grouped mean shift in the Z-residuals against an unused covariate may indicate a missing covariate, interaction, or nonlinear effect, and can be assessed with a scatterplot with a smooth trend or a grouped ANOVA test.

We have added a discussion of this use in Section 2.5 of the revised manuscript, where we also note that it should be treated as exploratory: when many predictors are screened, multiple testing bias should be taken into account.

Question 4. The general style of the writing and the way the results are presented and discussed need improvement.

Response: Thank you for the following specific suggestions which have been incorporated in the revised manuscript and have greatly strengthened this paper.

Question 4(a) I believe the authors should present the exact steps of their approach compactly in a simple algorithm. Since the authors are providing a general procedure, having a clear algorithm would significantly facilitate revisiting the paper, if one needs it.

Response: Thank you for this helpful suggestion. We agree that the exact steps of the proposed approach should be easy for readers to revisit and implement, and we have addressed this point in two ways.

First, **Table 1** in Section 2.6 already presents the algorithm: it lays out, step by step, the computing workflow of the proposed Z-residual diagnostics alongside that of PPC with parameter-wise Z-residuals, highlighting the key operational distinction between the proposed "summarize-first, then-test" workflow and the "test-first, then-summarize" workflow of PPCs. To make this algorithmic summary more visible to readers, we have revised the table caption to "Computational Procedures of PPC and Z-residual Diagnostics" and the section title accordingly.

Second, to facilitate implementation, we have added a new section "Data and Software Availability," which points readers to the accompanying `Zresidual` R package. The package includes:

- a generic function `Zresidual` for computing Z-residuals, with built-in support for `brms` model objects, Cox proportional hazards and parametric survival models from the `survival` package, and standard GLMs from `stats`;
- high extensibility to custom model-fitting objects via user-supplied RSP functions, through the `log_cdf` and `log_pmf` arguments;
- a comprehensive suite of graphical and numerical diagnostic tools for evaluating Bayesian regression models;
- the processed dolphin-sighting data and
- compiled HTML vignettes and Quarto sources for reproducing the results of all the simulation studies and the real-data analysis as reported in this paper, as well as many other vignettes such as an animated illustration of the parameter-wise Z-residual.

An anonymized version is available at https://figshare.com/s/00ac6b406e451c76e8e1 for the double-blind review; the package will be published publicly on GitHub and submitted to CRAN upon acceptance. Once this package is downloaded, one can start with the file "index.html" in "docs" directory to preview the pre-compiled HTML vignettes.

Question 4(b) The paper contains a lot of abbreviations and combinations of hyphens (PPP, PPC, NB, HNB, Post-MSP, RSP_i^{CV}, $Z^{\mathrm{ISCV-RSP-SW}}$, etc.), which are sometimes hard to follow. While these are summarized on the title page and have a rationale, the readability can be improved by limiting their use.

Response: We agree that the original manuscript used too many abbreviations and that this affected readability. In the revised manuscript, we have reviewed all abbreviations and reduced their use where possible. In particular, low-frequency abbreviations, such as CDF, PMF, CPO, CV, LOOCV, and PSIS-CV, are now written out in full or avoided when they are not needed. We also write terms such as goodness-of-fit, posterior predictive p-value, probability integral transform, hurdle Poisson, and Bartlett test in full in the main text when space allows.

At the same time, some abbreviations are difficult to remove completely because they refer to core diagnostic methods or model classes that are used repeatedly in formulas, figures, and tables. For example, RSP, PPC, ISCV, NB, HNB, SW, and AOV appear frequently in method labels and figure/table annotations. Writing these terms in full throughout the manuscript would substantially increase the length of the paper and make several figure labels and tables difficult to read. Therefore, we retain a small set of core abbreviations, define them clearly at first use, and use them consistently throughout the manuscript. We have also revised figure captions, table captions, and section headings to make them more self-contained, so that readers do not need to repeatedly refer back to the abbreviation list.

We have revised the notation and wording throughout the main text, especially Sections 2–4, and updated figure and table captions to reduce low-frequency abbreviations and improve readability.

Question 4(c) The paper only provides a high-level summary of the theoretical results. It would be better to report the core of some of the statements (such as Theorem S2.2 and S2.5) in the main manuscript, while leaving the list of assumptions in the supplement.

Response: Thank you for this helpful suggestion. Following it, we now restate the two fundamental theorems in Section 2.7 of the revised manuscript: the theorem establishing the total variation convergence of the LOO predictive distribution to the true data-generating distribution, and the theorem establishing the asymptotic independence and uniformity of the cross-validated RSPs (Theorems S2.2 and S2.5 in the original submission; their numbers have changed in the revision because additional sections were added to the Supplementary Materials). The theorems are stated in the main text with informal one-line summaries of the regularity conditions, while the full lists of assumptions and the proofs remain in the supplement, and the main-text numbering is kept consistent with the supplement. We agree that these technical statements substantially clarify the verbal description of the asymptotic results and make the theoretical foundation of the proposed diagnostics easier to grasp without consulting the supplement.

References

Liu, Y. & Xie, J. (2020), 'Cauchy combination test: a powerful test with analytic p-value calculation under arbitrary dependency structures', *Journal of the American Statistical Association* **115**(529), 393–402.

Response to Reviewer D

July 6, 2026

We sincerely thank you for your time, valuable feedback, and general appreciation of the paper. Guided by the collective insights from you and other referees, we have substantially revised the manuscript. We have responded to your comments item by item, and your constructive suggestions have greatly improved both the methodology and the presentation. In this detailed response, we quote several updated passages for your reference (please note that some of these quoted excerpts may have been further refined in the final manuscript to ensure perfect flow and accuracy). In the revised manuscript, revised text is highlighted in orange; for substantially rewritten sections, only the section title is highlighted rather than the entire text.

This paper proposes a new general-purpose residual diagnostic framework for Bayesian models based on randomized survival probabilities. The authors construct Z-residuals that are asymptotically independent and standard normal under correct model specification. They establish asymptotic calibration for both posterior and cross-validated versions of the residuals. They also demonstrate the testing power of their approach through simulation studies. The paper shows how standard model diagnostic tools, such as QQ plots and residual-versus-covariate plots, can be applied using these Z-residuals.

Response: We sincerely thank the reviewer for this excellent and highly accurate summary of our manuscript.

1. **Interpretation and intuition of the Z-residual:** In linear regression, residuals have a clear and intuitive interpretation as the difference between the observed outcome and the structural part of the model. This interpretation allows practitioners to directly link diagnostic patterns to model misspecification. In the Bayesian diagnostics framework considered in this paper, the interpretation of the proposed Z-residual is less transparent. It would be helpful if the authors could provide a more intuitive explanation of what the value of the Z-residual represents, in a manner analogous to classical residuals in linear models. Additionally, illustrations of applying the negative transformation $-\Phi^{-1}(\cdot)$ and an explanation of how this transformation helps diagnostic interpretation would be useful.

Response: Thank you for this helpful suggestion. We agree that the interpretation of Z-residuals should be explained more intuitively.

We have revised Section 2.2, immediately after Eq. (2), to include this intuitive interpretation and to clarify the role of the negative normal-score transformation, which is quoted as below:

This uniformity allows us to further transform the RSP into a standard normal residual via the inverse-survival function, $-\Phi^{-1}(\cdot)$, of the $N(0,1)$ distribution:

$$z_i^{\mathrm{RSP}}(y_i \mid \boldsymbol{\theta}) = -\Phi^{-1}(\mathrm{RSP}_i(y_i \mid \boldsymbol{\theta})). \tag{1}$$

In words, this is the standard normal value that shares the same upper-tail area (RSP) as the observed y_i. Under the true model, these transformed quantities are independent $N(0,1)$ variables. We term this a "**Z-residual**" because Z conventionally denotes a standard normal random variable. The negative sign preserves the traditional residual direction, ensuring that observations larger than expected yield positive residuals. Conceptually, a Z-residual reflects the distance from y_i to its predictive median, rescaled for skewness and smoothed for discreteness. Consequently, a residual of ± 2 conveys the exact same degree of extremeness regardless of the underlying predictive distribution's shape.

2. **Practical guidance for choosing Z-residuals:** The paper introduces both posterior Z-residuals and ISCV Z-residuals, but it is not clear under what practical circumstances one should be preferred over the other. Providing guidance or recommendations—for example, in terms of sample size or type of model—would enhance the applicability of the proposed methodology.

Response: Thank you for this helpful comment. We agree that practical guidance is needed for choosing between posterior and importance-sampling cross-validatory Z-residuals.

We have revised Section 2.4 to add the following clarification:

> In practice, we generally recommend posterior Z-residuals as the default tool for routine diagnostics of models without highly localized latent effects.. While the cross-validatory construction is theoretically appealing because it alleviates the direct self-influence of y_i on its own predictive distribution, it does not completely eliminate dependence in resultant LOO Z-residuals, as the training sets $\boldsymbol{y}_{-i}$ and $\boldsymbol{y}_{-j}$ share $n-2$ observations. Furthermore, ISCV Z-residuals are not exactly standard normal in finite samples—their variance is greater than 1 for high-leverage observations, akin to why frequentist ordinary least squares relies on studentized residuals. In our simulations, this structural variance, combined with importance-sampling approximation errors, resulted in higher Type I error rates for ISCV Z-residuals. Conversely, posterior Z-residuals are surprisingly more stable at smaller sample sizes, and their double-use bias vanishes asymptotically as they converge to standard normal (as justified in the appendix). Nevertheless, because ISCV computation is fast, it is good practice to compute both, as a substantial discrepancy between the two is itself informative—typically flagging highly influential observations or unstable importance-sampling approximations (e.g., large Pareto k diagnostic values).

Finally, ISCV Z-residuals remain distinctly superior for models with observation-specific latent variables (Li et al. 2016, Millar 2018), as including y_i during fitting exerts an overwhelming influence on its corresponding latent effect.

In short, we recommend posterior Z-residuals as the default for routine checking. They are straightforward to compute, highly stable in finite samples, and free from importance-weighting errors. ISCV Z-residuals serve as a valuable sensitivity check when finite-sample self-influence is a concern—such as in highly flexible models or when specific observations heavily dictate the posterior.Because ISCV computation is fast, we advise calculating both. If they yield consistent results, the diagnostic conclusions are robust. A substantial discrepancy between the two is itself informative: it typically flags highly influential observations or unstable importance-sampling approximations (e.g., large Pareto k diagnostic values). In such cases, analysts should inspect these influential data points and the reliability of the importance weights before finalizing their assessment.

3. In RSP, the random variable U_i adds an additional layer of randomness to the diagnostic procedure. This raises questions regarding the stability and reproducibility of diagnostic results. It would be useful for the authors to discuss how the variability induced by this randomization affects diagnostic conclusions, and whether any strategies can be used to control this source of variability.

Response: Thank you for this important comment. Because auxiliary randomization is required for discrete outcomes, addressing the resulting test variability is essential. To ensure stability and reproducibility, we resample the auxiliary uniforms for a fixed dataset and model, generating a distribution of replicated diagnostic p-values.

Visually, diagnostic stability can be assessed via histograms of these replicated p-values. Quantitatively, we derive a distribution-free upper-bound p-value based on the median of these replications, yielding a principled, single-number summary that formally accounts for the auxiliary randomness, as reported in Sec. 2.8.

Furthermore, we have expanded Section 5 (Conclusion and Discussion) to provide practical guidance on two complementary strategies for obtaining stable single-number summaries:

1. **Parameter-wise posterior predictive p-values (PPPs):** Our simulations demonstrate that parameter-wise Z-residual PPPs mitigate the notorious conservatism of traditional χ^2 PPCs. Because they average over the MCMC draws, they remain highly stable despite the auxiliary $\boldsymbol{U}$ drawn at each iteration, serving as a reliable secondary confirmation of model misfit.

2. **Simulation-based calibration:** For exact calibration, analysts can simulate replicated datasets from the posterior predictive distribution to approximate the sampling distribution of the median p-value. This avoids the computational burden of rerunning MCMC for each replicate.

Finally, we note that aggregating MCMC-wise p-values via the Cauchy combination rule (Liu & Xie 2020) represents a highly promising strategy for handling auxiliary diagnostic variability, which we plan to explore in future research.

4. In the simulation study 3.2, is it possible to gain insights from any of the diagnostic results to guide users to choose the correct model family? In particular, beyond indicating that the working family is misspecified, can the diagnostic results provide interpretable patterns that suggest which alternative families may be more appropriate?

Response: Thank you for this insightful question. Yes, diagnostic results can reveal interpretable patterns that guide users toward a more appropriate family, provided that graphical and numerical tools are evaluated jointly. While diagnostics pinpoint the *direction* of model improvement, predictive information criteria are ultimately required to confirm the final model choice.

We have expanded Supplementary Section S1.2 to detail how specific visual clues suggest distinct alternative families:

- **QQ plot departures:** Systematic tail deviations strongly indicate overarching distributional misspecification, suggesting alternatives like a Negative Binomial or a heavier-tailed error distribution.
- **Zero-inflation patterns:** In count models, if scatterplots reveal that residuals for zero observations cluster distinctly away from non-zero observations, it points users toward hurdle or zero-inflated architectures.
- **Covariate trends:** A pronounced trend in a LOWESS curve or a structural shift in grouped boxplots against a specific covariate usually flags a localized functional-form error (e.g., a missing nonlinear term) rather than a family misspecification.

Once these diagnostic clues point the analyst toward plausible alternatives, the final selection should be determined by comparing criteria such as the leave-one-out expected log predictive density (LOO `elpd`).

We also note an important caveat, as discussed in Section S1.2: severe misspecification in the response family can sometimes cascade and cause false rejections in localized functional-form checks (like AOV tests). Because of these cascading effects, a robust workflow requires synthesizing multiple diagnostics. Developing tests with differential sensitivity to distinct misspecification sources is an active area of our ongoing research. To maintain focus on the fundamental properties of Z-residuals relative to PPCs in this manuscript, we defer the detailed separation of zero-model and count-model misspecification diagnostics to a forthcoming paper.

5. Most of the numerical illustrations focus on Poisson count data. To further support the claimed generality of the framework, it would be valuable to include additional examples involving other types of discrete outcomes, such as ordinal data, to demonstrate that the proposed diagnostics perform across different data types.

Response: Thank you for this insightful suggestion. We entirely agree with your underlying premise: it is crucial to demonstrate that the proposed diagnostics perform effectively across a diverse range of data types.

To achieve this goal and strengthen the manuscript, we opted to expand the empirical scope by introducing a continuous data setting (the inverse Gaussian regression in Section 3.2), rather than adding another discrete case like ordinal data. This choice serves a dual purpose. First, it addresses your call to demonstrate the framework's generality beyond count data. Second, because continuous data does not require auxiliary randomization, it allows us to perfectly isolate and highlight our core methodological contribution: the ability of the "summarize-first, then-test" paradigm to detect localized functional-form misspecifications (using covariate-specific plots and AOV tests) without the confounding variability of discrete data.

We believe this addition effectively broadens the empirical scope while powerfully contrasting the observation-level precision of Z-residuals with conventional omnibus PPCs. Because our framework indeed handles ordinal data seamlessly (requiring only standard model-implied survival probabilities and probability masses), we will feature ordinal regression examples prominently in our upcoming software package vignettes.

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Z-residuals for Diagnosing Bayesian Models

Abstract

Residual diagnostics are fundamental for assessing model adequacy in frequentist regression, yet analogous tools remain largely unavailable in Bayesian statistics. Standard approaches, such as posterior predictive checks (PPCs), often yield conservative p-values and rely heavily on arbitrary test statistics. We propose a versatile residual framework for Bayesian diagnostics based on *randomized survival probabilities* (RSPs), which are uniformly distributed under the true model. Utilizing a "summarize-first, then-test" strategy, we integrate over model parameters to obtain posterior or cross-validated RSPs, subsequently transforming them into *Z-residuals* that approximately follow $N(0, 1)$. We show that, under correct specification, these Z-residuals are asymptotically $N(0, 1)$, ensuring uniformly distributed test p-values. This overcomes the persistent conservatism of conventional PPCs. Simulation studies demonstrate that Z-residual diagnostics achieve precise calibration, revealing that covariate-specific ANOVA tests yield substantially higher power than overall goodness-of-fit checks for detecting structural misspecifications, including nonlinear covariate effects and incorrect distributional families. An application to a dolphin sighting dataset illustrates the method's practical utility by revealing a nonlinear model with enhanced predictive performance. Ultimately, the proposed Z-residual offers a theoretically grounded, well-calibrated, and highly versatile framework for Bayesian model diagnostics. Online supplementary material is available.

Keywords: Z-residual, Bayesian model checking, Model diagnostics, Posterior predictive probability, Randomized survival probability, Goodness-of-fit

List of Abbreviations: **AOV**, analysis of variance; **CV**, cross-validation/cross-validatory; **elpd**, expected log point-wise predictive density; **HNB**, hurdle negative binomial; **HPC**, holdout predictive check; **IG**, inverse Gaussian; **ISCV**, importance-sampling cross-validation; **LOO**, leave-one-out; **MCMC**, Markov chain Monte Carlo; **MSP**, middle-value survival probability; **NB**, negative binomial; **PPC**, posterior predictive check; **PPP**, posterior predictive p-value; **QQ**, quantile–quantile **RSP**, randomized survival probability; **SW**, Shapiro–Wilk; **WAIC**, Watanabe–Akaike Information Criterion, **UPC**, Uniform Parametrization Checks

1 Introduction

Verifying model correctness remains a central challenge in modern statistics. Traditionally, residual diagnostics have played a pivotal role in regression modeling, serving as indispensable tools for evaluating the adequacy of fitted models. They enable analysts to visualize model fit through graphical methods such as scatterplots and quantile–quantile (QQ) plots, revealing patterns or deviations that may indicate model deficiencies (Anscombe 1973). Beyond visualization, observation-level residuals facilitate a rich suite of formal statistical tests designed to pinpoint a variety of structural flaws. These include, but are not limited to, tests for detecting misspecified functional forms (Ramsey 1969), the Box-Ljung test for temporal autocorrelation (Ljung & Box 1978), Moran's I statistic for spatial dependencies (Moran 1950), and the Breusch-Pagan and Bartlett tests for heteroskedasticity (Breusch & Pagan 1979, Bartlett 1937), among many others. Rather than merely confirming adequacy through a single goodness-of-fit metric, these granular diagnostics provide actionable insights for refining the model, ultimately leading to more reliable and interpretable inferences (Cook & Weisberg 1982, Draper & Smith 1998). Despite their foundational importance in frequentist statistics, this comprehensive suite of observation-level residual diagnostic tools remains largely unavailable for general Bayesian modeling (see, for example, recent discussions by Liu & Zhang (2018), Liu et al. (2025), and Moran et al. (2024)).

To date, posterior predictive checks (PPC) (Rubin 1984, Gelman et al. 1996, Gabry et al. 2019, Säilynoja et al. 2021) remain among the most widely used tools for assessing the adequacy of Bayesian models. Their primary objective is to evaluate whether the fitted model is consistent with the observed data, $\boldsymbol{y}$, by comparing features of the observed data with those of replicated datasets, $\boldsymbol{y}^{\text{rep}}$, drawn from the posterior predictive distribution. This comparison is often summarized by a PPC p-value, defined as $\text{PPP} = P(T(\boldsymbol{y}^{\text{rep}}, \boldsymbol{\theta}) > T(\boldsymbol{y}, \boldsymbol{\theta}) \mid \boldsymbol{y})$, where $T(\boldsymbol{y}, \boldsymbol{\theta})$ is a user-specified discrepancy statistic.

Despite their popularity, PPC suffer from several critical limitations. Fundamentally, they operate on a *test-first, then-summarize* paradigm, evaluating discrepancy statistics for each MCMC draw before summarizing the resulting p-values. Because the lack of clear guidelines for selecting the test statistic T renders the choice fundamentally arbitrary, practitioners often resort to simplistic statistics or focus exclusively on the marginal distribution of replicated responses. However, relying solely on these marginal checks is insufficient; such an approach often masks subtle structural flaws—for example covariate functional form misspecifications—that only become apparent when the relationships between residuals and specific covariates are explicitly examined. Furthermore, evaluating a dataset-level summary statistic on the exact same data used for fitting exacerbates the data double-use bias. Consequently, PPP—especially those using the χ^2 statistic—are rarely uniform under the null; instead, they tend to cluster around 0.5, which severely degrades their statistical power to detect underlying model deficiencies (Robins et al. 2000, Bayarri & Berger 2000, Sinharay & Stern 2003, Gelman 2013). While alternative methods have been proposed to mitigate non-uniformity—such as sampled predictive p-values based on pivotal quantities (Johnson 2007) or holdout predictive checks (HPC) (Moran et al. 2024)—they introduce new trade-offs, suffering from instability due to a random draw of MCMC sample or reduced statistical power due to smaller effective sample sizes, respectively.

In this paper, we introduce a versatile diagnostic framework for Bayesian models called the Z-residual to address the aforementioned critical gap—the need for tools for conducting granular residual *diagnostics* in addition to overall model *checking*. Our framework is rooted in assessing the discrepancy of each y_i and the predictive distribution of y_i via examining the distribution of the tail (survival) probability of y_i and the pattern of these survival probabilities versus covariates. The foundational quantity is *randomized survival probability* (RSP). For an observed y_i, the parameter-wise RSP is defined as $\mathrm{RSP}_i(y_i \mid \boldsymbol{\theta}) = S_i(y_i \mid \boldsymbol{\theta}) + U_i\, p_i(y_i \mid \boldsymbol{\theta})$, where $S_i(y_i \mid \boldsymbol{\theta})$ and $p_i(y_i \mid \boldsymbol{\theta})$ are the survival and probability mass functions given covariates

$\boldsymbol{x}_i$, and $U_i \sim \text{Unif}(0, 1)$. The parameter-wise Z-residual is then defined by mapping the RSP through the inverse-survival function of standard normal: $z_i^{\text{RSP}}(y_i \mid \boldsymbol{\theta}) = -\Phi^{-1}[\text{RSP}_i(y_i \mid \boldsymbol{\theta})]$.

For continuous responses, the auxiliary randomization U_i is unnecessary, and the RSP simplifies directly to the standard probability integral transform (Devroye 1986, Angus 1994), whose uniformity has previously been explored for Bayesian model checking (Conn et al. 2018). In Bayesian contexts, the RSP is also closely related to the conditional predictive ordinate (CPO) (Gelfand et al. 1992, Seillier-Moiseiwitsch 1993, Seillier-Moiseiwitsch & Dawid 1993, Gelfand & Dey 1994, Marshall & Spiegelhalter 2007, Li et al. 2017). CPOs typically employ the non-random mid-value survival probability ($U_i = 0.5$) to evaluate outlier extremeness without referencing the underlying distribution. However, the randomization in the RSP is strictly necessary for discrete responses to ensure exact uniform distribution on $[0, 1]$ for parameter-wise RSPs; see detailed discussions in the context of frequentist count regression models (Frühwirth-Schnatter 1996, Dunn & Smyth 1996, Feng et al. 2020).

In practice, we must account for the finite-sample uncertainty of $\boldsymbol{\theta}$. The construction of our residuals is straightforward, employing a "summarize-first, then-test" paradigm which is distinct to the "test-first, then-summarize" approaches of PPC, HPC, and recently proposed uniform parametrization checks (Covington & Miller 2025). We construct **posterior RSPs** and **ISCV RSPs** by averaging parameter-wise RSPs over the posterior of $\boldsymbol{\theta}$ and the leave-one-out (LOO) posterior of $\boldsymbol{\theta}$ (via an importance sampling approximation), respectively. The posterior or ISCV RSPs are then transformed into Z-residuals using the inverse-surival function of $N(0, 1)$. This framework establishes a single set of observation-level residuals that approximately follow a standard normal distribution, which can subsequently be analyzed using the graphical and numerical tools previously devised for ordinary linear regression.

To firmly ground this methodology, we provide a rigorous theoretical validation demonstrating that both the posterior and LOOCV RSPs are asymptotically uniformly distributed from a strictly frequentist perspective. This calibration guarantees that the resultant Z-residuals are asymptotically standard normal, bridging the gap between frequentist structural tests and Bayesian modeling. Furthermore, to address the variability introduced by the auxiliary randomization U_i for discrete responses, we introduce a novel, distribution-free upper bound to reliably summarize replicated p-values.

Despite its conceptual simplicity, posterior and ISCV Z-residuals rest upon solid theoretical foundations and offer clear practical advantages, as highlighted below:

- **Asymptotic Calibration:** We prove that the diagnostic p-values based on posterior and ISCV Z-residuals are asymptotically $\mathrm{Unif}(0, 1)$. This is theoretically justified by the fact that the posterior and LOO predictive distributions of y_i converge to their true counterparts when $n \to \infty$. This stands in sharp contrast to PPC, whose p-values remain poorly calibrated even as $n \to \infty$ (Robins et al. 2000).
- **Diagnostic Richness and Versatility:** The approximate normality of Z-residuals enables a broad suite of diagnostic tools—analogous to Pearson residuals—including goodness-of-fit assessments, functional-form checks, and heteroscedasticity detection. Once computed, a single set of Z-residuals can be reused for multiple graphical and numerical diagnostics. Our empirical examples show that specific diagnostics, such as nonlinearity tests, can be substantially more powerful in detecting model misspecification than overall goodness-of-fit tests.
- **No Reduction in Test Sample Size:** Z-residual diagnostics utilize the full sample size n, thereby preserving statistical power, especially for small or moderate datasets. This contrasts with the holdout predictive checks of Moran et al. (2024), which mitigate data double-use bias but sacrifice statistical power due to a smaller effective sample size.

We empirically evaluate the performance of Z-residuals in two simulation settings focused on detecting nonlinear covariate effects: a continuous inverse Gaussian regression and a discrete hurdle negative binomial regression. In both settings, the correctly specified and misspecified models use the same response family but differ in the functional form of a key covariate, allowing the diagnostics to be compared with PPC and HPC based on analogous discrepancy statistics. Our studies demonstrate that covariate-specific AOV tests based on posterior and ISCV RSP Z-residuals remain well calibrated under the true model and achieve higher power than omnibus SW or χ^2 checks and the corresponding PPC/HPC methods. Applied to a real dolphin-sighting dataset, the proposed Z-residual identifies a nonlinear HNB model that satisfies the diagnostic checks and achieves the best LOO predictive performance.

The remainder of this paper is organized as follows. Section 2 presents the proposed Z-residual framework. Section 3 reports simulation results evaluating calibration and power. Section 4 applies Z-residuals to an environmental dataset to demonstrate practical value. Section 5 concludes with a summary and future directions. The **supplementary materials** provide the formal theoretical framework and rigorous proofs establishing the asymptotic properties of Z-residuals, as well as the distribution-free upper bounds for replicated p-values. Additionally, they report further simulation studies demonstrating the versatility of Z-residuals in detecting response-family and mixed-model misspecifications, while highlighting the necessity of the auxiliary randomization U_i for discrete responses.

2 Z-residuals

2.1 Preliminaries and Notations

We model a response variable y_i given a vector of covariates $\boldsymbol{x}_i$ *independently* using a distribution with parameters $\boldsymbol{\theta}$. This distribution is characterized by its survival function $S_i(y_i|\boldsymbol{x}_i, \boldsymbol{\theta}) = P(Y_i > y \mid \boldsymbol{x}_i, \boldsymbol{\theta})$ and probability mass function $p_i(y_i|\boldsymbol{x}_i, \boldsymbol{\theta}) = P(Y_i = y \mid \boldsymbol{x}_i, \boldsymbol{\theta})$. The density function (in a measure-theoretic sense) for this distribution is denoted by $f_i(y_i|\boldsymbol{x}_i, \boldsymbol{\theta})$. The subscript i indicates that the distribution can differ for each observation, such as in a regression context. For notational simplicity, we will hereafter drop the explicit conditioning on $\boldsymbol{x}_i$ and write these functions as $S_i(y_i|\boldsymbol{\theta})$, $p_i(y_i|\boldsymbol{\theta})$, and $f_i(y_i|\boldsymbol{\theta})$.

We denote the prior density for the parameter vector $\boldsymbol{\theta}$ as $\pi(\boldsymbol{\theta})$. Let $\mathcal{Y}_i$ denote the sample space of Y_i, and let $\mathcal{Y}^{(n)} = \prod_{i=1}^{n} \mathcal{Y}_i$. The observed data vector is $\mathbf{y} = (y_1, \ldots, y_n) \in \mathcal{Y}^{(n)}$.

Given the full data $\boldsymbol{y} = (y_1, \ldots, y_n)$, the full-data posterior density is $\pi(\boldsymbol{\theta} \,|\, \boldsymbol{y})$. For LOO procedures, we use the data $\boldsymbol{y}_{-i}$, which excludes the i-th observation, to form the LOO posterior density $\pi(\boldsymbol{\theta} \,|\, \boldsymbol{y}_{-i})$. Finally, the expectations taken with respect to the full and LOO posteriors are denoted by $\mathbb{E}_{\boldsymbol{\theta} \,|\, \boldsymbol{y}}(\cdot)$ and $\mathbb{E}_{\boldsymbol{\theta} \,|\, \boldsymbol{y}_{-i}}(\cdot)$, respectively.

Throughout this article, we adopt the following notations. Uppercase letters denote random variables $(Y_i, \mathbf{Y})$ and lowercase letters denote their observed values $(y_i, \boldsymbol{y})$. Boldface distinguishes vectors (e.g., $\boldsymbol{y}$) from scalars. Finally, $\boldsymbol{y}^{\text{rep}}$ indicates replicated data drawn from the posterior predictive distribution as opposed to the observed data $\boldsymbol{y}^{\text{obs}}$ for contrast.

2.2 Parameter-wise RSP and Z-residuals

The foundation of our method is the **randomized survival probability (RSP)**, a transformation that converts any observation into a standard uniform variable, even if its distribution has discrete probability mass. The RSP of an observation y_i is calculated by adding a uniformly distributed random component, $U_i \sim \text{Unif}(0, 1)$, to its survival probability S_i:

$$\text{RSP}_i(y_i \,|\, \boldsymbol{\theta}) = S_i(y_i \,|\, \boldsymbol{\theta}) + U_i \cdot p_i(y_i \,|\, \boldsymbol{\theta}) \tag{1}$$

For continuous responses, $p_i(y_i \mid \theta) = 0$, the auxiliary randomization is unnecessary and $(\text{RSP}_i(y_i \mid \theta) = S_i(y_i \mid \theta))$.

A deterministic alternative is the **middle-value survival probability (MSP)**, which replaces the random component U_i in the RSP with the constant 0.5: $\text{MSP}_i(y_i \,|\, \boldsymbol{\theta}) = S_i(y_i \,|\, \boldsymbol{\theta}) + 0.5 \cdot p_i(y_i \,|\, \boldsymbol{\theta})$. While the MSP, as a probability, is useful for flagging potential outliers (Marshall & Spiegelhalter 2007, Li et al. 2017), as we discuss above, the randomization U_i is necessary to produce a truly uniform RSP for discrete y_i.

The randomization is needed when the response distribution has non-zero probability mass at the observed y_i. Discrete y_i can be regarded as an incomplete observation of a continuous random variable, hence, an injected U_i is necessary to recover it. Specifically, the survival function has a jump at y_i, and without randomization all observations with the same discrete value would be mapped to the same survival probability. The auxiliary variable U_i randomly maps y_i to a random number in the probability interval at this jump and thereby restoring the uniform reference distribution under the true model. A key property of this transformation is that if the model and parameters $\boldsymbol{\theta}$ are correct and the observations are independent, the resulting RSPs are independent standard uniform distribution; see Theorem S3.1 for a formal proof and a graphical illustration using a toy example. Detailed illustrations of the uniformity can be found in Dunn & Smyth (1996), Feng et al. (2020).

This uniformity allows us to further transform the RSP into a standard normal residual via the inverse-survival function, $-\Phi^{-1}(\cdot)$, of the $N(0,1)$ distribution:

$$z_i^{\mathrm{RSP}}(y_i \mid \boldsymbol{\theta}) = -\Phi^{-1}(\mathrm{RSP}_i(y_i \mid \boldsymbol{\theta})). \tag{2}$$

In words, this is the standard normal value with the same upper-tail area (RSP) as the observed y_i. Under the true model, these transformed quantities are independent $N(0,1)$ variables. We term this a "**Z-residual**" because Z conventionally denotes a standard normal random variable. The negative sign preserves the traditional residual direction, ensuring that observations larger than expected yield positive residuals. Conceptually, a Z-residual reflects the distance from y_i to its predictive median, rescaled for skewness and smoothed for discreteness. Consequently, a residual of ± 2 conveys the exact same degree of extremeness regardless of the underlying predictive distribution's shape.

It is noteworthy that the RSP and the randomized probability integral transform (Smith 1985) are statistically identical. Using the identities $S_i(y_i \mid \boldsymbol{\theta}) = 1 - F_i(y_i \mid \boldsymbol{\theta})$ and $p_i(y_i \mid \boldsymbol{\theta}) = F_i(y_i \mid \boldsymbol{\theta}) - F_i(y_i - \mid \boldsymbol{\theta})$, we can show the identity: $1 - \mathrm{RSP}_i(y_i \mid \boldsymbol{\theta}) = F_i(y_i - \mid \boldsymbol{\theta}) + (1 - U_i) \cdot p_i(y_i \mid \boldsymbol{\theta})$, where $F_i(y_i - \mid \boldsymbol{\theta}) = P(Y_i < y_i \mid \boldsymbol{\theta})$ is the left-hand limit of the cumulative distribution function F. Since $U_i' = 1 - U_i$ is also a standard uniform random variable, this identity shows that $1 - \mathrm{RSP}_i$ is distributionally equivalent to the randomized probability integral transform.

2.3 Posterior Z-residuals

In the Bayesian paradigm, parameters $\boldsymbol{\theta}$ are unknown and are described by a posterior distribution $\pi(\boldsymbol{\theta} \mid \boldsymbol{y})$. To account for this uncertainty, we can average the point-wise RSP over the posterior distribution. We define the **posterior RSP** as:

$$\mathrm{RSP}_i^{\mathrm{Post}}(y_i) = \int \mathrm{RSP}_i(y_i \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta} \mid \boldsymbol{y})d\boldsymbol{\theta} = \mathbb{E}_{\boldsymbol{\theta} \mid \boldsymbol{y}}(S_i(y_i \mid \boldsymbol{\theta})) + U_i \cdot \mathbb{E}_{\boldsymbol{\theta} \mid \boldsymbol{y}}(p_i(y_i \mid \boldsymbol{\theta})) \quad (3)$$

In practice, given a sample of T draws, $\{\boldsymbol{\theta}^{(t)}\}_{t=1}^{T}$, from the posterior via Markov chain Monte Carlo (MCMC) or other methods, we estimate this integral with a Monte Carlo average:

$$\widehat{\mathrm{RSP}}_i^{\mathrm{Post}}(y_i) = \frac{\sum_{t=1}^{T} \mathrm{RSP}_i(y_i \mid \boldsymbol{\theta}^{(t)})}{T} \quad (4)$$

The **posterior RSP Z-residual** is then defined as:

$$z_i^{\text{Post-RSP}}(y_i) = -\Phi^{-1}(\widehat{\mathrm{RSP}}_i^{\mathrm{Post}}(y_i)) \quad (5)$$

The posterior MSP Z-residual, $z_i^{\text{Post-MSP}}(y_i)$, is defined analogously by substituting the MSP for the RSP in the equations above or fixing $U_i = 0.5$.

2.4 Importance-Sampling Cross-Validatory (ISCV) Z-residuals

While simple to compute, the posterior Z-residual uses the data $\boldsymbol{y}$ twice: once to form the posterior $\pi(\boldsymbol{\theta} \mid \boldsymbol{y})$ and again in the calculation for y_i. This double-use may reduce its power to detect model misfit in finite-sample cases. To address this potential limitation, we consider the method of LOO cross-validation. The **cross-validatory RSP**, $\mathrm{RSP}_i^{\mathrm{CV}}(y_i)$, is averaged over the posterior distribution conditioned on all data except y_i, denoted $\boldsymbol{y}_{-i}$:

$$\mathrm{RSP}_i^{\mathrm{CV}}(y_i) = \int \mathrm{RSP}_i(y_i \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta} \mid \boldsymbol{y}_{-i})d\boldsymbol{\theta} = \mathbb{E}_{\boldsymbol{\theta} \mid \boldsymbol{y}_{-i}}(S_i(y_i \mid \boldsymbol{\theta})) + U_i \cdot \mathbb{E}_{\boldsymbol{\theta} \mid \boldsymbol{y}_{-i}}(p_i(y_i \mid \boldsymbol{\theta})) \quad (6)$$

Calculating this directly would require re-running the MCMC analysis n times, which is computationally prohibitive. Instead, we efficiently approximate it using the full-data posterior sample via importance sampling reweighting. The importance weights are given by the ratio of the target LOO posterior to the full-data posterior, which simplifies to the inverse of the likelihood due to the left-out observation (Li et al. 2016):

$$w_i^{\mathrm{IS}}(\boldsymbol{\theta}) = \frac{1}{f_i(y_i \mid \boldsymbol{\theta})} \propto \frac{\pi(\boldsymbol{\theta} \mid \boldsymbol{y}_{-i})}{\pi(\boldsymbol{\theta} \mid \boldsymbol{y})} \quad (7)$$

The **ISCV RSP** is then computed as a weighted average using the MCMC samples:

$$\widehat{\text{RSP}}_i^{\text{ISCV}}(y_i) = \frac{\sum_{t=1}^{T} \left[\text{RSP}_i(y_i \,|\, \boldsymbol{\theta}^{(t)}) w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)})\right]}{\sum_{t=1}^{T} w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)})} \tag{8}$$

Finally, the **ISCV RSP Z-residual** is defined as:

$$z_i^{\text{ISCV-RSP}}(y_i) = -\Phi^{-1}(\widehat{\text{RSP}}_i^{\text{ISCV}}(y_i)) \tag{9}$$

The ISCV MSP Z-residual, $z_i^{\text{ISCV-MSP}}(y_i)$, is calculated similarly with the U_i fixed at 0.5.

In practice, we generally recommend posterior Z-residuals as the default tool for routine diagnostics of models without highly localized latent effects.. While the cross-validatory construction is theoretically appealing because it alleviates the direct self-influence of y_i on its own predictive distribution, ISCV Z-residuals are not exactly standard normal in finite samples—their variance is greater than 1 for high-leverage observations, akin to why frequentist ordinary least squares relies on studentized residuals. Furthermore, this construction does not completely eliminate dependence in the resulting LOO Z-residuals, as the training sets $\boldsymbol{y}_{-i}$ and $\boldsymbol{y}_{-j}$ share $n-2$ observations. In our simulations, this variance inflation and residual dependence, combined with importance-sampling approximation errors, resulted in higher Type I error rates for ISCV Z-residuals. Conversely, posterior Z-residuals are surprisingly more stable at smaller sample sizes, and their double-use bias vanishes asymptotically as they converge to standard normal. Nevertheless, because ISCV computation is fast, it is good practice to compute both, as a substantial discrepancy between the two is itself informative—typically flagging highly influential observations or unstable importance-sampling approximations (e.g., large Pareto k diagnostic values). Finally, ISCV Z-residuals remain distinctly superior for models with observation-specific latent variables (Li et al. 2016, Millar 2018), as including y_i during fitting exerts an overwhelming influence on its corresponding latent effect.

2.5 Model Diagnostics with Z-residuals

Once computed, Z-residuals serve as the traditional regression residuals for an iterative process of model refinement. The construction of Z-residuals does not depend on a particular test, hence, they can be evaluated using a set of graphical and numerical tools.

This iterative workflow typically begins with marginal checking to assess the overall distributional fit. Practitioners can check whether the Z-residuals approximately follow a normal reference distribution visually using a QQ plot, and numerically using a normality test such as the SW test. For marginal checking, we also explored a χ^2 test, adjusting the degrees of freedom to account for posterior variance shrinkage. However, we do not recommend this test. As shown in our simulation studies, it might perform poorly due to severe sensitivity to residual variance deflation and inflation. The SW test is a better-calibrated choice.

In addition to marginal checks, diagnostics should employ targeted checks against covariates (or linear predictors) to explore specific functional-form misspecifications. Residual-versus-covariate scatterplots should be structureless. For a discrete covariate, any systematic trend can be formally evaluated by directly applying an AOV test across its natural categories to check for grouped mean changes, and Bartlett's test for heteroskedasticity. For a continuous covariate, we achieve this by first dividing its range into k equally spaced intervals and then applying the same tests; see examples of grouped Z-residuals in Fig. 8(b) and 9(b). Because this grouped AOV approach does not assume a specific non-linear function, it provides a broadly powerful test for arbitrary non-linearity. Furthermore, because it exhibits low sensitivity to the choice of k, we recommend fixing the bin count to a standard default of $k = 10$. Additionally, residual-vs-index plots can reveal isolated outliers, clustering, or disparate observations in two-part models, while residual dependence can be assessed using Durbin-Watson and Moran's I statistics. As another exploratory step, Z-residuals can be plotted and tested against variables (eg. spatial variables) not included in the fitted model to identify potentially omitted covariate effects.

It is worth noting that we do not prescribe a rigid order for model diagnostics. Rather, effective model diagnostics should be regarded as a synthesis of diagnostic information aimed at identifying structural problems. *For example, severe mean-structure misspecifications can* *cascade and cause failures in the marginal QQ and SW checks.* By jointly evaluating these marginal metrics alongside covariate-specific plots and AOV tests, the analyst can better untangle these cascading effects and identify the true cause of the misfit.

Finally, we wish to clarify that multiple comparison bias is generally not a primary concern in this context. The use of statistical tests in model diagnostics differs fundamentally from traditional scientific hypothesis testing. In scientific discovery, a significant result is the goal, and multiple testing inflates the risk of false discoveries. In model diagnostics, however, the goal is a well-fitting model, indicated by *non-significant* p-values. An inflated Type I error rate in this exploratory phase merely makes the diagnostic scrutiny more rigorous. A "significant" model deficiency is not a final conclusion, but rather a trigger for iterative refinement. Once a refined model yields non-significant p-values, it is then compared against previous iterations using independent predictive criteria—such as WAIC or LOOCV (Vehtari et al. 2017)—to ensure genuine improvement. From a pragmatic modeling perspective, the cost of a false positive in model diagnostics is minimal: if a falsely flagged deficiency leads to a proposed remedy that fails to improve predictive utility, the analyst simply discards the remedy and retains the original model.

However, we note two important exceptions: if Z-residuals are used to hunt for correlated features among a large pool of unused covariates, or if a vast battery of pre-specified tests is deployed to criticize a model, the procedure remains subject to multiple testing bias. In such scenarios, the results must be interpreted with caution and accompanied by robust multiplicity corrections. Appropriate strategies include established false discovery rate controls (Benjamini & Hochberg 1995), E-value calibration techniques (Vovk & Wang 2021, Ramdas et al. 2023), or data-splitting methods that reserve a hold-out set for final model evaluation.

2.6 Computational Procedures of Z-residuals and Z-residual-PPC

We can conduct a PPC using *parameter-wise* Z-residuals to construct the discrepancy. For each MCMC draw $\boldsymbol{\theta}^{(t)}$, we first compute $\{z_i^{\mathrm{RSP}}(y_i^{\mathrm{obs}}, \boldsymbol{\theta}^{(t)})\}_{i=1}^n$ and $\{z_i^{\mathrm{RSP}}(y_i^{\mathrm{rep}}, \boldsymbol{\theta}^{(t)})\}_{i=1}^n$. Let

$\mathcal{P}(\boldsymbol{y}, \boldsymbol{\theta})$ be a function that computes a test p-value. For each MCMC draw $\boldsymbol{\theta}^{(t)}$, we compute a p-value for the observed data, $\mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)})$, and for each replicated dataset, $\mathcal{P}(\boldsymbol{y}_t^{\text{rep}}, \boldsymbol{\theta}^{(t)})$.

The PPC p-value is estimated by the proportion of draws where the p-value from the replicated data is smaller than the p-value from the observed data:

$$\widehat{\text{PPP}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\Big\{ \mathcal{P}(\boldsymbol{y}_t^{\text{rep}}, \boldsymbol{\theta}^{(t)}) < \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)}) \Big\}. \tag{10}$$

A key simplification for computing PPP follows from a basic testing principle (Meng 1994). Conditional on a parameter value $\boldsymbol{\theta}^{(t)}$, a replicate $\boldsymbol{y}^{\text{rep}}$ is generated from the model specified by $\boldsymbol{\theta}^{(t)}$, which is the true parameter for $\boldsymbol{y}^{\text{rep}}$. Therefore, if $\mathcal{P}(\boldsymbol{y}, \boldsymbol{\theta}^{(t)})$ is a well-calibrated test p-value conditional on $\boldsymbol{\theta}^{(t)}$, then $\mathcal{P}(\boldsymbol{y}^{\text{rep}}, \boldsymbol{\theta}^{(t)})$ is *exactly* uniform on $(0, 1)$. For a fixed observed value $\mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)})$, this implies that the conditional probability of the replicated p-value being smaller than the observed p-value is simply equal to the observed p-value itself: $P\Big(\mathcal{P}(\boldsymbol{Y}^{\text{rep}}, \boldsymbol{\theta}^{(t)}) < \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)}) \,|\, \boldsymbol{\theta}^{(t)}\Big) = \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)})$. By the law of iterated expectations,

$$P\Big(\mathcal{P}(\boldsymbol{Y}^{\text{rep}}, \boldsymbol{\Theta}) < \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\Theta}) \,|\, \boldsymbol{y}^{\text{obs}}\Big) = \mathbb{E}_{\boldsymbol{\theta}\,|\,\boldsymbol{y}^{\text{obs}}}\Big[\mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta})\Big]. \tag{11}$$

Therefore, we need not replicate datasets for computing PPP with the RSP Z–residuals. The p-value is simply the average of the p-values computed for each MCMC draw:

$$\widehat{\text{PPP}}^{\text{RSP}} = \frac{1}{T}\sum_{t=1}^{T} \mathcal{P}(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}^{(t)}). \tag{12}$$

The identity (12) holds for any PPC and HPC when the computed p-value for each $\boldsymbol{\theta}^{(t)}$ truly follows Unif$(0, 1)$ (Meng 1994), a condition satisfied by the tests based on parameter-wise Z-residuals. For HPC with parameter-wise Z-residuals, we can also compute the p-values using (12), except that the MCMC sample of $\boldsymbol{\theta}$ is drawn from an isolated training sample. Because the null distribution of MSP Z-residuals is unknown, identity (11) does not apply; we instead compute PPP via (10) using replicated datasets.

The key difference between this PPC method and diagnostics using Z-residuals is the order of operations. The PPC method follows a **"test first, then summarize"** workflow, where a test is performed on the set of parameter-wise Z-residuals for each draw $\boldsymbol{\theta}^{(t)}$. Conversely, diagnostics with Z-residuals follow a **"summarize first, then test"** workflow. They first

Table 1: Computational Procedures of Z-residuals and Z-residual-PPC

PPC with Parameter-wise Z-residuals

	y_1	y_2	$\cdots$	y_n		
$\boldsymbol{\theta}^{(1)}$	$z_1^{\text{RSP}}(y_1 \mid \boldsymbol{\theta}^{(1)})$	$z_2^{\text{RSP}}(y_2 \mid \boldsymbol{\theta}^{(1)})$	$\cdots$	$z_n^{\text{RSP}}(y_n \mid \boldsymbol{\theta}^{(1)})$	$\xrightarrow{\text{Test}}$	p_1^{obs}
$\boldsymbol{\theta}^{(2)}$	$z_1^{\text{RSP}}(y_1 \mid \boldsymbol{\theta}^{(2)})$	$z_2^{\text{RSP}}(y_2 \mid \boldsymbol{\theta}^{(2)})$	$\cdots$	$z_n^{\text{RSP}}(y_n \mid \boldsymbol{\theta}^{(2)})$	$\xrightarrow{\text{Test}}$	p_2^{obs}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$		$\vdots$
$\boldsymbol{\theta}^{(T)}$	$z_1^{\text{RSP}}(y_1 \mid \boldsymbol{\theta}^{(T)})$	$z_2^{\text{RSP}}(y_2 \mid \boldsymbol{\theta}^{(T)})$	$\cdots$	$z_n^{\text{RSP}}(y_n \mid \boldsymbol{\theta}^{(T)})$	$\xrightarrow{\text{Test}}$	p_T^{obs}
						$\downarrow$ PPP

Z-Residual Diagnostics

	y_1	y_2	$\cdots$	y_n		
$\boldsymbol{\theta}^{(1)}$	$\text{RSP}_1(y_1 \mid \boldsymbol{\theta}^{(1)})$	$\text{RSP}_2(y_2 \mid \boldsymbol{\theta}^{(1)})$	$\cdots$	$\text{RSP}_n(y_n \mid \boldsymbol{\theta}^{(1)})$		
$\boldsymbol{\theta}^{(2)}$	$\text{RSP}_1(y_1 \mid \boldsymbol{\theta}^{(2)})$	$\text{RSP}_2(y_2 \mid \boldsymbol{\theta}^{(2)})$	$\cdots$	$\text{RSP}_n(y_n \mid \boldsymbol{\theta}^{(2)})$		
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$		
$\boldsymbol{\theta}^{(T)}$	$\text{RSP}_1(y_1 \mid \boldsymbol{\theta}^{(T)})$	$\text{RSP}_2(y_2 \mid \boldsymbol{\theta}^{(T)})$	$\cdots$	$\text{RSP}_n(y_n \mid \boldsymbol{\theta}^{(T)})$		
z_i^{RSP}	$\downarrow$ $z_1^{\text{RSP}}(y_1)$	$\downarrow$ $z_2^{\text{RSP}}(y_2)$	$\cdots$	$\downarrow$ $z_n^{\text{RSP}}(y_n)$	$\xrightarrow{\text{Test}}$	Z^{RSP} p-value

summarize over the MCMC draws to obtain the posterior or ISCV RSPs, which are then used to create a *single* set of Z-residuals. Finally, a diagnostic test is applied to Z-residuals. Table 1 gives their computing procedures for showing the difference of these approaches.

2.7 Asymptotic Normality and Independence of Z-residuals

The results in supplementary §S3 establish the asymptotic properties that make Z-residuals a reliable basis for Bayesian model diagnostics. The foundation is the convergence of the LOO predictive distribution to the true data-generating distribution, which we restate here informally; the full regularity conditions are given in the Supplementary Materials.

Theorem S3.2 (Total variation convergence of the LOO predictive distribution, informal statement)**.** *Let $Y_1, \ldots, Y_{n+1}$ be independent with true densities $f_i^{(*)}$, and suppose the model $\{f_i(\cdot \mid \theta) : \theta \in \Theta\}$ with prior Π is well-specified, so that the true parameter set $W_0 = \{\theta \in \Theta : f_i(\cdot \mid \theta) = f_i^{(*)} \text{ for all } i\}$ is non-empty. Assume standard regularity conditions ensuring posterior consistency and predictive continuity (specifically, conditions (i)–(v) of supplementary §S3). Under these conditions, the LOO predictive distribution $P_i^{(n),\text{CV}}$ converges to the true distribution $P_i^{(*)}$ in total variation, in probability: for each fixed i,*

$$\left\| P_i^{(n),\text{CV}} - P_i^{(*)} \right\|_{\text{TV}} \xrightarrow{P} 0 \quad \text{as } n \to \infty. \tag{13}$$

Because each observation's RSP is computed from a predictive distribution that converges to the truth, the RSPs themselves inherit the correct limiting behavior, not only marginally but also jointly across observations:

Theorem S3.5 (Asymptotic independence of cross-validated RSPs)**.** *Under the assumptions of Theorem S3.2, for any fixed pair of distinct indices $i \neq j$, the cross-validated RSPs converge jointly in distribution to a pair of independent standard uniform random variables: as $n \to \infty$,*

$$\left(\mathrm{RSP}_i^{(n),\mathrm{CV}},\, \mathrm{RSP}_j^{(n),\mathrm{CV}}\right) \;\Rightarrow\; \left(\mathrm{Uniform}(0,1),\, \mathrm{Uniform}(0,1)\right), \tag{14}$$

where the two limiting uniform variables are independent.

The same argument extends to the posterior construction: posterior RSPs are likewise asymptotically independent and uniformly distributed (Theorems S3.4 and S3.6). Consequently, in large samples both posterior and ISCV Z-residuals behave approximately as a sample from the $N(0,1)$ distribution, so that well-behaved diagnostic procedures applied to them are asymptotically well-calibrated. We say "approximately" because both constructions incur pre-asymptotic error: the RSPs are averaged over the finite-sample posterior or LOO distribution of $\boldsymbol{\theta}$ rather than evaluated at the true $\boldsymbol{\theta}$, and ISCV incurs additional error from importance sampling.

These results also clarify the theoretical distinction between Z-residual diagnostics and standard PPC, which lies in how much an observation y_i^{obs} influences its own check.

- **Tests on Z-residuals:** The influence of any single y_i^{obs} on its own predictive distribution vanishes as $n \to \infty$: by Theorem S3.4, the full-data posterior predictive law $P_i^{(n)}(\cdot \mid \boldsymbol{y})$ and the LOO predictive law $P_i^{(n),\mathrm{CV}}(\cdot \mid \boldsymbol{y}_{-i})$ become asymptotically indistinguishable, and both converge to the true law $P_i^{(*)}$. Each RSP is therefore asymptotically evaluated against the true distribution, and the resulting Z-residuals follow the $N(0,1)$ distribution.
- **PPC:** In contrast, a PPC evaluates a dataset-level discrepancy statistic $T(\mathbf{y}, \boldsymbol{\theta})$. Because the **entire** observed dataset $\mathbf{y}$ is conditioned upon to generate the reference distribution,

the simulated replicates are systematically biased toward the observed data. This dataset-level double use renders PPC p-values conservative, even in the asymptotic regime (Robins et al. 2000). Specifically, as the sample size $n \to \infty$, the downward bias in the expected discrepancy does not vanish; rather, it converges to an $\mathcal{O}(1)$ constant dictated by the parameter dimension, p. Conceptually, this phenomenon is analogous to the in-sample optimism formalized by Wilks' theorem, where maximum likelihood estimation absorbs sample noise, inducing a non-vanishing p-dimensional upward bias in the maximized log-likelihood. Similarly, the double use of data in PPC reduces the expected discrepancy by p degrees of freedom. Consequently, this bias scales with model complexity, p, and does not vanish even when $n \to \infty$.

2.8 A Distribution-Free Upper-Bound Summary of Replicated p-values of the Same Dataset

For continuous data, a Z-residual test p-value is a deterministic function of the data. For discrete data, however, randomization is used to create a valid test, hence, the resulting p-value, which we denote $\mathcal{P}(\boldsymbol{y}, \boldsymbol{U})$, is a random variable of auxiliary uniform draws $\boldsymbol{U} = (U_1, \ldots, U_n)$. To account for this variability, we can generate replicated p-values by holding the data $\boldsymbol{y}$ fixed while repeatedly redrawing the auxiliary randomness. For J independent sets of draws, $\boldsymbol{U}^{(1)}, \ldots, \boldsymbol{U}^{(J)}$, we compute: $p_j(\boldsymbol{y}) = \mathcal{P}(\boldsymbol{y}, \boldsymbol{U}^{(j)}), j = 1, \ldots, J$. A histogram of these replicated p-values, $\{p_j(\boldsymbol{y})\}_{j=1}^{J}$, is informative for model checking. A large proportion of small p-values indicates potential model misfit; see two comparative histograms in Fig. 8 (c)(d).

While this distribution is informative, it is impractical for formal reporting, which requires a single number. Let $p_{(1)}(\boldsymbol{y}) \leq \cdots \leq p_{(J)}(\boldsymbol{y})$ denote the order statistics of the replicated p-values. Under the null hypothesis, any single p-value, $\mathcal{P}(\boldsymbol{Y}, \boldsymbol{U})$, is asymptotically uniform on $(0, 1)$. This property allows us to derive a simple and conservative upper-bound for the p-value of an order statistic. For any fixed rank $r \in \{1, \ldots, J\}$ and threshold $t \in [0, 1]$, the probability that the order statistic $p_{(r)}(\boldsymbol{Y})$ falls below t is bounded by:

$$P\{p_{(r)}(\boldsymbol{Y}) \leq t\} \leq H_{J,r} \cdot t, \text{ where, } H_{J,r} := \sup_{p\in(0,1]} \frac{P\{\text{Bin}(J,p) \geq r\}}{p}. \tag{15}$$

Here, $\text{Bin}(J,p)$ is a binomial random variable with J trials and success probability p. This inequality immediately yields an upper-bound p-value:

$$\mathcal{P}^{\text{ub}}_{(r)}(\boldsymbol{y}) := \min\{1, H_{J,r} \cdot p_{(r)}(\boldsymbol{y})\}, \tag{16}$$

which is super-uniform under the null—$P\{\mathcal{P}^{\text{ub}}_{(r)}(\boldsymbol{Y}) \leq \alpha\} \leq \alpha$ for any $\alpha \in (0,1)$. Notably, this bound is distribution-free, hence, widely applicable. The proof is provided in Theorem S3.8.

The value of $H_{J,r}$ can be computed using numerical optimization. For the median ($r = \lceil J/2 \rceil$), the constant is well-behaved, with $H_{J,r} < 2$ for all J. For example, $H_{100,50} \approx 1.65$, and $H_{500,250} \approx 1.80$. This yield a better bound than 2 for the median of correlated sample (Johnson 2007, Yuan & Johnson 2012), which is derived with Markov inequality (see the remark of Theorem S3.8. While conservative, this bound provides a useful and computationally efficient summary for replicated test p-values generated from the same dataset with different $\boldsymbol{U}$.

3 Simulation Studies

3.1 Notations of Diagnostic Tests

To compare diagnostic calibration and power in the following simulations, we denote Z-residual tests as Z^{type}-TEST, where type (**Post-RSP** or **ISCV-RSP**) indicates the residual construction method and TEST specifies the applied statistical test. For example, $Z^{\text{Post-RSP}}$-SW applies the SW test to posterior RSP residuals, while $Z^{\text{ISCV-RSP}}$-χ^2 applies the χ^2 goodness-of-fit test to ISCV RSP residuals. We relegate MSP-based diagnostics (**Post-MSP** and **ISCV-MSP**) to a supplementary comparison because their use of a deterministic midpoint ($U_i = 0.5$) yields non-uniform Z-residuals for discrete data, thereby justifying our primary focus on RSP-based residuals. We compare two PPC classes: $\mathbf{PPC}_{Z^{\text{type}}\text{-TEST}}$, which uses statistical tests on Z-residuals as the discrepancy measure (e.g., $\mathbf{PPC}_{Z^{\text{RSP}}\text{-SW}}$ uses the SW p-value on RSP residuals), and a traditional benchmark, $\mathbf{PPC}_{y-\chi^2}$, using the χ^2 discrepancy $\chi^2(\boldsymbol{y},\boldsymbol{\theta}) = \sum_{i=1}^{n} \frac{(y_i - \mu_i(\boldsymbol{\theta}))^2}{\sigma_i^2(\boldsymbol{\theta})}$, where $\mu_i(\boldsymbol{\theta})$ and $\sigma_i^2(\boldsymbol{\theta})$ are the mean and variance of y_i.

3.2 Detecting Nonlinearity in Inverse Gaussian Regression

In this section, we consider a continuous inverse Gaussian (IG) regression. Because the response is continuous, auxiliary randomization is unnecessary, isolating the ability of our summarize-first Z-residual workflow to detect functional-form misspecification. For $n \in \{50, 100, 200\}$, we generated 500 datasets with covariates $X_1^{(i)} \sim \text{Unif}(0.5, 15.5)$, binary $X_2^{(i)}, X_3^{(i)} \sim \text{Binom}(1, 0.5)$, and five standard normal predictors $\boldsymbol{Z}^{(i)} = (Z_1^{(i)}, \ldots, Z_5^{(i)})^\top$ to increase dimensionality. Defining piecewise quadratic terms $B^{(i)} = (X_1^{(i)} - c)^2 I(X_1^{(i)} < c)$ and $A^{(i)} = (X_1^{(i)} - c)^2 I(X_1^{(i)} \geq c)$ with $c = 7.95$, we simulated responses from $Y_i \sim \text{IG}(\mu_i, 25)$ where $\log(\mu_i) = -0.80 + 0.012B^{(i)} + 0.006A^{(i)} + 0.40X_2^{(i)} - 0.25X_3^{(i)} + \boldsymbol{\gamma}^\top \boldsymbol{Z}^{(i)}$, with $\boldsymbol{\gamma} = (0.12, -0.10, 0.08, -0.06, 0.05)^\top$. These parameters were chosen to generate datasets with moderate non-linearity relative to response dispersion. For each dataset, we used `brms` to fit two IG models: a correctly specified piecewise model containing $B^{(i)}$ and $A^{(i)}$, and a misspecified linear model replacing these terms with a simple linear effect of $X_1^{(i)}$. Both models use all other covariates used in the data generating process.

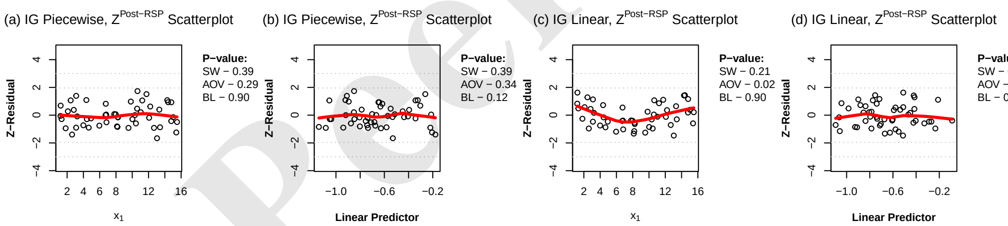

Figure 1: Scatterplots of posterior RSP Z-residuals for the continuous IG simulation with sample size $n = 50$. Panels (a) and (b) show the correctly specified piecewise IG model, while Panels (c) and (d) show the misspecified linear IG model.

To demonstrate how Z-residuals can detect misspecified functional forms in this continuous setting, we show posterior RSP Z-residuals from a representative dataset with $n = 50$ in Figure 1. For the correctly specified piecewise IG model, the residuals are randomly scattered around zero against both X_1 and the fitted linear predictor, with no discernible trend (Figures 1(a) and (b)). In contrast, under the misspecified linear IG model, the residual plot against X_1 displays a clear nonlinear pattern, reflecting the true piecewise quadratic covariate effect (Figure 1(c)). Notably, no such pattern is visible in the plot against the fitted linear predictor (Figure 1(d)): the fitted predictor reduces all covariates to a single variable, in which the misspecified linear approximation to the effect of X_1 is blended with the remaining covariate effects, thereby masking the residual curvature associated with X_1. These results illustrate that covariate-specific Z-residual plots are essential for detecting subtle functional-form misspecifications.

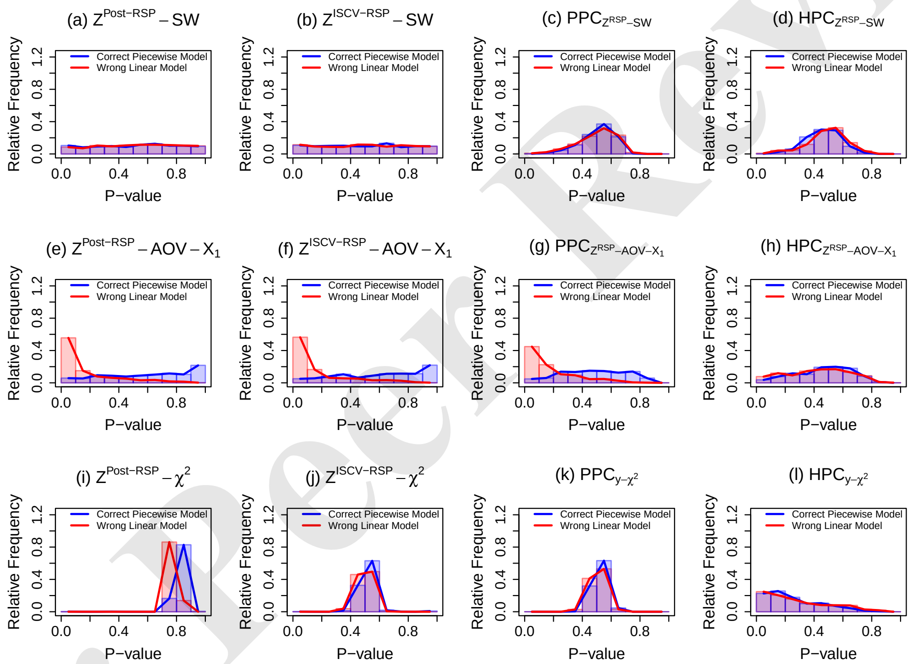

Figure 2: Empirical p-value distributions for various diagnostic methods in the IG simulation with $n = 50$. The blue histograms represent the distributions under the correctly specified piecewise IG model, and the red histograms show the distributions under the misspecified linear IG model. The HPC tests use a splitting ratio of $4 : 1$ for training and testing.

We then examine the empirical distributions of diagnostic p-values based on 500 replications with $n = 50$ (Figure 2). Under the correct piecewise model, the p-values of the SW and AOV tests based on posterior and ISCV RSP Z-residuals (panels (a), (b), (e), and (f)) are spread roughly uniformly over the unit interval, indicating good calibration. The AOV tests against X_1 (panels (e) and (f)) show the sharpest contrast between models: their p-values concentrate strongly near zero under the wrong linear model, as expected, since the misspecification occurs at the functional form of X_1, which the AOV diagnostic directly targets. The SW p-values (panels (a) and (b)) remain nearly uniform under both models, because the wrong model does not substantially distort the marginal normality of the residuals. The Z-residual χ^2 tests are poorly calibrated even under the correct model, peaking around 0.8 (posterior, panel (i)) or 0.5 (ISCV, panel (j)) under both models, implying deflated Type-I error rates and little sensitivity to the misspecification. The PPC and HPC diagnostics show similar pathologies: panels (c), (d), and (k) are mound-shaped around 0.5 under both models, while $\text{HPC}_{y\text{-}\chi^2}$ (panel (l)) is skewed toward small p-values under both. Only PPC with the Z-AOV statistic (panel (g)) shifts noticeably toward zero under the wrong model, though less sharply than panels (e) and (f); the corresponding HPC (panel (h)) yields nearly overlapping distributions. In summary, the covariate-specific AOV tests based on posterior and ISCV RSP Z-residuals give the clearest and best-calibrated signal: near-uniform p-values under the correct model and strong concentration near zero under the wrong one.

Figure 3 compares the empirical rejection rates across sample sizes $n = 50, 100, 200$, based on 500 replications and a nominal significance level of 0.05. Numerical rejection rates are reported in Supplementary Table S3. Under the correctly specified piecewise model, the posterior and ISCV RSP Z-residual tests maintain rejection rates close to the nominal level for the SW and AOV diagnostics (panels (a) and (b)), indicating good calibration and valid Type-I error control. Under the misspecified linear model, the SW tests (panel (a)) have little power, with rejection rates below 0.13 even at $n = 200$, confirming that an overall normality check is insensitive to this localized nonlinear mean misspecification. The χ^2-based tests (panel (c)) fail on calibration itself: the rejection rates of the Z-residual and PPC χ^2 tests are far below the nominal level under both models (at most 0.002), while the rejection rate

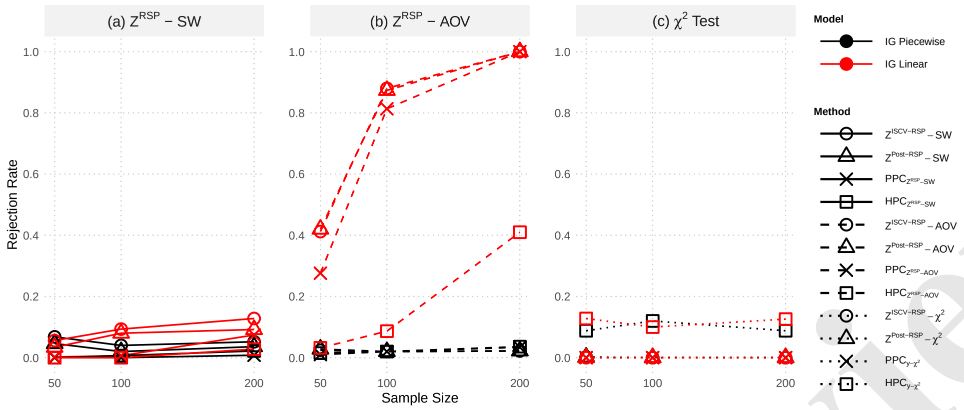

Figure 3: Empirical model rejection rates for various diagnostic methods under different sample sizes ($n = 50, 100, 200$) in the continuous IG simulation, with a nominal significance level of 0.05. The rejection rates under the correctly specified piecewise IG model are shown in black, and the rates under the misspecified linear IG model are shown in red. The HPC tests use a splitting ratio of $4:1$ for training and testing.

of $\text{HPC}_{y\text{-}\chi^2}$ is 0.09–0.13 under both, so the nominal cutoff is invalid in either direction and the two models are barely separated.

In contrast, the Z-residual AOV tests against X_1 (panel (b)) combine calibration with power: rejection rates stay near-nominal under the correct model and rise sharply with sample size under the wrong one. In addition, the Z-residual AOV tests have higher rejection rates than the corresponding PPC test at $n = 50$ (0.42 versus 0.28) and $n = 100$ (0.88 versus 0.81), and all three reach complete rejection at $n = 200$. The HPC AOV test is markedly less powerful, with a rejection rate of only 0.41 at $n = 200$.

3.3 Detecting Nonlinear Covariate Effects in HNB Models

In this section, we conducted a second simulation study focusing on a hurdle negative binomial (HNB) model. To evaluate the detection of nonlinear mean misspecification in count data, we generated 500 datasets for each sample size $n \in \{50, 100, 200\}$. Each dataset included covariates $X_1^{(i)} \sim \text{Unif}(0, 15)$, binary $X_2^{(i)} \sim \text{Binom}(1, 0.5)$, and five standard normal predictors $\boldsymbol{Z}^{(i)} = (Z_1^{(i)}, \ldots, Z_5^{(i)})^\top$ to increase dimensionality. The responses were simulated from an HNB distribution where the count mean μ_i and the zero-probability π_i^o

were specified directly using the true coefficients: $\log(\mu_i) = -1 + 6\log(X_1^{(i)}) + X_2^{(i)}$ and $\text{logit}(\pi_i^o) = -1 - X_1^{(i)} - 5X_2^{(i)}$. This parameterization yielded an overall zero proportion of $\pi^0 \approx 30\%$. For each dataset, we fitted two Bayesian HNB models using `brms`: a correctly specified **Logarithmic** model with $\log(X_1^{(i)})$ in the count component, and a misspecified **Linear** model that incorrectly replaced this term with a linear term of $X_1^{(i)}$. Both models retained the correct zero-model structure and used all other covariates.

To demonstrate the covariate-specific diagnostic, we show posterior RSP Z-residuals from a representative dataset in Figure 4. Under the specified logarithmic HNB model, the residuals show no discernible trend against X_1, and the grouped boxplots are approximately homogeneous, supported by a non-significant AOV test with p-value > 0.05 (Figures 4(a) and (b)). In contrast, the misspecified linear HNB model produces a clear residual pattern and pronounced grouped mean differences across X_1, detected by the AOV test with p-value < 0.01 (Figures 4(c) and (d)). These results show that Z-residual plots and covariate-specific grouped tests can identify misspecified functional forms in discrete HNB models.

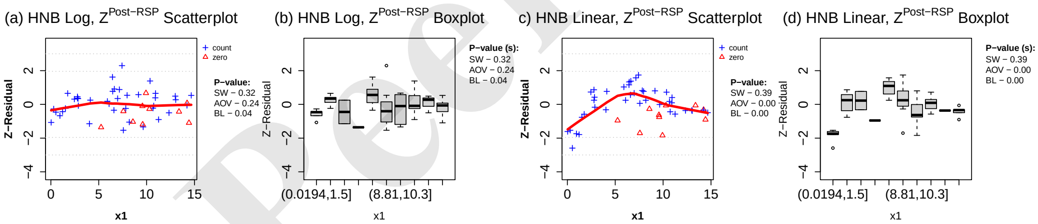

Figure 4: Visual diagnostics of posterior RSP Z-residuals for HNB models with logarithmic (true) and linear (wrong) covariate effects, based on a representative dataset with sample size $n = 50$. Panels (a) and (c) display scatterplots of the posterior RSP Z-residuals against covariate X_1, with positive count observations shown as blue crosses and zero observations shown as red triangles. Panels (b) and (d) present the boxplots of Z-residuals grouped by X_1.

We then examine the empirical distributions of diagnostic p-values using replicated datasets, as displayed in Figure 5. The histograms are based on 500 replications with $n = 50$ and an overall zero proportion of $\pi^o \approx 30\%$. Consistent with the continuous IG simulation, the covariate-specific AOV diagnostics provide the clearest signal. The SW tests on posterior and ISCV RSP Z-residuals are approximately calibrated under the true logarithmic HNB model, but their separation between the true and wrong models is weaker than that of the AOV tests (Figures 5(a) and (b)). In contrast, the AOV tests against X_1 based on posterior and ISCV Z-residuals are well calibrated under the true model and highly powerful under the misspecified linear model, with p-values heavily concentrated near zero (Figures 5(e) and (f)).

The corresponding PPC and HPC methods are more poorly calibrated and less powerful. The PPC and HPC checks based on SW have p-values concentrated around 0.5 and provide little separation (Figures 5(c) and (d)). The PPC AOV check has high power under the wrong model, but its null p-values are again biased toward 0.5 rather than uniformly distributed (Figure 5(g)); the HPC AOV check shows only weak separation (Figure 5(h)). The χ^2-based checks are less targeted to this functional-form error and do not distinguish the two models as clearly as the Z-residual AOV diagnostics (Figures 5(i)–(l)). Overall, the Z-residual-based AOV tests are the most powerful and well-calibrated in this discrete setting.

Finally, Figure 6 compares model rejection rates across $n = 50, 100, 200$. The corresponding numerical rejection rates are reported in Supplementary Table S4. The SW tests based on posterior and ISCV RSP Z-residuals maintain Type I error rates near 0.05 and show increasing power under the misspecified linear model, whereas the corresponding PPC and HPC checks have rejection rates close to zero or provide much weaker separation (Figure 6(a)). The χ^2-based checks are also less informative: the Z-residual χ^2 tests show some power but remain well below the AOV tests, while the PPC and HPC χ^2 checks are generally weak or unstable (Figure 6(c)). The AOV-based diagnostics provide the clearest results (Figure 6(b)): the posterior and ISCV Z-residual AOV tests maintain low rejection rates under the true model and achieve high power under the misspecified linear model, reaching nearly complete rejection for $n \geq 100$. The PPC AOV check also has high power but is lower at $n = 50$, whereas the HPC AOV check is less powerful at small sample sizes. These rejection-rate results confirm that covariate-specific AOV tests on Z-residuals provide the strongest diagnostic signal for

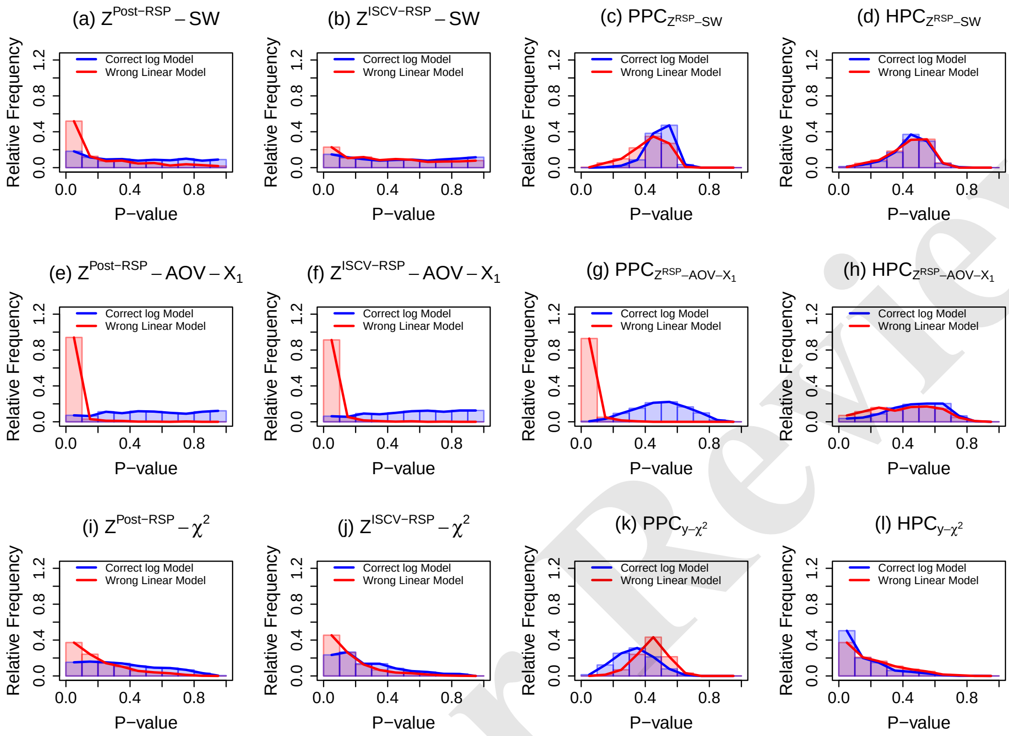

Figure 5: Empirical p-value distributions for various diagnostic methods in the nonlinear HNB simulation with sample size $n = 50$ and overall zero proportion $\pi^o \approx 30\%$. The blue histograms represent the distributions under the correctly specified logarithmic HNB model, and the red histograms represent the distributions under the misspecified linear HNB model. The HPC tests use a splitting ratio of 4 : 1 for training and testing.

nonlinear functional-form misspecification in this discrete HNB simulation.

In summary, the simulation results show that the AOV tests on Z-residuals give significantly higher power than overall goodness-of-fit tests such as SW and χ^2 tests in detecting misspecified covariate functional forms. The PPC method with AOV test can also achieve high power. However, visual diagnostics of Z-residuals, such as the scatterplots and boxplots in Figure 4, can inform the selection of an appropriate test for a PPC.

A supplementary sensitivity analysis (Section S1.2) fits NB-log and NB-linear models to the same HNB-generated data to examine how response-family misspecification interacts with

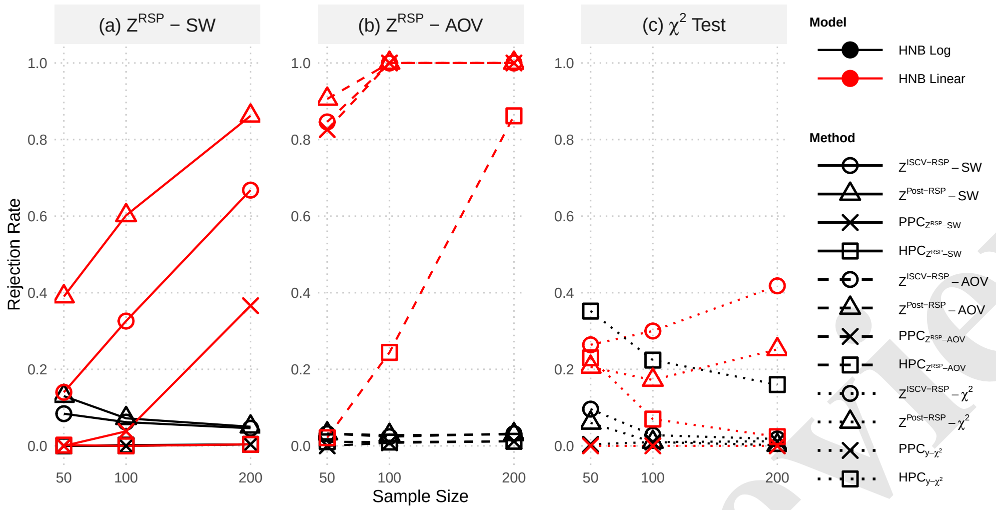

Figure 6: Empirical model rejection rates for various diagnostic methods under different sample sizes ($n = 50, 100, 200$), with a nominal significance level of 0.05. The rejection rates under the true logarithmic HNB model (Type I error) are shown in black, and the rates under the wrong linear HNB model (power) are shown in red. The HPC tests use a splitting ratio of 4 : 1 for training and testing.

the residual-versus-X_1 diagnostics. While a functional-form deficiency remains detectable under the misspecified NB family, the results highlight the critical power of a joint analysis. Severe misspecification in one model component can cascade into diagnostics targeting another—for instance, an incorrect response family can distort the implied mean structure—meaning no single test provides a complete picture. By evaluating numerical and graphical residual diagnostics jointly, practitioners gain significantly more leverage to disentangle these interacting model deficiencies, even though isolating the exact source of misfit remains challenging; see Section S1.2 for a detailed discussion.

4 Applications to a Zero-inflated Dataset

This section applies Z-residual diagnostics in parallel with PPC methods to a real dataset. We analyze the dolphin-sighting data from Chapter 7 of Zuur & Ieno (2021), which draws on publicly available sources including Jackson-Ricketts et al. (2020). The dataset contains $n = 936$ observations. The response is the count of dolphin sightings; covariates include water depth (`depth`), water temperature (`temp`), distance to the river mouth (`dr`, indexing the estuarine–coastal gradient), `salinity`, `turbidity`, survey `year` and a location identifier. Because zeros are prevalent (about 75%), we fit four regression families: hurdle Poisson, hurdle negative binomial (HNB), Poisson, and negative binomial (NB). We then assess overall goodness-of-fit and the functional forms of covariate effects using posterior Z–residual diagnostics alongside PPC methods.

We first fit each family with *linear* covariate effects and evaluate overall goodness-of-fit using **posterior RSP Z–residuals**. Figure 7 shows QQ–plots of posterior RSP Z–residuals. The QQ-plots of the HNB and NB models adhere closely to the diagonal lines, suggesting adequate fit, whereas the QQ-plots of the hurdle Poisson and Poisson models deviate markedly from the diagonal lines, especially in their upper tails. The $Z^{\text{Post-RSP}}$-SW p-values beside the QQ-plots also confirm numerically these impressions: HNB and NB models having p-values ≥ 0.05 (no rejection) versus hurdle Poisson and Poisson both having < 0.01 p-values (clear rejection). It is clear that the Poisson family cannot model well the strong overdispersion and the hurdle modification does not improve it.

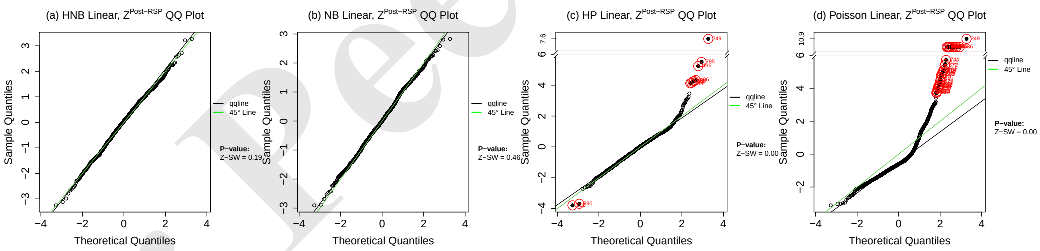

Figure 7: QQ plots of posterior RSP Z-residuals for the (a) HNB, (b) NB, (c) hurdle Poisson, and (d) Poisson models, each fitted in linear form to the dolphin-sighting dataset.

Given the overall adequacy of HNB and NB, we next examine covariate functional forms using posterior Z-residual diagnostics, focusing on the AOV F test ($Z^{\text{Post-RSP}}$-AOV). As shown in Figure 8 for the linear HNB model, plotting Z-residuals against `depth` reveals a clear U-shape (Figure 8a) and heterogeneous boxplots across `depth` groups (Figure 8b), indicating a potential misspecification of the functional form for `depth`. The $Z^{\text{Post-RSP}}$-AOV test yields a p-value < 0.01, strongly rejecting the hypothesis of equal Z-residual means across the `depth` groups. To examine robustness with respect to the auxiliary randomization U (for discrete outcomes), we drew 500 sets of $\boldsymbol{U}$, computed the $Z^{\text{Post-RSP}}$-SW and $Z^{\text{Post-RSP}}$-AOV p-values for each set, and summarized the replicated p-values using the conservative upper-bound for the median (Sec. 2; Eq. (16)). Figures 8(c) and (d) display the replicated p-value histograms, with red lines marking the upper-bounds. For $Z^{\text{Post-RSP}}$-AOV (d), most p-values fall below 0.05, with a median upper-bound of 0.012, revealing a significant nonlinear effect of `depth`. The p-values for $\text{PPC}_{Z^{\text{RSP}}\text{-SW}}$ and $\text{PPC}_{Z^{\text{RSP}}\text{-AOV}}$ in Table 2 support this finding. In contrast, the $Z^{\text{Post-RSP}}$-SW test (c) yields p-values mostly above 0.05, with a median upper-bound of 0.790 (Table 2). This finding supports the conclusion from Section 3.3: overall goodness-of-fit tests such as SW can lack power for detecting nonlinearity.

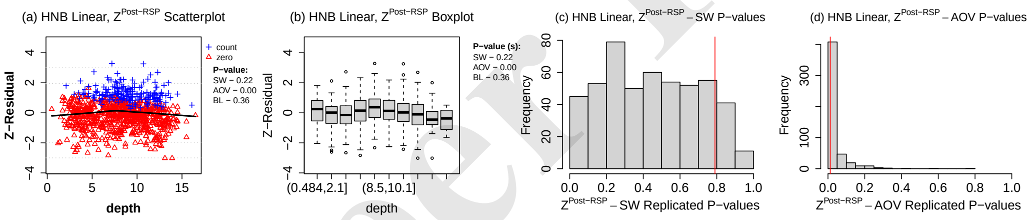

Figure 8: Posterior RSP Z-residual diagnostics for nonlinearity in the linear HNB model for the dolphin-sighting data. (a): scatterplot and (b): grouped boxplots of Z-residuals versus `depth`, with non-zeros as blue crosses and zeros as red triangles. (c, d): histograms of 500 replicated p-values for the $Z^{\text{Post-RSP}}$-SW and $Z^{\text{Post-RSP}}$-AOV tests, respectively; red lines mark the upper-bound for the median.

After rejecting a linear term for `depth`, we specified a piecewise quadratic effect to better capture the nonlinearity. This choice was informed by the scatterplot in Figure 8(a), which shows a clear turning point at $\texttt{depth} = 7.95$. We created two corresponding variables: $\texttt{below7.95} = \{(\texttt{depth} - 7.95)^2\}\,\mathbf{1}(\texttt{depth} < 7.95)$ and $\texttt{above7.95} = \{(\texttt{depth} - 7.95)^2\}\,\mathbf{1}(\texttt{depth} \geq 7.95)$. After refitting the HNB model with these two variables, the diagnostic results in Figure 9 confirmed a substantially improved fit. The scatterplot (a) and boxplots (b) now display randomly scattered, homogeneous Z-residuals with no visible patterns, indicating that the nonlinear effect of `depth` has been successfully modeled. The large $Z^{\text{Post-RSP}}$-AOV p-value of 0.88 numerically supports this conclusion. Furthermore, the histograms (c, d) show that the replicated p-values for both the $Z^{\text{Post-RSP}}$-SW and $Z^{\text{Post-RSP}}$-AOV tests are now mostly above 0.05. As summarized in the Nonlinear panel of Table 2, the Z-residuals for both the nonlinear HNB and NB models yield large median upper-bound p-values; for instance, the Z-SW and Z-AOV tests applied to the HNB model resulted in p-values of 0.969 and 1.000, respectively. The PPC tests using Z-SW and Z-AOV also yield large p-values for these two models. Finally, a comparison of the models using the LOO expected log point-wise predictive density (elpd), shown in the `elpd` column of Table 2, confirms that the HNB model with a nonlinear `depth` effect significantly improves the fit over the linear version and also outperforms the NB models.

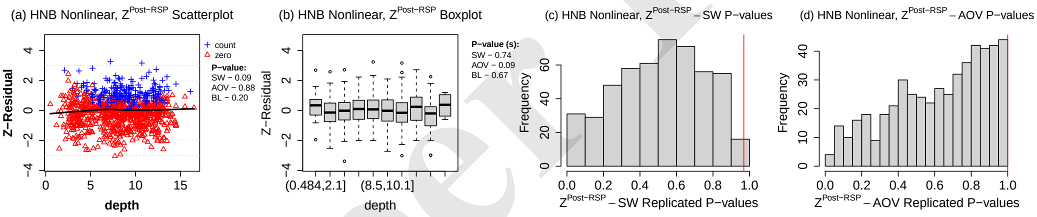

Figure 9: Diagnostics based on posterior RSP Z-residuals for the HNB nonlinear model fitted to the dolphin-sighting data. Panels (a) and (b) show a scatterplot and a boxplot of the posterior RSP Z-residuals against `depth`, distinguishing counts observations (blue crosses) from zeros observations (red triangles). Panels (c) and (d) display histograms of 500 replicated p-values for $Z^{\text{Post-RSP}}$-SW and $Z^{\text{Post-RSP}}$-AOV p-values.

The PPC$_{y-\chi^2}$ results (Table 2, PPC columns) differ from the other diagnostic results in this example. This test rejects both the linear HNB ($p = 0.017$) and the nonlinear HNB ($p = 0.024$) while accepting the NB variants. Because both Z-residual diagnostics and PPC Z-residual tests indicate adequate fit for the nonlinear HNB and NB models, we adjudicate using out-of-sample performance via LOO expected log predictive density (elpd). The LOO comparison across all six candidates ranks the nonlinear HNB first ($\Delta_{\text{elpd}} = 0$), followed by linear HNB (-20.4, se $= 6.8$), nonlinear NB (-34.1, se $= 9.9$), linear NB (-43.0, se $= 10.0$), and then Poisson-based models. Thus, while both HNB and NB families can fit adequately once the nonlinearity is addressed, the nonlinear HNB has the highest `elpd` among the fitted candidate models.

Table 2: Diagnostic p-values and LOO comparisons for the dolphin-sighting data. Rows are grouped into four linear models (HNB, NB, hurdle Poisson, Poisson) and two nonlinear models (HNB, NB). For posterior and ISCV Z-residual tests, we report median upper-bound p-values (eq. (16)); all Z-AOV tests are conducted against `depth`. The last three columns report Δ_{elpd} relative to the nonlinear HNB model, its standard error, and $\Pr(\Delta_{\text{elpd}} \geq 0)$.The HPC tests use a splitting ratio of $4:1$ for training and testing.

	Post. Z-Residual Tests			**ISCV Z-Residual Tests**			**PPC Tests**			**HPC Tests**			**LOO Comparison**		
Family	Z-SW	Z-AOV	Z-χ^2	Z-SW	Z-AOV	Z-χ^2	Z-SW	Z-AOV	$y-\chi^2$	Z-SW	Z-AOV	$y-\chi^2$	Δ_{elpd}	SE	$\Pr(\Delta_{\text{elpd}} \geq 0)$
Linear models															
HNB	0.790	0.012	0.823	0.395	0.013	0.316	0.674	0.035	0.017	0.609	0.369	0.250	-19.9	6.8	0.002
NB	0.299	0.010	0.948	0.880	0.009	0.520	0.392	0.026	0.538	0.530	0.403	0.691	-42.5	9.9	< 0.001
HP	< 0.01	0.020	< 0.01	< 0.01	0.022	< 0.01	< 0.01	0.041	< 0.01	< 0.01	0.324	< 0.01	-118.6	34.4	< 0.001
Poisson	< 0.01	0.012	< 0.01	< 0.01	0.015	< 0.01	< 0.01	0.013	< 0.01	< 0.01	0.340	< 0.01	-803.3	101.1	< 0.001
Nonlinear models															
HNB	1.000	1.000	0.732	0.534	1.000	0.262	0.616	0.575	0.024	0.593	0.518	0.348	0.0	0.0	Ref.
NB	0.337	0.760	0.905	0.031	0.650	0.372	0.149	0.372	0.169	0.478	0.538	0.755	-34.0	9.9	< 0.001

Two factors plausibly explain the $\text{PPC}_{y-\chi^2}$ rejections. First, posterior predictive p-values (PPP) are not uniform under the null because the data are used twice. For some discrepancy measures and replication schemes, they can even be anti-conservative (under-dispersed), inflating the Type I error rate rather than merely reducing power, as reported by (e.g., Berkhof et al. 2000, Rubin-Delanchy & Lawson 2014). Second, the response variable is extremely sparse ($\sim 75\%$ zeros, with most positive counts ≤ 5), yielding many cases with very small model-expected counts. The $\text{PPC}_{y-\chi^2}$ checks are then dominated by these cases, which magnify innocuous random fluctuations and trigger spurious rejections; similar inflation has been observed in other incomplete or sparse-data settings (e.g., Xu et al. 2016). We conducted small-scale simulation studies with datasets generated from HNB models with a large overall proportion of zeros ($\approx 70\%$) and observed that the $\text{PPC}_{y-\chi^2}$ test produced similarly high rejection rates (nearly 100%) for these correctly specified HNB models.

We further conducted two sensitivity checks for the real-data conclusions. First, refitting the candidate models under baseline, tighter, and more diffuse priors yielded qualitatively unchanged Z-residual, PPC, and HPC conclusions (Section S2.2, Table S6, and Figure S6). Second, repeating the depth-based grouped diagnostic with $k \in \{5, 10, 15, 20\}$ equally spaced intervals showed that the conclusion was not driven by the binning choice: the linear HNB model consistently retained residual mean differences across depth groups, whereas the nonlinear HNB model did not (Section S2.3, Table S7, and Figure S7).

Full regression coefficients for the nonlinear HNB model are available in Table S5. These results show the effect of `depth` is primarily on the model's hurdle (zero-inflation) component. The estimated coefficients for the hurdle part are $\hat{\beta}_{\texttt{hu\_below7.95}} = 0.12$ (CI: [0.08, 0.16]) and $\hat{\beta}_{\texttt{hu\_above7.95}} = 0.03$ (CI: [0.01, 0.05]). This means the odds of a zero count increase by factors of roughly 1.13 and 1.03 for each unit increase in $(\texttt{depth} - 7.95)^2$ below and above the 7.95m threshold, respectively. Conversely, the coefficients for the count component are not significant (their credible intervals include zero), which suggests that expected positive counts don't vary substantially with `depth`. Together, these results quantify the U-shaped relationship seen in diagnostics: the probability of a zero count is minimized around 7.95m and increases away from this point, especially in shallower waters.

5 Conclusions and Discussions

In this paper, we have introduced a versatile Bayesian residual diagnostic based on RSPs. Mapping posterior or ISCV RSPs through the standard normal quantile function yields Z-residuals that support familiar graphical and quantitative diagnostics. The distinctive feature of our approach is its *summarize-first, then-test* workflow: rather than testing conditionally on each parameter draw and then summarizing the resulting p-values, we integrate the RSPs over the posterior or LOOCV distribution to produce a *single* set of residuals that are asymptotically distributed as $N(0, 1)$ under a correctly specified model.

In two simulation studies with misspecified covariate functional forms in IG and HNB models, the proposed Z-residual-based tests yielded well-calibrated Type-I error rates and greater power for rejecting misspecified models than traditional PPC and HPC. The simulations also showed that relying solely on the marginal residual distribution is often insufficient: omnibus tests such as the Shapiro–Wilk (SW) and χ^2 tests masked structural flaws that covariate-specific Z–AOV tests readily detected, underscoring the value of graphical Z-residual summaries as exploratory tools that expose unknown deficiencies and guide formal testing. Moreover, by averaging over MCMC draws, Z-residual-based PPC remained stable despite the auxiliary $\boldsymbol{U}$ drawn at each iteration, mitigating the double-use-of-data conservatism of traditional χ^2 PPC. Finally, applying the framework to the dolphin-sighting dataset exposed a nonlinear depth effect, guiding a revision to a nonlinear HNB model that passed all diagnostic checks and improved predictive performance as measured by LOO `elpd`.

The core strength of the Z-residual framework lies in assessing the approximated predictive distribution (posterior or ISCV) against the actually observed y_i. Because this predictive distribution converges to the true data-generating distribution as the sample size increases, our method yields observation-level residuals that are asymptotically normal under the null. However, this strict observation-level focus entails an inherent **limitation**: the method is largely insensitive to non-observation-level components, such as latent variables, hierarchical effects, and priors. Consequently, the proposed Z-residuals are less effective at detecting misspecification within a model's internal structure.

To address this limitation, a highly promising direction for future work is to extend the RSP framework to the evaluation of internal model components, drawing inspiration from Uniform Parametrization Checks (UPC) (Covington & Miller 2025). For a single posterior draw $\boldsymbol{\theta}^{(t)}$, UPC shares a fundamental mathematical equivalence with RSPs: its observation-level uniform variables are distributionally identical to $1 - \text{RSP}_i(y_i \mid \boldsymbol{\theta}^{(t)})$. The two frameworks diverge, however, in how they aggregate over the posterior. UPC adopts a "test-first, then-summarize" approach, combining MCMC-wise p-values via the Cauchy combination rule (Liu & Xie 2020)—a highly innovative strategy, although its approximation error may not fully vanish asymptotically. Combining UPC's novel perspective on internal random components with the asymptotic stability of our "summarize-first" RSP framework offers an interesting avenue for diagnosing latent structures. For this combination to succeed, the primary methodological challenge will be to formally define what constitutes an "observed" value for these internal components, so that their corresponding RSPs can be validly constructed and assessed.

Beyond extending the methodology to the assessment of latent variables, several practical enhancements could further strengthen the current Z-residual framework. One immediate challenge arises when applying Z-residuals to *discrete* responses: the auxiliary randomness $\boldsymbol{U}$ causes the resulting test p-values to vary across random seeds. Because practitioners often require a single representative value, we recommend re-randomizing $\boldsymbol{U}$ multiple times and summarizing the replicated p-values with an order statistic $p_{(r)}$, such as the median. To this end, we derived a distribution-free, super-uniform upper bound $H_{J,r}\, p_{(r)}$, which provides an easily computable single-number summary of statistical significance.

As a complementary single-number summary, analysts can also use parameter-wise Z-residual posterior predictive p-values. Whereas traditional χ^2 PPC are notoriously conservative and underpowered, our simulations demonstrate that PPP based on parameter-wise Z-residuals effectively mitigate this conservatism. Moreover, these PPP remain highly stable despite the auxiliary $\boldsymbol{U}$ drawn at each MCMC iteration, making them a reliable secondary confirmation of model misfit. For analysts seeking an exactly calibrated single-number p-value for discrete data, a simulation-based approach could be employed: by simulating T replicated datasets $\{\boldsymbol{y}_{(t)}^{\text{rep}}\}_{t=1}^{T}$ from the posterior predictive distribution, one can approximate the sampling distribution of the median p-value $p_{(r)}$ and compute a calibrated p-value as the proportion of replicates yielding a smaller median p-value. Our results suggest that the posterior predictive distribution is an adequate proxy for the true data-generating model, likely obviating the need to rerun MCMC for each replicate.

Finally, future research can expand the suite of targeted Z-residual diagnostic tests with both high sensitivity and high specificity. Because models can be misspecified in numerous ways, no single diagnostic is sufficient on its own; a joint and iterative assessment across diverse tests is often needed to pinpoint the exact source of deficiency. The construction of a single, stable set of observational Z-residuals naturally facilitates such joint analyses. Building on this foundation, we also propose developing E-value adaptations of Z-residual diagnostics (Vovk & Wang 2021, Ramdas et al. 2023, Grünwald et al. 2024). Because E-values retain rigorous error-control guarantees when combined, constructing residual-based e-statistics offers a highly promising methodology for sequential and multiple Bayesian model criticism.

Supplementary Materials

The online supplementary materials provide the formal theoretical framework and rigorous proofs establishing the asymptotic properties of Z-residuals, as well as the distribution-free upper bounds for replicated p-values. Additionally, they report further simulation studies demonstrating the versatility of Z-residuals in detecting response-family and mixed-model misspecifications, while highlighting the necessity of the auxiliary randomization U_i for discrete responses.

Data and Software Availability

The `Zresidual` R package—which provides a comprehensive suite of graphical and numerical diagnostic tools for evaluating Bayesian regression models and is highly extensible with custom RSP functions—is available at https://figshare.com/s/00ac6b406e451c76e8e1 (anonymized version, to be replaced by Github/CRAN links). The package also includes the processed dolphin-sighting data.

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Supplementary Material for "Z-residuals for Diagnosing Bayesian Models"

July 6, 2026

Abstract

This document contains supplementary simulations, extended applied analyses, and mathematical proofs detailing the theoretical properties of Z-residuals. The material is organized as follows:

- **Section S1 (Additional Simulation Studies):** Presents simulation studies evaluating wrong families (comparing HNB versus NB models) and mixed misspecifications in both families and covariate functional forms. It also provides the full tabular rejection rates for the main-text simulations.
- **Section S2 (Additional Results of the Analysis of the Dolphin-Sighting Data):** Details further applied analyses, including posterior summaries for the selected nonlinear HNB model, robustness checks via prior and binning sensitivity analyses, and a numerical illustration of replicated p-values.
- **Section S3 (Asymptotic Properties of Z-residuals):** Provides the formal theoretical framework, including illustrations and rigorous proofs for the uniformity of Randomised Survival Probabilities (RSPs) under the true model, the asymptotic independence and uniformity of cross-validated and posterior RSPs, the asymptotic uniformity of tests based on Z-residuals, and distribution-free upper bounds for the order statistics of replicated p-values.

S1 Additional Simulation Studies

S1.1 Simulation Studies with Wrong Families: HNB versus NB

To assess the capability of the proposed Z-residuals in identifying misspecified distributional families, we conducted a simulation study involving the generation of data from a HNB model, followed by fitting both the true HNB model and a misspecified NB. Each synthetic dataset was constructed by first sampling a covariate $X_i \sim N(0,1)$ and five independent noise covariates $Z_{i,j} \sim N(0,1)$ for $j = 1, \dots, 5$. The zero-component probability, π_i^o, was then determined by $\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_i$, and the mean of the count component, μ_i, by $\log(\mu_i) = \alpha_0 + \alpha_1 X_i$, with true regression coefficients set as $\alpha_0 = 2$, $\alpha_1 = 8$, $\beta_0 = -1$, and $\beta_1 = -1$, which were used to generate the final response $y_i \sim \text{HNB}(\mu_i, \pi_i^o, \gamma = 4)$. We generated 500 datasets for each sample size $n \in \{50, 100, 200\}$, with an the overall zero proportions $\pi^0 \approx 30\%$. For each simulated dataset, two Bayesian models were fitted using the `brms` R package: **the true HNB model**, specified as $y_i \sim \text{HNB}(\mu_i, \pi_i^o, \gamma)$ with $\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j}$ and $\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_i + \sum_{j=1}^5 b_j Z_{i,j}$; and **a misspecified NB model**, formulated as $y_i \sim \text{NB}(\mu_i, \gamma)$ with $\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j}$. The inclusion of noise variables in the linear predictors ensures the models' complexity and a more realistic testing environment for the diagnostic tools.

To demonstrate the performance of Z-residuals as a graphical diagnostic tool, we present scatterplots, QQ and marginal distribution plots in Figure S1 for a representative dataset with a sample size of $n = 100$ and overall zero proportion of $\pi^o \approx 30\%$. The three rows provide complementary graphical summaries: the residual-versus-index plots display observation-level residual patterns, the QQ plots assess agreement with the standard normal reference distribution, and the marginal distribution plots show histograms of the Z-residuals with the empirical kernel density and the $N(0,1)$ reference density overlaid. For the two-part HNB model, the index plots also distinguish zero observations from positive count observations. When the true HNB model is fitted, the posterior RSP Z-residuals are randomly scattered without any discernible pattern (Figure S1(a)). Their corresponding QQ plot aligns well with the diagonal line (Figure S1(e)), and their marginal distribution is close to the standard normal reference density (Figure S1(i)). The SW test also yields a p-value greater than 0.05. This behavior is consistent with the theoretical expectation of a random sample from a standard normal distribution for a correctly specified model. In contrast, the QQ plot for the MSP Z-residuals (Figure S1(f)) and its marginal distribution plot (Figure S1(j)) deviate visibly from the standard normal reference, resulting in an SW p-value of 0.01 that would incorrectly reject the true model. For the misspecified NB model, the QQ plots and marginal distribution plots for the RSP and MSP Z-residuals (Figures S1(g), S1(h), S1(k), and S1(l)) exhibit substantial departures from the standard normal reference distribution, with SW p-values less than 0.05, clearly indicating poor model fit. These graphical diagnostics show that RSP Z-residuals can distinguish the correctly specified HNB model from the misspecified NB model. Notably, the MSP-based Z-residuals fail to approximate a normal distribution when the data are fitted with the true HNB model.

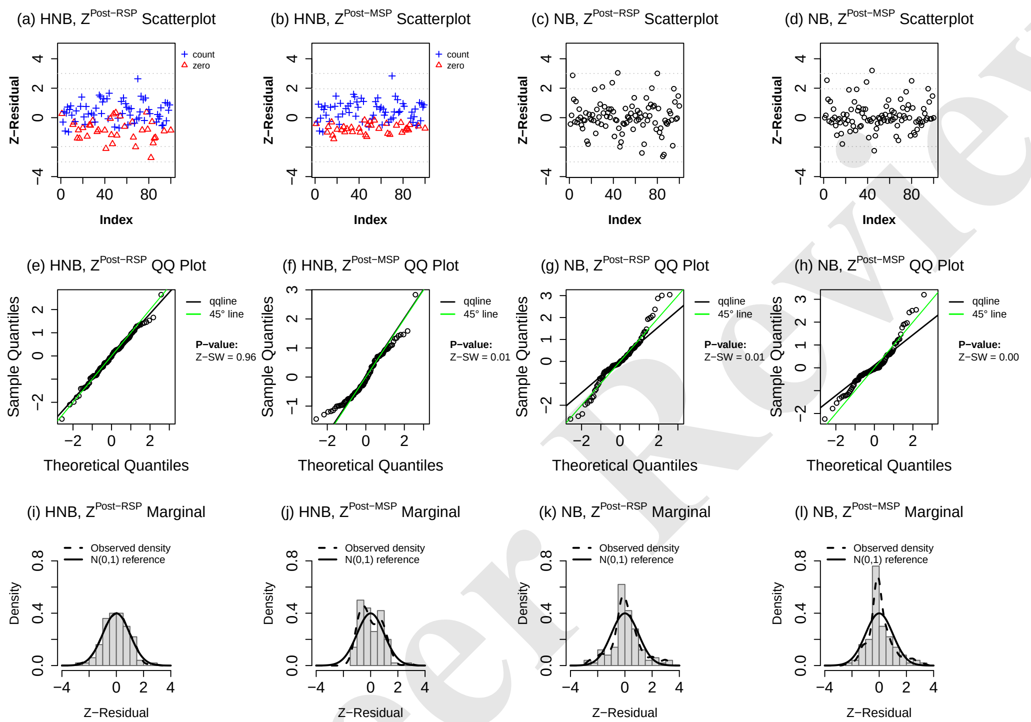

Figure S1: Visual diagnostics with posterior RSP and MSP Z-residuals for HNB and NB models. The first row displays residual-versus-index plots, with positive count observations and zero observations marked separately for the HNB fits. The second row presents QQ plots of the Z-residuals against the standard normal reference distribution, with Shapiro–Wilk p-values reported in the panels. The third row shows the marginal distributions of the Z-residuals, with histograms, empirical kernel densities, and the $N(0, 1)$ reference density overlaid.

To further evaluate the performance of all diagnostic tests introduced in Section 2, we examine the empirical distributions of their p-values using replicated datasets, as displayed in Figure S2. The histograms are based on 500 replications with a sample size of $n = 100$ and a zero proportion $\pi^o \approx 30\%$. For each diagnostic method, we show two distributions: a blue histogram representing the p-values when the true HNB model is fitted (the null hypothesis), and a red histogram showing the p-values when a misspecified NB model is fitted (the alternative hypothesis). A well-calibrated diagnostic test should produce a p-value distribution that is approximately uniform under the null hypothesis, while a powerful test should have a distribution heavily concentrated towards zero under the alternative hypothesis.

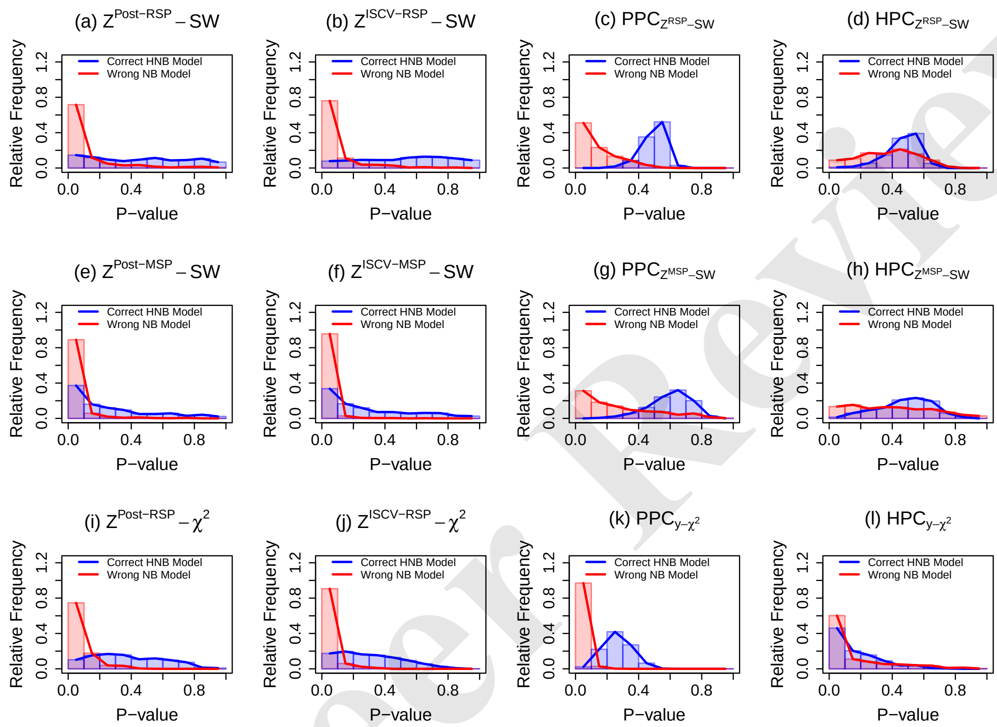

Figure S2: Empirical distributions of p-values for various diagnostic methods under the true HNB model (blue histograms) and the wrong NB model (red histograms), based on 500 replications with sample size $n = 100$ and zero proportion $\pi^o \approx 30\%$.

We compare the p-value distributions from Z-residual diagnostics and posterior predictive checks and holdout predictive checks using similar parameter-wise tests. As shown in Figures S2(a) and (b), the RSP-based Z-residual method with the SW test produces well-calibrated, uniform p-values distributions under the true model and exhibits high power under the wrong model, with p-values heavily concentrated near zero; in contrast, the PPC with the same SW test statistic (Figure S2(c)) exhibits a severe conservative bias—with p-values concentrated around 0.5 under the true model—and has noticeably lower power under the misspecified NB model. The corresponding HPC check is also conservative and shows even weaker separation between the true HNB and wrong NB models (Figure S2(d)). Similarly, Figures S2(e) and (f) show that MSP-based posterior and ISCV Z-residual diagnostics can be poorly calibrated under the true discrete model, while Figures S2(g) and (h) show that the corresponding PPC and HPC methods based on MSP Z-residuals are also highly conservative and significantly less powerful than the corresponding MSP-based Z-residual diagnostics. The p-values from a PPC with the χ^2 statistic are also severely biased toward 0.5 under the true HNB model. However, this approach is highly powerful in this scenario and more powerful than the χ^2 test applied to RSP-based posterior and ISCV Z-residuals (Figures S2(i)–(k)). The corresponding HPC with the $y-\chi^2$ statistic is less stable, producing many small p-values even under the true HNB model in this sparse count setting (Figure S2(l)). In summary, these results confirm the asymptotic findings in Section 2.7: tests based on RSP Z-residuals are asymptotically calibrated, whereas PPC methods are not, and HPC methods can lose power or become unstable depending on the statistic, although a PPC with a well-chosen test statistic (χ^2 here) could achieve high power in some scenarios.

We also compare the performance of different types of Z-residual. The RSP Z-residuals (Figure S2 (a), (b), (i), and (j)) consistently produce p-value distributions that are approximately uniform under the true HNB model, indicating robust calibration regardless of whether the posterior or ISCV method is used, which confirms the asymptotic analysis in Section 2.7. Under the misspecified NB model (red histograms), their p-values are heavily concentrated towards zero, demonstrating high power. In contrast, the MSP Z-residuals methods (Figure S2 (d) and (e)) exhibit a clear lack of uniformity under the true model, with p-values biased towards larger values. More precisely, the MSP-based posterior and ISCV diagnostics are shown in Figures S2(e) and (f), while the PPC and HPC versions based on MSP Z-residuals are shown in Figures S2(g) and (h). Although these methods also show high power, their poor calibration under the null hypothesis makes them less reliable than their RSP counterparts.

Finally, we assess model rejection rates for different sample sizes ($n = 50, 100, 200$), with results broken down by test statistic in Figure S3. An ideal test should maintain a Type I error rate near 0.05 under the true model while having high power (rejection rate near 1) under a misspecified model. Figure S3(a) shows that when using an SW test on RSP Z-residuals, both the posterior and ISCV approaches maintain Type I error rates close to the nominal 0.05 level, demonstrating robust calibration across all sample sizes. In contrast, the corresponding PPC is overly conservative, with Type I error rates near zero and significantly lower power (e.g., ~ 0.036 ($-\times-$) vs. ~ 0.310 ($-\triangle-$) for $n = 50$) (Table S1). The corresponding HPC check is also conservative and has even lower power, especially for small and moderate sample sizes. Under the wrong model, the posterior and ISCV RSP Z-residual SW tests show increasing power, with rejection rates approaching 1 as n reaches 200, whereas the PPC improves more slowly and the HPC remains much less powerful.

Figure S3(b) presents the results for MSP-based methods. SW tests applied to MSP Z-residuals (both posterior and ISCV) are poorly calibrated, showing unacceptably high Type

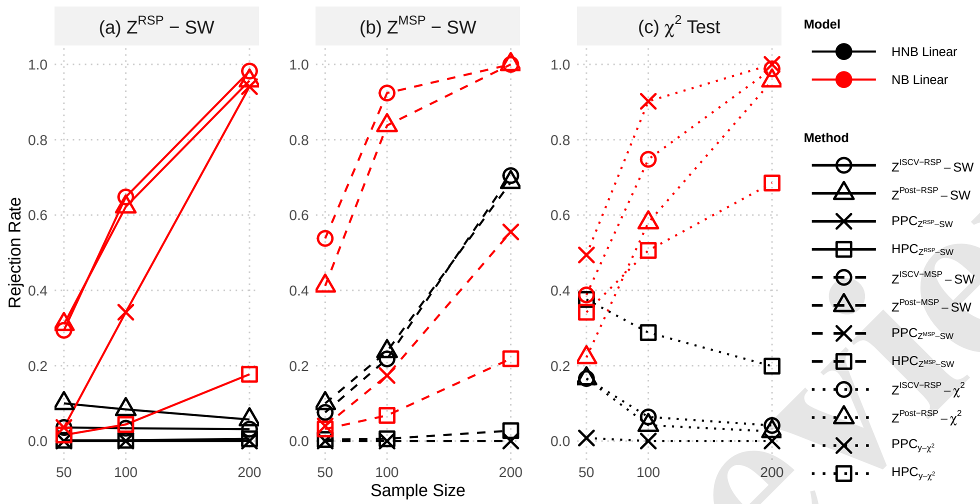

Figure S3: Empirical model rejection rates for various diagnostic methods under different sample sizes ($n = 50, 100, 200$), based on 500 replications with a nominal significance level of 0.05. The rejection rates under the true HNB model (Type I error rates) are shown in black, and the rates under the wrong NB model (power) are shown in red.

Table S1: Empirical rejection rates for the HNB versus NB wrong-family simulation. The nominal significance level is 0.05. Rejection rates under the correctly specified HNB model estimate type I error rates, whereas rejection rates under the misspecified NB model estimate power. Monte Carlo standard errors can be computed as $\sqrt{\hat{p}(1-\hat{p})/R}$; with $R = 500$, the Monte Carlo standard error is approximately 0.010 for a rejection rate near 0.05.

Method	$n = 50$		$n = 100$		$n = 200$	
	HNB	NB	HNB	NB	HNB	NB
$Z^{\text{Post-RSP}}$–SW	0.100	0.310	0.084	0.622	0.057	0.957
$Z^{\text{ISCV-RSP}}$–SW	0.036	0.294	0.034	0.648	0.031	0.982
$\text{PPC}_{Z^{\text{RSP}}\text{–SW}}$	$<$ 0.001	0.036	$<$ 0.001	0.342	$<$ 0.001	0.941
$\text{HPC}_{Z^{\text{RSP}}\text{–SW}}$	0.002	0.016	0.002	0.044	0.006	0.177
$Z^{\text{Post-MSP}}$–SW	0.100	0.412	0.238	0.838	0.687	1.000
$Z^{\text{ISCV-MSP}}$–SW	0.076	0.538	0.218	0.924	0.705	1.000
$\text{PPC}_{Z^{\text{MSP}}\text{–SW}}$	$<$ 0.001	0.040	$<$ 0.001	0.174	$<$ 0.001	0.555
$\text{HPC}_{Z^{\text{MSP}}\text{–SW}}$	0.004	0.032	0.006	0.068	0.028	0.219
$Z^{\text{Post-RSP}}$–χ^2	0.166	0.222	0.042	0.580	0.026	0.957
$Z^{\text{ISCV-RSP}}$–χ^2	0.166	0.388	0.064	0.748	0.041	0.988
$\text{PPC}_{y\text{–}\chi^2}$	0.008	0.494	$<$ 0.001	0.902	$<$ 0.001	1.000
$\text{HPC}_{y\text{–}\chi^2}$	0.376	0.342	0.288	0.506	0.199	0.685

Note. HNB denotes the correctly specified hurdle negative binomial model, and NB denotes the misspecified negative binomial model. SW denotes the Shapiro–Wilk normality test, and χ^2 denotes the omnibus chi-square test. HPC denotes holdout predictive checks.

I error rates under the true model. Despite their high power, this makes them unreliable for formal hypothesis testing. The corresponding PPC and HPC methods are again overly conservative and have very low power.

Figure S3(c) illustrates the methods using a χ^2 statistic. The χ^2 tests based on RSP-based Z-residuals (both posterior and ISCV) maintain Type I error rates close to the nominal 0.05 level for larger sample sizes, although some finite-sample deviation is visible at $n = 50$. This contrasts with the PPC with a χ^2 statistic, which is conservatively biased with a Type I error rate near zero, although it achieves the highest power in these scenarios. In contrast, the corresponding HPC with the χ^2 statistic is unstable in this sparse count setting, showing inflated rejection rates even under the correctly specified HNB model; therefore, its apparent power should be interpreted cautiously. The corresponding numerical rejection rates are reported in Table S1.

S1.2 Diagnostics for Mixed Misspecifications in Response Family and Covariate Functional Form Using the Synthetic Dataset from §3.3

To demonstrate how a joint analysis of graphical and numerical Z-residual diagnostics can disentangle multiple sources of model deficiency, we conducted an additional simulation study using the synthetic dataset ($n = 50$) introduced in Figure 4 of the main text. Four fitted models were compared, representing all combinations of correct and misspecified components: HNB-log (fully correct), HNB-linear (misspecified functional form), NB-log (misspecified response family), and NB-linear (misspecified in both).

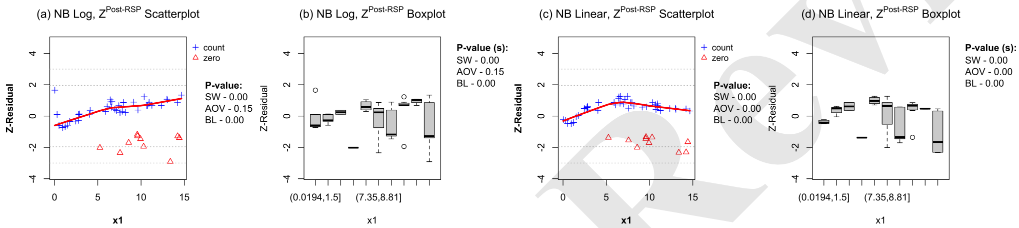

Figure S4: Posterior RSP Z-residual diagnostics for the NB models fitted to the HNB-generated dataset from Section 3.3 of the main text. Panels (a) and (b) display the NB-log model, which misspecifies the response family while maintaining the correct logarithmic functional form for X_1. Panels (c) and (d) illustrate the NB-linear model, which misspecifies both the response family and the functional form of X_1. Scatterplots plot posterior RSP Z-residuals against X_1, with triangles denoting zero-count observations; boxplots group the residuals across X_1 intervals.

Visual diagnostics reveal the distinct consequences of each misspecification type and guide subsequent model refinement. Figure S4 presents the residual-versus-X_1 scatterplots and grouped boxplots for the two NB models. For the NB-log model (panels a and b), the Z-residual LOWESS curve exhibits a nearly linear upward trend with respect to X_1. This trend has a clear statistical interpretation: in the true HNB process, the hurdle component causes the probability of a positive count to increase with X_1. Fitting a single NB family forces this increasing probability into the NB mean structure, necessitating an additional linear term in X_1 beyond the specified $\log X_1$ effect. The observed residual trend accurately isolates this missing component. Furthermore, the Z-residuals of the zero counts (triangles) are distinctly separated from those of the positive counts. This divergence indicates that the NB family fails to capture the excess zeros, directly motivating a zero-modified specification such as the hurdle NB. The NB-linear model (panels c and d) inherits these family-related deficiencies while additionally displaying a pronounced curvature in the mean trend, alongside location shifts across X_1 intervals. These features reflect the increased severity of the functional-form error.

Table S2: Z-residual diagnostic p-values for the response-family sensitivity analysis, using the HNB-generated dataset from Section 3.3 of the main text. The HNB-log and HNB-linear models were evaluated in the main text; the NB-log and NB-linear models are introduced here to assess response-family misspecification. Because the response is discrete and RSP Z-residuals depend on auxiliary randomization, we report median upper-bound p-values (Eq. (16)) for the posterior and ISCV RSP Z-residual tests. The AOV and Bartlett (BL) tests are evaluated against X_1. The logarithmic form is marked "Wrong" for the NB family because the X_1-dependent hurdle probability is improperly absorbed into the NB mean structure (see text).

			Posterior RSP Z-residuals				ISCV RSP Z-residuals			
Model	Family	Functional form	SW	AOV-X_1	BL-X_1	χ^2	SW	AOV-X_1	BL-X_1	χ^2
HNB-log	HNB (True)	log (True)	0.904	0.630	0.120	0.884	0.595	0.767	0.089	0.606
HNB-linear	HNB (True)	linear (Wrong)	0.282	< 0.001	0.002	0.165	0.700	< 0.001	0.017	0.140
NB-log	NB (Wrong)	log (Wrong)	0.003	0.137	0.001	0.363	0.044	0.281	0.001	0.178
NB-linear	NB (Wrong)	linear (Wrong)	< 0.001	0.007	< 0.001	0.415	< 0.001	0.004	< 0.001	0.368

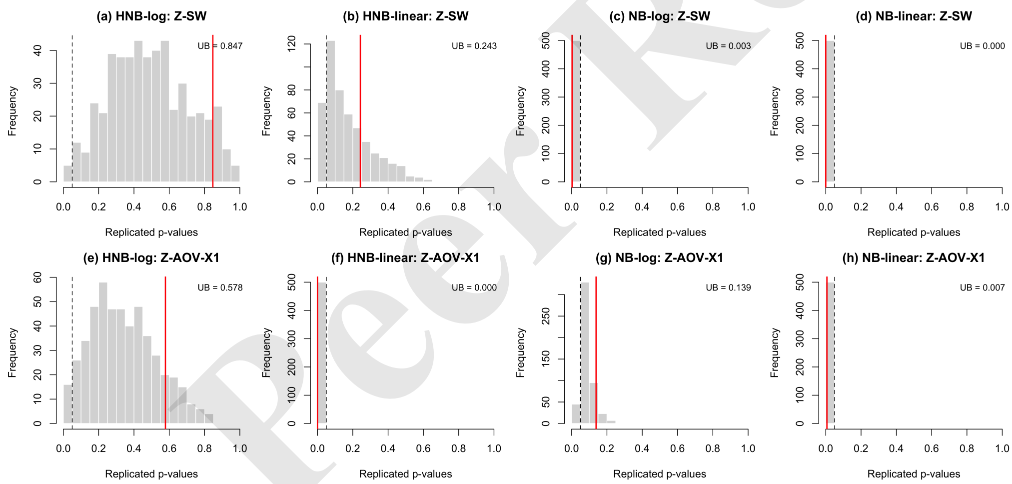

Figure S5: Histograms of 500 replicated Z-residual test p-values for the four fitted models. These were obtained by re-randomizing the auxiliary uniform variables, $\boldsymbol{U}$, used to construct the RSPs for the discrete response. The top row presents the SW tests, and the bottom row displays the AOV tests against X_1. Red vertical lines denote the corresponding median upper-bound p-values (Eq. (16)) from Table S2, and dashed vertical lines indicate the 0.05 significance level.

Numerical diagnostics corroborate and refine these visual findings. Figure S5 displays the distributions of the 500 replicated p-values, which are summarized in Table S2 via their median upper bounds. For the correctly specified HNB-log model, all tests appropriately fail to reject the null hypothesis. Conversely, for the HNB-linear model, the overall SW test is non-significant, but the AOV test against X_1 strongly rejects the model, effectively isolating the localized functional-form error.

The NB-log model illustrates the consequences of an incorrect response family. The SW and Bartlett tests strongly reject the model, reflecting a fundamental distributional mismatch. The AOV test against X_1 yields larger median upper-bound p-values (0.137 and 0.281). However, two considerations caution against interpreting these values as evidence of an adequate functional form. First, the upper-bound p-value (Eq. (16)) is inherently conservative: the replicated p-values for this test (Figure S5g) concentrate within the $(0.05, 0.1)$ interval, revealing non-negligible evidence of a residual mean pattern against X_1 (as an aside, in supplementary simulations with other datasets, this upper-bound p-value frequently drops below 0.05). Second, this residual mean pattern is theoretically expected under the true data-generating process. Because the hurdle component in the HNB process causes the probability of a positive count to increase with X_1, fitting a single NB family absorbs this dynamic into the NB mean structure. Consequently, the log-linear form alone fails to correctly specify the conditional mean. The moderate AOV signal thus represents a valid, albeit attenuated, detection of this induced mean distortion rather than a false alarm. This interpretation is corroborated by the near-linear residual trend in Figure S4a. Finally, the NB-linear model is decisively rejected by both the SW and AOV tests, simultaneously flagging the overall distributional failure and the severe functional-form error.

In summary, a joint evaluation of these diagnostics allows practitioners to better pinpoint a model's deficiencies, even though isolating the exact source of misspecification remains challenging. The overall SW test is highly sensitive to response-family misspecifications that alter the global residual distribution, while residual-versus-covariate scatterplots and grouped AOV tests are tuned to localized covariate functional-form errors. These two error sources are not entirely separable, however. As demonstrated by the NB-log example, a misspecified response family can itself distort the implied mean structure, causing family errors to legitimately propagate into covariate-specific tests. Given these interactions, a robust diagnostic workflow must combine multiple graphical and numerical residual diagnostics, supplemented by model comparison metrics such as information criteria. Developing Z-residual tests and visualization methods that offer both high sensitivity and specificity to particular forms of model misspecification remains a crucial direction for future research. The versatility of Z-residuals provides a strong foundation for this ongoing effort.

S1.3 Tabular Rejection Rates of Main-text Simulations

To make the empirical type I error and power explicit, Tables S3 and S4 detail the numerical rejection rates corresponding to Figures 3 and 6. Based on $R = 500$ replications, Monte Carlo standard errors ($\sqrt{\hat{p}(1-\hat{p})/R}$) range from roughly 0.010 (95% margin of error ≈ 0.019) at a 0.05 rejection rate, up to 0.022 (margin of error ≈ 0.044) at a 0.50 rejection rate.

Table S3: Rejection rates ($\alpha = 0.05$) for the continuous IG simulation (Section 3.2), showing type I error ("Correct" piecewise model) and power ("Wrong" linear model). Tests: Shapiro–Wilk (SW), grouped ANOVA (AOV-X_1), and omnibus chi-square (χ^2).

Method	$n = 50$		$n = 100$		$n = 200$	
	Correct	Wrong	Correct	Wrong	Correct	Wrong
$Z^{\text{Post-RSP}}$–SW	0.046	0.034	0.020	0.080	0.036	0.092
$Z^{\text{ISCV-RSP}}$–SW	0.068	0.056	0.040	0.093	0.052	0.128
$\text{PPC}_{Z^{\text{RSP}}\text{–SW}}$	< 0.001	0.002	< 0.001	0.007	0.008	0.074
$\text{HPC}_{Z^{\text{RSP}}\text{–SW}}$	< 0.001	< 0.001	0.007	< 0.001	0.022	0.028
$Z^{\text{Post-RSP}}$–AOV-X_1	0.028	0.420	0.020	0.873	0.022	1.000
$Z^{\text{ISCV-RSP}}$–AOV-X_1	0.024	0.412	0.020	0.880	0.022	1.000
$\text{PPC}_{Z^{\text{RSP}}\text{–AOV-}X_1}$	0.010	0.276	0.020	0.813	0.034	1.000
$\text{HPC}_{Z^{\text{RSP}}\text{–AOV-}X_1}$	0.015	0.032	0.020	0.087	0.036	0.410
$Z^{\text{Post-RSP}}$–χ^2	0.002	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$Z^{\text{ISCV-RSP}}$–χ^2	0.002	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$\text{PPC}_{y\text{–}\chi^2}$	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$\text{HPC}_{y\text{–}\chi^2}$	0.088	0.128	0.120	0.100	0.088	0.126

Table S4: Rejection rates ($\alpha = 0.05$) for the nonlinear HNB simulation (Section 3.3), showing type I error ("Correct" logarithmic model) and power ("Wrong" linear model). Test abbreviations follow Table S3.

Method	$n = 50$		$n = 100$		$n = 200$	
	Correct	Wrong	Correct	Wrong	Correct	Wrong
$Z^{\text{Post-RSP}}$–SW	0.130	0.390	0.072	0.602	0.050	0.862
$Z^{\text{ISCV-RSP}}$–SW	0.084	0.140	0.062	0.326	0.046	0.668
$\text{PPC}_{Z^{\text{RSP}}\text{–SW}}$	< 0.001	< 0.001	< 0.001	0.038	0.004	0.366
$\text{HPC}_{Z^{\text{RSP}}\text{–SW}}$	< 0.001	0.002	0.002	< 0.001	0.004	0.004
$Z^{\text{Post-RSP}}$–AOV-X_1	0.032	0.906	0.028	1.000	0.030	1.000
$Z^{\text{ISCV-RSP}}$–AOV-X_1	0.032	0.846	0.024	1.000	0.032	1.000
$\text{PPC}_{Z^{\text{RSP}}\text{–AOV-}X_1}$	< 0.001	0.826	0.008	1.000	0.012	1.000
$\text{HPC}_{Z^{\text{RSP}}\text{–AOV-}X_1}$	0.011	0.021	0.010	0.244	0.012	0.862
$Z^{\text{Post-RSP}}$–χ^2	0.060	0.206	0.010	0.172	0.002	0.252
$Z^{\text{ISCV-RSP}}$–χ^2	0.096	0.264	0.028	0.300	0.018	0.418
$\text{PPC}_{y\text{–}\chi^2}$	0.004	< 0.001	0.010	< 0.001	0.012	< 0.001
$\text{HPC}_{y\text{–}\chi^2}$	0.352	0.230	0.224	0.070	0.160	0.024

S2 Additional Results of the Dolphin-sighting Data

S2.1 Posterior Summaries for the Selected Nonlinear HNB Model

Table S5: Regression coefficients for the selected nonlinear HNB model fitted to the dolphin-sighting data. Columns report posterior mean (Estimate), Monte Carlo SE (Est.Error), 95% credible interval, and $\hat{R}$.

	Estimate	Est.Error	l-95% CI	u-95% CI	$\hat{R}$
Intercept	9.16	6.61	-3.55	22.14	1.00
hu_Intercept	-11.25	6.62	-24.35	1.81	1.00
below7.95	-0.00	0.02	-0.05	0.04	1.00
above7.95	-0.00	0.01	-0.02	0.02	1.00
temp	-0.37	0.13	-0.63	-0.13	1.00
dr	0.00	0.03	-0.05	0.05	1.00
salinity	0.08	0.19	-0.30	0.45	1.00
turbidity	0.14	0.13	-0.11	0.39	1.00
year2	-0.52	0.41	-1.33	0.29	1.00
year3	0.34	0.34	-0.33	0.99	1.00
year4	1.98	1.49	-0.83	5.06	1.00
year5	0.43	0.66	-0.84	1.74	1.00
year6	-1.60	0.54	-2.65	-0.55	1.00
hu_below7.95	0.12	0.02	0.08	0.16	1.00
hu_above7.95	0.03	0.01	0.01	0.05	1.00
hu_temp	-0.16	0.12	-0.39	0.08	1.00
hu_dr	0.01	0.02	-0.04	0.06	1.00
hu_salinity	0.54	0.19	0.18	0.92	1.00
hu_turbidity	0.03	0.13	-0.23	0.29	1.00
hu_year2	-1.41	0.39	-2.17	-0.65	1.00
hu_year3	-0.37	0.33	-1.03	0.28	1.00
hu_year4	3.24	1.39	0.72	6.19	1.00
hu_year5	0.89	0.62	-0.33	2.09	1.00
hu_year6	-2.29	0.50	-3.29	-1.31	1.00

S2.2 Prior Sensitivity Analysis

Because Z-residuals are constructed from posterior or cross-validatory predictive distributions, the resulting diagnostics may depend on the prior specification, especially when the prior is informative or the sample size is small. We therefore conducted a prior sensitivity analysis for the dolphin-sighting data.

We compared three prior specifications: the baseline prior used in the main analysis, a tighter prior, and a more diffuse prior. The tight setting used smaller prior standard deviations for regression coefficients and intercepts, while the diffuse setting used larger prior standard deviations. The three prior specifications are summarized in Table S6.

Table S6: Prior specifications used in the prior sensitivity analysis for the dolphin-sighting data. Gamma priors use the shape-rate parameterization.

Prior component	Baseline	Tight	Diffuse
Regression coefficients	Normal$(0, 2)$	Normal$(0, 1)$	Normal$(0, 5)$
Intercepts	Normal$(0, 5)$	Normal$(0, 2.5)$	Normal$(0, 10)$
Dispersion/shape parameter	Gamma$(2, 0.5)$	Gamma$(4, 1)$	Gamma$(2, 0.1)$

Figure S6 shows the diagnostic p-values under the three prior specifications for the linear HNB, linear NB, nonlinear HNB, and nonlinear NB models. Overall, the diagnostic conclusions are qualitatively stable across the prior choices. The AOV diagnostic against depth continues to give small p-values for the linear HNB and NB models, indicating remaining nonlinear structure in depth, whereas the corresponding p-values are large for the nonlinear HNB and NB models. The Shapiro–Wilk and χ^2-type diagnostics also show similar qualitative patterns across the three prior settings. These results suggest that the main real-data conclusions are not driven by the particular baseline prior specification.

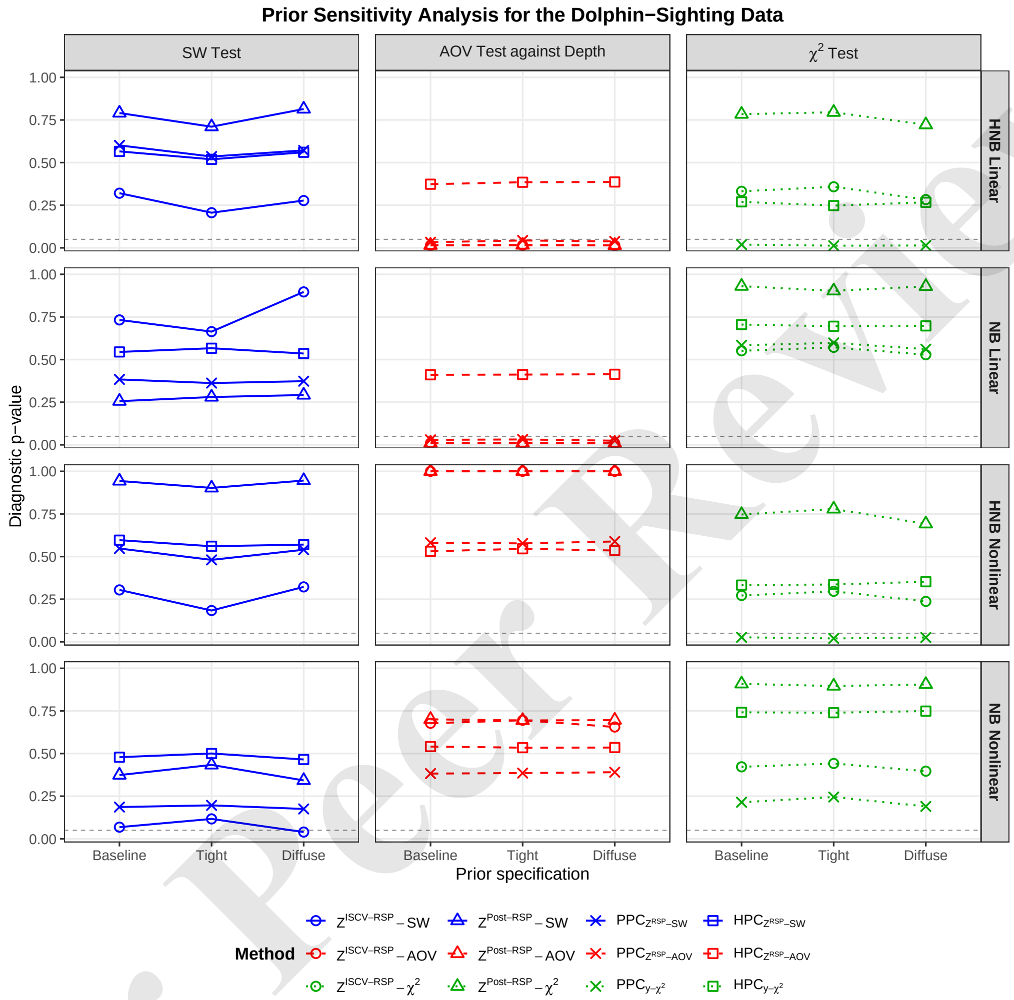

Figure S6: Prior sensitivity analysis for the dolphin-sighting data. Diagnostic p-values are shown for the linear HNB, linear NB, nonlinear HNB, and nonlinear NB models under three prior specifications: baseline, tight, and diffuse. Columns correspond to the Shapiro–Wilk test, the ANOVA test against depth, and the omnibus chi-square test. The horizontal dashed line marks the nominal 0.05 level. The qualitative diagnostic conclusions are stable across the three prior specifications: the ANOVA diagnostic continues to reject the linear HNB and NB models, while the nonlinear models are not rejected by the ANOVA diagnostic.

S2.3 Sensitivity of the Z-AOV Test to the Number of Groups

Figure S7 examines the sensitivity of the Z-AOV diagnostic to the number of intervals (k) of the variable `depth`. We recreated the grouped boxplots and AOV tests using $k \in \{5, 10, 15, 20\}$ equally spaced intervals. To account for the discrete response, we repeated the auxiliary randomization $J = 500$ times for each k, summarizing the resulting p-values via the conservative upper-bound p-value for the median (Eq. 16).

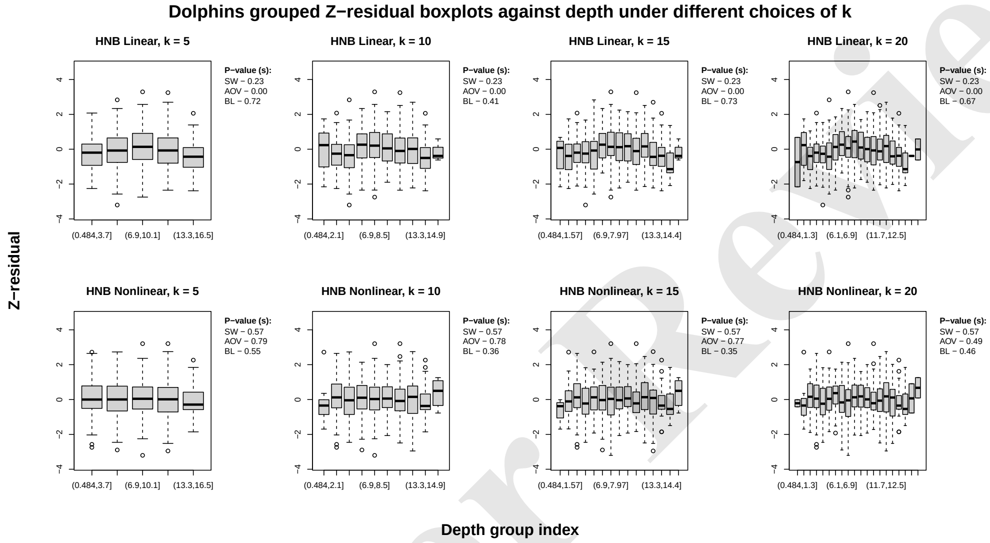

Figure S7: Grouped posterior RSP Z-residual boxplots against depth for the linear (top) and nonlinear (bottom) HNB models, evaluated under varying numbers of equally spaced intervals (k).

Table S7 reports these upper-bound p-values. Across all k, the linear HNB model consistently exhibits a residual mean pattern against depth, whereas the nonlinear model shows no remaining grouped differences. This confirms our depth diagnostic conclusions are robust to the default choice of $k = 10$.

Table S7: Sensitivity of the Z-AOV test to the number of depth groups (k) for the dolphin-sighting data. Smaller p-values indicate stronger evidence of residual mean differences. The default choice used in the main text is $k = 10$.

Model	$k = 5$	$k = 10$	$k = 15$	$k = 20$
Linear HNB	0.006	0.015	0.047	0.039
Nonlinear HNB	1.000	1.000	1.000	1.000

S3 Asymptotic Properties of Z-residuals

S3.1 Illustration of the Uniformity of RSP

We illustrate the uniformity of RSP using a toy example. We consider a Hurdle Exponential model, for which the survival function, $S(y)$, and probability mass function (PMF), $p(y)$, are defined as:

$$S(y) = \begin{cases} 1 & \text{if } y < 0 \\ (1-p_0)\cdot e^{-\lambda y} & \text{if } y \geq 0 \end{cases} \quad \text{and} \quad p(y) = \begin{cases} p_0 & \text{if } y = 0 \\ 0 & \text{if } y \neq 0 \end{cases} \tag{S1}$$

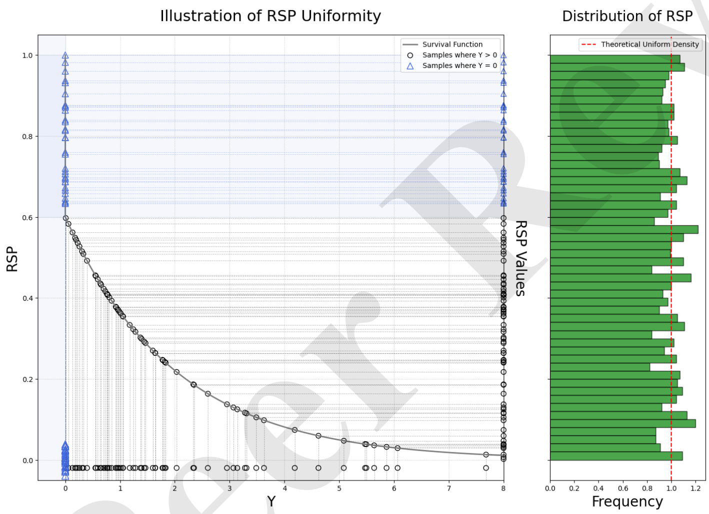

Figure S8: The plot displays the survival function alongside randomized survival probabilities (RSP) for a discrete distribution. The histogram on the right shows the distribution of 5000 RSP values.

Figure S8 (left panel) shows the survival function for this model, which has a discrete jump at $y = 0$ corresponding to the probability mass p_0. To illustrate the theorem, we generated 5000 observations (y_i) from this distribution and computed the Randomized Survival Probability (RSP) for each. The calculation depends on the value of the observation. For an observation from the continuous part ($y_i > 0$), where the probability mass $p(y_i) = 0$, the RSP is simply its survival probability, $\text{RSP}_i = S(y_i)$; this deterministic mapping is shown by the hollow circles. For an observation from the discrete part ($y_i = 0$), the RSP is calculated as $\text{RSP}_i = S(0) + U_i \cdot p_0$, using a random draw $U_i \sim \text{Unif}(0, 1)$. As shown by the triangles, this process randomly maps each observation where $y_i = 0$ to a uniform value within the interval $[S(0), 1]$, effectively distributing the probability mass p_0 across this range (the shaded blue region). The right panel of Figure S8 displays the histogram of all 5000 calculated RSP values, which closely follows a uniform distribution on $(0, 1)$.

S3.2 Proof of the Uniformity of RSP Under the True Model

Theorem S3.1 (Uniformity of the Randomized Survival Probability)**.** *Let Y be any random variable on $\mathbb{R}$ with survival function $S(y) = P(Y > y)$ and probability mass function $p(y) = P(Y = y)$, which is 0 when $S(y)$ is continuous at y. Let $U \sim$ Uniform$(0, 1)$ be independent of Y. Then the Randomized Survival Probability (RSP), defined as*

$$V = S(Y) + U \cdot p(Y), \tag{S2}$$

is a standard uniform random variable, i.e., $V \sim$ Uniform$(0, 1)$.

Proof: To prove that V is a standard uniform random variable, we will show that for any interval $A \subseteq [0, 1]$, the probability $P(V \in A)$ is equal to its length (Lebesgue measure), $\lambda(A)$. The proof proceeds in four steps.

Step 1 (*Setup and Conditional Expectation*)**.** We express the probability $P(V \in A)$ using the law of total expectation:

$$P(V \in A) = \mathbb{E}[\mathbf{1}_A(V)] = \mathbb{E}\left[\mathbb{E}[\mathbf{1}_A(S(Y) + U \cdot p(Y)) \mid Y]\right] \tag{S3}$$

Let's analyze the inner conditional expectation, which we denote $g(y) = E_U[\mathbf{1}_A(S(y) + U \cdot p(y))]$. For any fixed y, the conditional random variable $V \mid Y = y$ is uniformly distributed on the interval $I_y^V = [S(y), S(y) + p(y)]$. Let $D = \{y_k : p(y_k) > 0\}$ be the set of discrete mass points of Y.

- **For a discrete point** $y_k \in D$, the interval $I_{y_k}^V$ has length $p(y_k) > 0$. The conditional expectation is the probability that a uniform variable on this interval falls in A:

$$g(y_k) = \frac{\lambda(A \cap I_{y_k}^V)}{\lambda(I_{y_k}^V)} = \frac{\lambda(A \cap I_{y_k}^V)}{p(y_k)} \tag{S4}$$

- **For a continuous point** $y \notin D$, we have $p(y) = 0$. The interval I_y^V collapses to a single point, and the conditional expectation becomes:

$$g(y) = \mathbb{E}[\mathbf{1}_A(S(y))] = \mathbf{1}_A(S(y)) \tag{S5}$$

Step 2 (*Contribution from Discrete Points*)**.** We now evaluate the outer expectation by first integrating over the discrete part of Y's distribution. This contribution is a sum:

$$\int_D g(y)\,dF(y) = \sum_{y_k \in D} g(y_k)p(y_k) = \sum_{y_k \in D} \left(\frac{\lambda(A \cap I^V_{y_k})}{p(y_k)} \right) p(y_k) = \sum_{y_k \in D} \lambda(A \cap I^V_{y_k}) \tag{S6}$$

Step 3 (*Contribution from Continuous Points*)**.** Next, we integrate over the continuous part of the distribution, $C = \mathbb{R} \setminus D$. Let the density function on this set be $f_c(y)$.

$$\int_C g(y)\,dF(y) = \int_C \mathbf{1}_A(S(y)) f_c(y)\,dy \tag{S7}$$

We perform a change of variables from y to $v = S(y)$, for which the differential is $dv = -S'(y)dy = -(-f_c(y))dy = f_c(y)dy$. As y sweeps across the set C, the variable $v = S(y)$ sweeps across the set $I^V_C = [0,1] \setminus \bigcup_k I^V_{y_k}$. The integral becomes:

$$\int_{I^V_C} \mathbf{1}_A(v)\,dv = \lambda(A \cap I^V_C) \tag{S8}$$

Step 4 (*Conclusion*)**.** Finally, we add the contributions from the discrete and continuous parts:

$$P(V \in A) = \sum_{y_k \in D} \lambda(A \cap I^V_{y_k}) + \lambda(A \cap I^V_C) \tag{S9}$$

The sets $\{I^V_{y_k}\}_{y_k \in D}$ and I^V_C form a partition of the unit interval $[0,1]$. Therefore, their intersections with the set A form a partition of A. The sum of the lengths of these disjoint intersecting sets is simply the total length of A:

$$P(V \in A) = \lambda\left(\left(\bigcup_k (A \cap I^V_{y_k}) \right) \cup (A \cap I^V_C) \right) = \lambda(A) \tag{S10}$$

Since $P(V \in A) = \lambda(A)$ for any Borel set $A \subseteq [0,1]$, the random variable V is, by definition, uniformly distributed on $[0,1]$. □

S3.3 Asymptotic Uniformity of the CV and Posterior RSP

Theorem S3.2 (Total Variation Convergence of the LOO Predictive Distribution)**.** *Let $Y_1, \ldots, Y_{n+1}$ be independent random variables with true probability laws $P_i^{(*)}$ possessing densities $f_i^{(*)}$. Consider a statistical model $\{f_i(\cdot \mid \theta) : \theta \in \Theta\}$ and a prior law Π. Assume the model is well-specified, meaning the set of true parameters*

$$W_0 = \{\theta \in \Theta : f_i(\cdot \mid \theta) = f_i^{(*)} \textit{ for all } i\} \tag{S11}$$

is non-empty. Let $\theta^\star$ be any representative element from W_0.

For each i, let $\Pi(\cdot \mid \mathbf{Y}_{-i})$ be the posterior law based on the LOO sample $\mathbf{Y}_{-i}$ of size n. The LOO predictive density is $f_i^{(n),\mathrm{CV}}(y) = \int_\Theta f_i(y \mid \theta)\, d\Pi(\theta \mid \mathbf{Y}_{-i})$. Let $\bar{K}_{n,-i}(\theta, \theta^\star)$ and $\bar{h}_{n,-i}(\theta, \theta^\star)$ be the average KL divergence and Hellinger distance from the true distribution. Assume the following conditions hold for some sequence $\varepsilon_n \downarrow 0$ and sieves $\Theta_n \subset \Theta$:

(i) *Measurability: The function $(y, \theta) \mapsto f_i(y \mid \theta)$ is measurable for each i.*

(ii) *Prior Mass Condition: The prior law assigns sufficient mass to Kullback-Leibler neighborhoods of the true parameter set W_0, i.e., for some constant $C > 0$:*

$$\Pi(\{\theta : \bar{K}_{n,-i}(\theta, \theta^\star) \le \varepsilon_n^2\}) \ge e^{-Cn\varepsilon_n^2}. \tag{S12}$$

(iii) *Existence of Tests: There exist uniformly consistent tests for Hellinger neighborhoods of W_0. That is, for every $\delta > 0$, there exists a sequence of tests ϕ_n and a constant $c > 0$ such that for large n:*

$$\mathbb{E}_{\theta^\star}\phi_n \le e^{-cn\varepsilon_n^2} \quad \textit{and} \quad \sup_{\{\theta \in \Theta_n : \bar{h}_{n,-i}(\theta,\theta^\star) > \delta\}} \mathbb{E}_\theta(1 - \phi_n) \le e^{-cn\varepsilon_n^2}. \tag{S13}$$

(iv) *Complexity Control: The parameter space has controlled complexity, meaning the entropy of the sieves does not grow too quickly and the prior mass outside the sieves is exponentially small:*

$$\log N(\varepsilon_n, \Theta_n, \bar{h}_{n,-i}) \le Cn\varepsilon_n^2 \quad \textit{and} \quad \Pi(\Theta_n^c) \le e^{-Dn\varepsilon_n^2}, \tag{S14}$$

for constants $C, D > 0$, where the term $N(\varepsilon_n, \Theta_n, \bar{h}_{n,-i})$ denotes the covering number indicating the complexity of a model, see Ghosal & van der Vaart (2007) for precise definition.

(v) *L^1 Continuity: For each fixed i, the map $\theta \mapsto f_i(\cdot \mid \theta)$ is continuous in the $L^1(\mu)$ norm on a neighborhood of the set W_0. That is, for every $\epsilon > 0$, there exists an open set U_ϵ containing W_0 such that $\sup_{\theta \in U_\epsilon} \|f_i(\cdot \mid \theta) - f_i^{(*)}\|_{L^1(\mu)} < \epsilon$.*

Then for each fixed i, as $n \to \infty$, the LOO predictive distribution converges to the true distribution in total variation, in probability:

$$\left\| P_i^{(n),\mathrm{CV}} - P_i^{(*)} \right\|_{\mathrm{TV}} \xrightarrow{P} 0. \tag{S15}$$

Proof: The cornerstone of the proof is posterior consistency with respect to the true parameter set W_0. Conditions (i)–(iv) are sufficient for Schwartz-type theorems (Ghosal &

van der Vaart 2007, Choi & Ramamoorthi 2008), which guarantee that the LOO posterior law $\Pi(\cdot \mid \mathbf{Y}_{-i})$ concentrates on Hellinger neighborhoods of the true density. In the parameter space, this means the posterior concentrates on the set of parameters that produce the true density, which is exactly W_0. Formally, for any open set U containing the set W_0, it follows that:

$$\Pi(U^c \mid \mathbf{Y}_{-i}) \xrightarrow{P} 0 \quad \text{as } n \to \infty. \tag{S16}$$

With this rigorous consistency result, we now prove the convergence of the predictive distribution. Let $\Delta_n := \|f_i^{(n),\mathrm{CV}} - f_i^{(*)}\|_{L^1(\mu)}$. Our goal is to show $\Delta_n \xrightarrow{P} 0$.

By the triangle inequality, we bound Δ_n by the posterior expectation of the L^1 error:

$$\Delta_n \le \int_\Theta \|f_i(\cdot \mid \theta) - f_i^{(*)}\|_{L^1(\mu)}\, d\Pi(\theta \mid \mathbf{Y}_{-i}). \tag{S17}$$

For any $\epsilon > 0$, Condition (v) guarantees the existence of an open neighborhood U_ϵ containing the entire set W_0 where $\|f_i(\cdot \mid \theta) - f_i^{(*)}\|_{L^1(\mu)} < \epsilon$. Decomposing the integral over this U_ϵ and its complement yields:

$$\Delta_n \le \int_{U_\epsilon} \epsilon\, d\Pi(\theta \mid \mathbf{Y}_{-i}) + \int_{U_\epsilon^c} 2\, d\Pi(\theta \mid \mathbf{Y}_{-i}) \le \epsilon + 2\,\Pi(U_\epsilon^c \mid \mathbf{Y}_{-i}). \tag{S18}$$

Since U_ϵ is an open set containing W_0, our consistency result from (S16) directly implies that $\Pi(U_\epsilon^c \mid \mathbf{Y}_{-i}) = o_P(1)$. Thus, for any arbitrarily small $\epsilon > 0$,

$$\Delta_n \le \epsilon + o_P(1). \tag{S19}$$

This implies that $\Delta_n \xrightarrow{P} 0$, which completes the proof. □

Theorem S3.3 (Asymptotic Uniformity of the Cross-Validated RSP)**.** *Under the same setup and conditions specified in Theorem S3.2, let $S_i^{(n),\mathrm{CV}}(\cdot \mid \mathbf{Y}_{-i})$ and $p_i^{(n),\mathrm{CV}}(\cdot \mid \mathbf{Y}_{-i})$ be the survival function and probability mass function, respectively, corresponding to the LOO predictive density $f_i^{(n),\mathrm{CV}}$.*

*The cross-validated Randomized Survival Probability (*RSP*) for observation Y_i is defined as:*

$$\mathrm{RSP}_i^{(n),\mathrm{CV}} = S_i^{(n),\mathrm{CV}}(Y_i \mid \mathbf{Y}_{-i}) + U_i \cdot p_i^{(n),\mathrm{CV}}(Y_i \mid \mathbf{Y}_{-i}), \quad \textit{where } U_i \overset{i.i.d.}{\sim} \mathrm{Unif}(0,1) \tag{S20}$$

Then, as $n \to \infty$, the cross-validated RSP converges in distribution to a standard uniform random variable:

$$\mathrm{RSP}_i^{(n),\mathrm{CV}} \Rightarrow \mathrm{Unif}(0,1). \tag{S21}$$

Proof: The proof is established in four parts. We first state the key result from Theorem S3.2, then show that the $\mathrm{RSP}_i^{(n),\mathrm{CV}}$ converges in probability to a known uniform random variable, and finally conclude with Slutsky's theorem.

Step 1 (*Convergence in Total Variation*)**.** The conditions assumed in Theorem S3.2 are sufficient to guarantee that the LOO predictive distribution, $P_i^{(n),\mathrm{CV}}$, converges to the true data-generating distribution, $P_i^{(*)}$, in total variation (TV) distance, in probability. That is,

$$\|P_i^{(n),\mathrm{CV}} - P_i^{(*)}\|_{\mathrm{TV}} \xrightarrow{P} 0 \quad \text{as} \quad n \to \infty. \tag{S22}$$

Step 2 (*Bounding the Difference Between the RSP and Its Limit*)**.** Our goal is to show that $\mathrm{RSP}_i^{(n),\mathrm{CV}}$ converges in distribution to a Unif0,1) random variable. We begin by defining this target variable as $V_i := S_i^{(*)}(Y_i) + U_i p_i^{(*)}(Y_i)$, where $S_i^{(*)}$ and $p_i^{(*)}$ are the survival function and PMF of the true distribution $P_i^{(*)}$. By the probability integral transform, V_i is exactly distributed as $\mathrm{Unif}(0,1)$.

We now bound the absolute difference between the RSP and its target, $D_n := |\mathrm{RSP}_i^{(n),\mathrm{CV}} - V_i|$, using the triangle inequality:

$$D_n \leq \left|S_i^{(n),\mathrm{CV}}(Y_i \mid \mathbf{Y}_{-i}) - S_i^{(*)}(Y_i)\right| + \left|p_i^{(n),\mathrm{CV}}(Y_i \mid \mathbf{Y}_{-i}) - p_i^{(*)}(Y_i)\right|. \tag{S23}$$

Step 3 (*Convergence of the Bounded Components*)**.** We now show that the bound in (S23) converges to zero in probability by leveraging the TV convergence from (S22).

- **Survival Term:** The maximum absolute difference between two survival functions is bounded by the total variation distance between their corresponding probability measures. Thus, for any value y,

$$|S_i^{(n),\mathrm{CV}}(y \mid \mathbf{Y}_{-i}) - S_i^{(*)}(y)| \leq \|P_i^{(n),\mathrm{CV}} - P_i^{(*)}\|_{\mathrm{TV}}. \tag{S24}$$

  Since the TV distance converges to zero in probability, the first term in the bound, evaluated at the random observation Y_i, also converges to zero in probability.

- **PMF Term:** Similarly, the absolute difference between two probability mass functions at any single point is also bounded by the TV distance. For continuous distributions, both $p_i^{(n),\mathrm{CV}}$ and $p_i^{(*)}$ are zero. In either case, the second term in the bound converges to zero in probability.

Since both terms on the right-hand side of (S23) converge to zero in probability, their sum, which is the upper bound for D_n, must also converge to zero in probability.

Step 4 (*Conclusion via Slutsky's Theorem*)**.** From the previous step, we have established that $|\mathrm{RSP}_i^{(n),\mathrm{CV}} - V_i| \xrightarrow{P} 0$, which is equivalent to the statement that the sequence of random variables $\mathrm{RSP}_i^{(n),\mathrm{CV}} - V_i$ converges to 0 in probability. By Slutsky's theorem, $\mathrm{RSP}_i^{(n),\mathrm{CV}}$ must converge in distribution to the same limit as V_i. Therefore, we conclude that:

$$\mathrm{RSP}_i^{(n),\mathrm{CV}} \;\Rightarrow\; \mathrm{Unif}(0,1). \tag{S25}$$

□

Theorem S3.4 (Asymptotic Uniformity of the Posterior RSP)**.** *Under the same setup and conditions specified in Theorem S3.2, let the full sample be $\mathbf{Y} = (Y_1, \ldots, Y_{n+1})$. For any fixed $i \in \{1, \ldots, n+1\}$, let the full-data posterior predictive distribution for Y_i be denoted $P_i^{(n+1),\mathrm{post}}$, with density $f_i^{(n+1),\mathrm{post}}(y_i \mid \mathbf{Y})$. Let $S_i^{(n+1),\mathrm{post}}$ and $p_i^{(n+1),\mathrm{post}}$ be the corresponding survival function and PMF.*

*The posterior Randomized Survival Probability (*RSP*) is defined as:*

$$\mathrm{RSP}_i^{(n+1),\mathrm{post}} \;=\; S_i^{(n+1),\mathrm{post}}(Y_i \mid \mathbf{Y}) \;+\; U_i \cdot p_i^{(n+1),\mathrm{post}}(Y_i \mid \mathbf{Y}) \tag{S26}$$

where $U_i \sim \mathrm{Unif}(0,1)$ is an independent random variable.

Then, for each fixed i, as $n \to \infty$, the posterior RSP converges in distribution to a standard uniform random variable:

$$\mathrm{RSP}_i^{(n+1),\mathrm{post}} \;\Rightarrow\; \mathrm{Unif}(0,1). \tag{S27}$$

Proof: The proof demonstrates that the posterior RSP is asymptotically equivalent to the cross-validated RSP from Theorem S3.3, whose asymptotic uniformity has already been established.

Step 1 (*Convergence of Predictive Distributions*)**.** The conditions of Theorem S3.2 ensure that any posterior predictive distribution based on a sufficiently large sample will converge to the true data-generating distribution. This applies to both the full-data posterior predictive (based on $n+1$ points) and the leave-one-out (LOO) predictive (based on n points). As $n \to \infty$, both converge to the true distribution $P_i^{(*)}$ in total variation:

$$\|P_i^{(n+1),\mathrm{post}} - P_i^{(*)}\|_{\mathrm{TV}} \xrightarrow{P} 0 \quad \text{and} \quad \|P_i^{(n),\mathrm{CV}} - P_i^{(*)}\|_{\mathrm{TV}} \xrightarrow{P} 0. \tag{S28}$$

By the triangle inequality, the distance between the full-data and LOO predictive distributions must therefore also converge to zero:

$$\|P_i^{(n+1),\text{post}} - P_i^{(n),\text{CV}}\|_{\text{TV}} \leq \|P_i^{(n+1),\text{post}} - P_i^{(*)}\|_{\text{TV}} + \|P_i^{(*)} - P_i^{(n),\text{CV}}\|_{\text{TV}} \xrightarrow{P} 0. \tag{S29}$$

Step 2 (*Asymptotic Equivalence of RSPs*)**.** Next, we show that the difference between the posterior RSP and the cross-validated RSP vanishes asymptotically. This difference, $D_n := \text{RSP}_i^{(n+1),\text{post}} - \text{RSP}_i^{(n),\text{CV}}$, can be bounded by the TV distance between the two predictive distributions:

$$\begin{aligned}
|D_n| &= \left|\left(S_i^{(n+1),\text{post}}(Y_i|\mathbf{Y}) - S_i^{(n),\text{CV}}(Y_i|\mathbf{Y}_{-i})\right) + U_i\left(p_i^{(n+1),\text{post}}(Y_i|\mathbf{Y}) - p_i^{(n),\text{CV}}(Y_i|\mathbf{Y}_{-i})\right)\right| \\
&\leq \left|S_i^{(n+1),\text{post}}(Y_i|\mathbf{Y}) - S_i^{(n),\text{CV}}(Y_i|\mathbf{Y}_{-i})\right| + \left|p_i^{(n+1),\text{post}}(Y_i|\mathbf{Y}) - p_i^{(n),\text{CV}}(Y_i|\mathbf{Y}_{-i})\right| \\
&\leq 2 \cdot \|P_i^{(n+1),\text{post}} - P_i^{(n),\text{CV}}\|_{\text{TV}}.
\end{aligned}$$

From the result in (S29), the TV distance on the right converges to zero in probability. Therefore, $|D_n| \xrightarrow{P} 0$, which gives us the key asymptotic equivalence:

$$\text{RSP}_i^{(n+1),\text{post}} - \text{RSP}_i^{(n),\text{CV}} \xrightarrow{P} 0. \tag{S30}$$

Step 3 (*Conclusion via Slutsky's Theorem*)**.** We can express the posterior RSP as the sum of two terms: the cross-validated RSP and a residual term that converges to zero in probability. From (S30), we have:

$$\text{RSP}_i^{(n+1),\text{post}} = \text{RSP}_i^{(n),\text{CV}} + o_P(1). \tag{S31}$$

We have already proven in Theorem S3.3 that the cross-validated RSP converges in distribution to a standard uniform variable:

$$\text{RSP}_i^{(n),\text{CV}} \Rightarrow \text{Unif}(0, 1). \tag{S32}$$

By Slutsky's Theorem, the sum of a sequence of random variables that converges in distribution and another that converges in probability to zero will converge in distribution to the same limit. Therefore, we conclude that:

$$\text{RSP}_i^{(n+1),\text{post}} \;\Rightarrow\; \text{Unif}(0, 1). \tag{S33}$$

□

S3.4 Asymptotic Independence of the Posterior and CV RSPs

Theorem S3.5 (Asymptotic Independence of Cross-Validated RSPs)**.** *Under the same assumptions as the previous theorem (for independent, non-i.i.d. data), for any fixed pair of distinct indices $i \neq j$, the random vector of cross-validated Randomized Survival Probabilities converges in distribution to a pair of independent standard uniform random variables. Formally, as the sample size $n \to \infty$:*

$$(RSP_i^{(n),CV}, RSP_j^{(n),CV}) \;\Rightarrow\; (Uniform(0,1), Uniform(0,1)) \tag{S34}$$

where the two uniform random variables in the limiting distribution are independent.

Proof: The proof strategy is to show that the random vector $(\mathrm{RSP}_i^{(n),\mathrm{CV}}, \mathrm{RSP}_j^{(n),\mathrm{CV}})$ converges in distribution to the vector of RSPs under the true parameters, (V_i, V_j). Since Y_i and Y_j are independent, the limiting variables V_i and V_j will be independent, establishing the theorem.

Step 1 (*Component-wise Convergence in Probability*)**.** In the proof of the previous theorem, we established that for any fixed i, the absolute difference between the estimated RSP and its true-parameter counterpart converges to zero in probability. Let $V_i := S_i^{(*)}(Y_i) + U_i p_i^{(*)}(Y_i)$ and $V_j := S_j^{(*)}(Y_j) + U_j p_j^{(*)}(Y_j)$. We have already shown:

$$|\mathrm{RSP}_i^{(n),\mathrm{CV}} - V_i| \xrightarrow{P} 0 \tag{S35}$$

and by identical logic for index j:

$$|\mathrm{RSP}_j^{(n),\mathrm{CV}} - V_j| \xrightarrow{P} 0. \tag{S36}$$

Step 2 (*Joint Convergence in Probability*)**.** Consider the 2-dimensional random vector formed by the differences:

$$\mathbf{D}_n = \left(\mathrm{RSP}_i^{(n),\mathrm{CV}} - V_i, \quad \mathrm{RSP}_j^{(n),\mathrm{CV}} - V_j\right). \tag{S37}$$

A random vector converges in probability to a constant vector if and only if each of its components converges in probability to the corresponding component of the constant vector (Shao 2008, Thm 1.9). Since we established in Step 1 that each component of $\mathbf{D}_n$ converges in probability to 0, the vector itself converges in probability to the zero vector:

$$\mathbf{D}_n \xrightarrow{P} (0,0). \tag{S38}$$

Step 3 (*Independence of the Limiting Variables*). The limiting random vector is (V_i, V_j). By definition, V_i is a function of only (Y_i, U_i) and V_j is a function of only (Y_j, U_j). By the theorem's initial setup, for $i \neq j$, the random variables Y_i and Y_j are independent. The auxiliary variables U_i and U_j are also independent of each other and of all Y variables. Since V_i and V_j are functions of disjoint sets of independent random variables, they are themselves independent. Each is marginally distributed as Uniform$(0, 1)$.

Step 4 (*Conclusion via Multivariate Slutsky's Theorem*). We can write the vector of estimated RSPs as the sum of the true RSP vector and the difference vector $\mathbf{D}_n$:

$$(\mathrm{RSP}_i^{(n),\mathrm{CV}}, \mathrm{RSP}_j^{(n),\mathrm{CV}}) = (V_i, V_j) + \mathbf{D}_n. \tag{S39}$$

We have established that the distribution of (V_i, V_j) is a pair of independent Uniform$(0, 1)$ variables, and that $\mathbf{D}_n \xrightarrow{P} (0, 0)$. By the multivariate version of Slutsky's theorem, the vector $(\mathrm{RSP}_i^{(n),\mathrm{CV}}, \mathrm{RSP}_j^{(n),\mathrm{CV}})$ must converge in distribution to the same limit as (V_i, V_j).

□

Theorem S3.6 (Asymptotic Independence of Posterior RSPs). *Under the same setup and conditions as in Theorem S3.2, for any fixed pair of distinct indices $i \neq j$, the random vector of posterior RSPs converges in distribution to a pair of independent standard uniform random variables.*

Formally, as $n \to \infty$:

$$(\mathrm{RSP}_i^{(n+1),\mathrm{post}}, \mathrm{RSP}_j^{(n+1),\mathrm{post}}) \;\Rightarrow\; (\mathrm{Unif}(0, 1), \mathrm{Unif}(0, 1)) \tag{S40}$$

where the two uniform random variables in the limiting distribution are independent.

Proof: The proof relies on the multivariate Slutsky's theorem. We first establish that the vector of posterior RSPs is asymptotically equivalent to the vector of cross-validated RSPs.

From the proof of Theorem S3.4, we know that for any single observation k, the difference between the posterior and cross-validated RSP converges to zero in probability: $\mathrm{RSP}_k^{(n+1),\mathrm{post}} - \mathrm{RSP}_k^{(n),\mathrm{CV}} \xrightarrow{P} 0$. Since convergence in probability for a random vector is equivalent to component-wise convergence (Shao 2008), the difference vector also converges in probability to the zero vector:

$$\left(\mathrm{RSP}_i^{(n+1),\mathrm{post}} - \mathrm{RSP}_i^{(n),\mathrm{CV}}, \quad \mathrm{RSP}_j^{(n+1),\mathrm{post}} - \mathrm{RSP}_j^{(n),\mathrm{CV}}\right) \xrightarrow{P} (0, 0). \tag{S41}$$

This establishes the asymptotic equivalence of the two RSP vectors. Next, we consider the limiting distribution of the cross-validated vector, $(\mathrm{RSP}_i^{(n),\mathrm{CV}}, \mathrm{RSP}_j^{(n),\mathrm{CV}})$. As shown in Theorem S3.3, each component converges in distribution to a standard uniform. Furthermore, $\mathrm{RSP}_i^{(n),\mathrm{CV}}$ is asymptotically a function of only Y_i, and $\mathrm{RSP}_j^{(n),\mathrm{CV}}$ is asymptotically a function of only Y_j. Because the original observations Y_i and Y_j are independent for $i \neq j$, the RSPs are asymptotically independent. Thus, the vector converges in distribution:

$$(\mathrm{RSP}_i^{(n),\mathrm{CV}}, \mathrm{RSP}_j^{(n),\mathrm{CV}}) \ \Rightarrow \ (\mathrm{Unif}(0,1), \mathrm{Unif}(0,1)), \tag{S42}$$

where the two uniform random variables in the limiting distribution are independent.

We can now write the posterior RSP vector as the sum of a vector that converges in distribution and a vector that converges in probability to zero. By the multivariate Slutsky's theorem, the posterior RSP vector must converge in distribution to the same limit, which concludes the proof. □

S3.5 Asymptotic Uniformity of Tests Based on Z-residuals

Theorem S3.7 (Asymptotic Uniformity of Tests Based on Z-Residuals)**.** *Let the assumptions of the preceding theorems hold, ensuring that a set of n Randomized Survival Probabilities, $\{\mathrm{RSP}_1, \ldots, \mathrm{RSP}_n\}$, are asymptotically independent and identically distributed (i.i.d.) as $\mathrm{Unif}(0,1)$. Define the Z-residuals as $Z_i = \Phi^{-1}(\mathrm{RSP}_i)$, where Φ^{-1} is the standard normal quantile function.*

Let $T^{(n)} = T(Z_1, \ldots, Z_n)$ be any test statistic that is a continuous function of the Z-residuals, and assume its distribution under the null hypothesis (where $Z_i^* \sim_{i.i.d.} N(0,1)$) is continuous. Let $\mathcal{P}^{(n)}$ be the p-value from this test.

Then, as $n \to \infty$, the p-value converges in distribution to a standard uniform random variable:

$$\mathcal{P}^{(n)} \ \Rightarrow \ \mathrm{Unif}(0,1). \tag{S43}$$

Proof: The proof follows a chain of transformations, where each step preserves the convergence in distribution.

By assumption, the vector of RSPs converges in distribution to a vector of i.i.d. standard uniform variables. Since the normal quantile function Φ^{-1} is continuous, the Contunuous Mapping Theorem (CMT) implies that the vector of Z-residuals $(Z_1, \ldots, Z_n)$ converges in distribution to a vector of i.i.d. standard normal variables, $(Z_1^*, \ldots, Z_n^*)$.

Next, since the test statistic $T(\cdot)$ is also a continuous function, a second application of the CMT shows that the test statistic $T^{(n)} = T(Z_1, \ldots, Z_n)$ converges in distribution to its null distribution, $T^{*(n)} = T(Z_1^*, \ldots, Z_n^*)$.

Finally, the p-value is the survival function of the null distribution evaluated at the test statistic, $\mathcal{P}^{(n)} = S_{T^{*(n)}}(T^{(n)})$. A fundamental result of probability theory states that if $X_n \Rightarrow X_0$ and the distribution of X_0 is continuous, then $S_{X_0}(X_n) \Rightarrow \text{Unif}(0,1)$. Applying this to our case, where $T^{(n)} \Rightarrow T^{*(n)}$ and the distribution of $T^{*(n)}$ is continuous, we conclude that:

$$\mathcal{P}^{(n)} \Rightarrow \text{Unif}(0,1). \tag{S44}$$

□

S3.6 Distribution-free Upper-bound p-value for the Order Statistics of Replicated p-values

Theorem S3.8. *Let $\mathcal{P}(\boldsymbol{y}, \boldsymbol{U}) \in [0,1]$ be a p-value computed from data $\boldsymbol{y}$ and auxiliary randomness $\boldsymbol{U}$. Assume:*

1. *$\mathcal{P}(\boldsymbol{Y}, \boldsymbol{U})$ is asymptotically Unif0,1) under the true model for the random data $\boldsymbol{Y}$.*
2. *Conditional on $\boldsymbol{Y}$, the auxiliary draws $\boldsymbol{U}^{(1)}, \ldots, \boldsymbol{U}^{(J)}$ are i.i.d. and independent of $\boldsymbol{Y}$.*

For $j = 1, \ldots, J$, define the replicated p-values $p_j(\boldsymbol{y}) = \mathcal{P}(\boldsymbol{y}, \boldsymbol{U}^{(j)})$ and let $p_{(r)}(\boldsymbol{y})$ be their r-th order statistic. Let $H_{J,r}$ be a calibration factor defined by:

$$H_{J,r} := \sup_{p \in (0,1]} \frac{P\{\text{Bin}(J,p) \geq r\}}{p} \tag{S45}$$

Then, for any threshold $t \in [0,1]$,

$$P\{p_{(r)}(\boldsymbol{Y}) \leq t\} \leq H_{J,r} \cdot t. \tag{S46}$$

Proof: The proof proceeds in two main steps: first analyzing the system conditional on the data, and second, averaging over the data randomness to obtain the final bound.

Step 1 (*Conditional Analysis and the Binomial Distribution*)**.** First, we analyze the system for a fixed dataset $\boldsymbol{Y} = \boldsymbol{y}$ and a given threshold $t \in [0,1]$. Let $P_t(\boldsymbol{y}) := P\{\mathcal{P}(\boldsymbol{y}, \boldsymbol{U}) \leq t \mid \boldsymbol{Y} = \boldsymbol{y}\}$ be the probability that a single replicated p-value is less than or equal to t.

Since the J replicates are independent conditional on the data (Assumption 2), they form a series of J Bernoulli trials with success probability $P_t(\boldsymbol{y})$. The event that the r-th order statistic is below the threshold, $\{p_{(r)}(\boldsymbol{y}) \leq t\}$, is equivalent to the event that **at least** r of these trials are successful.

Therefore, the conditional probability of this event is exactly the tail probability of a binomial distribution:

$$P\{p_{(r)}(\boldsymbol{y}) \le t \mid \boldsymbol{Y} = \boldsymbol{y}\} = P\{\text{Bin}(J, P_t(\boldsymbol{y})) \ge r\}. \tag{S47}$$

Step 2 (*Unconditional Bound via Expectation*)**.** To find the unconditional probability, we take the expectation of the result from the previous step over the distribution of the random data $\boldsymbol{Y}$:

$$P\{p_{(r)}(\boldsymbol{Y}) \le t\} = \text{E}_{\boldsymbol{Y}}\left[P\{\text{Bin}(J, P_t(\boldsymbol{Y})) \ge r\}\right]. \tag{S48}$$

By the definition of the constant $H_{J,r}$, we know that $P\{\text{Bin}(J,p) \ge r\} \le H_{J,r} \cdot p$ for all $p \in (0,1]$. Applying this inequality inside the expectation gives:

$$P\{p_{(r)}(\boldsymbol{Y}) \le t\} \le \text{E}_{\boldsymbol{Y}}[H_{J,r} \cdot P_t(\boldsymbol{Y})] = H_{J,r} \cdot \text{E}_{\boldsymbol{Y}}[P_t(\boldsymbol{Y})]. \tag{S49}$$

Finally, by the law of total expectation, the term $\text{E}_{\boldsymbol{Y}}[P_t(\boldsymbol{Y})]$ is simply the total probability $P\{\mathcal{P}(\boldsymbol{Y}, \boldsymbol{U}) \le t\}$. By Assumption 1, this probability is t because the p-value is uniformly distributed. Substituting this in yields the final bound:

$$P\{p_{(r)}(\boldsymbol{Y}) \le t\} \le H_{J,r} \cdot t. \tag{S50}$$

Remark: For the median ($r = \lceil J/2 \rceil$), the constant is well-behaved, with $H_{J,r} < 2$ for all J. For example, $H_{100,50} \approx 1.65$, and $H_{500,250} \approx 1.80$. This yield a better bound than 2 for the median of correlated sample (Johnson 2007, Yuan & Johnson 2012), which is derived with Markov inequality:

$$P\{p_{(r)}(\boldsymbol{y}) \le t \mid \boldsymbol{Y} = \boldsymbol{y}\} = P\{\text{Bin}(J, P_t(\boldsymbol{y})) \ge r\} < \mathbb{E}(\text{Bin}(J, P_t(\boldsymbol{y})))/r = J/r \tag{S51}$$

By the law of iterated expectation, $P\{p_{(r)}(\boldsymbol{Y}) \le t\} < J/r$. For meadian, $J/r \approx 2$. □

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